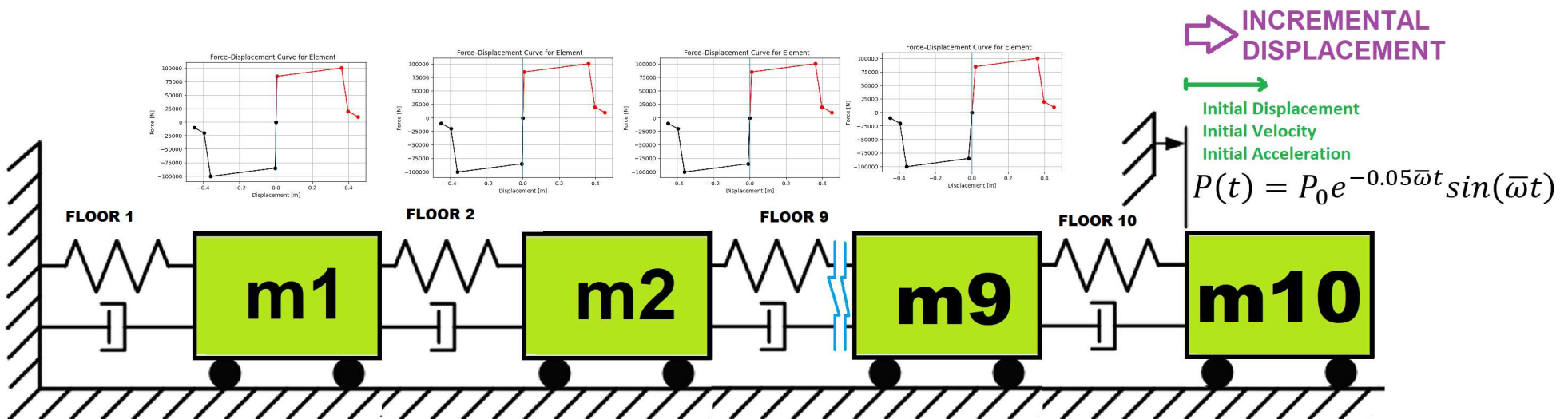


>> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<

**COMPREHENSIVE NONLINEAR
SEISMIC ASSESSMENT OF A MULTI-
DEGREE-FREEDOM (MDOF)
STRUCTURE: AN OPENSEES
FRAMEWORK FOR STATIC
PUSHOVER, CYCLIC DEGRADATION,
STATIC TIME-HISTORY AND
DYNAMIC TIME-HISTORY ANALYSIS**

THIS PROGRAM IS WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)



$$\text{Structural Ductility Damage Index} = \frac{\Delta_d - \Delta_y}{\Delta_u - \Delta_y}$$

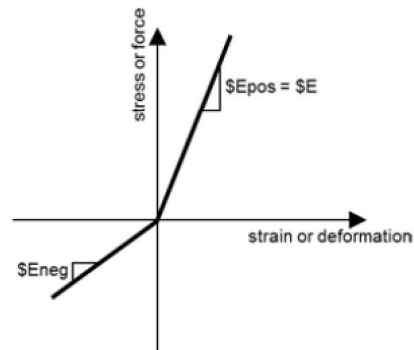
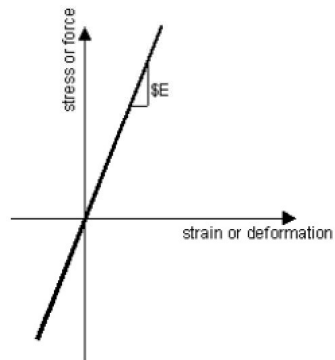
Δ_d = Lateral Displacement from Dynamic Analysis

Δ_y = Lateral Yield Displacement from Pushover Analysis

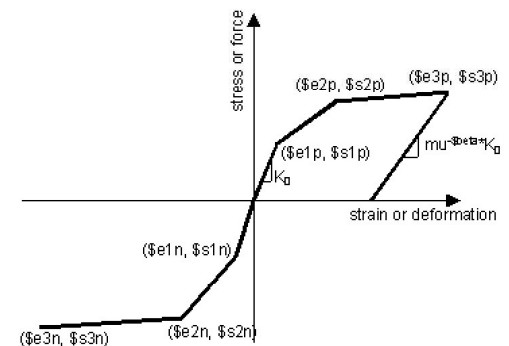
Δ_u = Lateral Ultimate Displacement from Pushover Analysis

$$P(t) = P_0 e^{-0.05\bar{\omega}t} \sin(\bar{\omega}t)$$

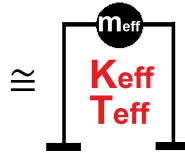
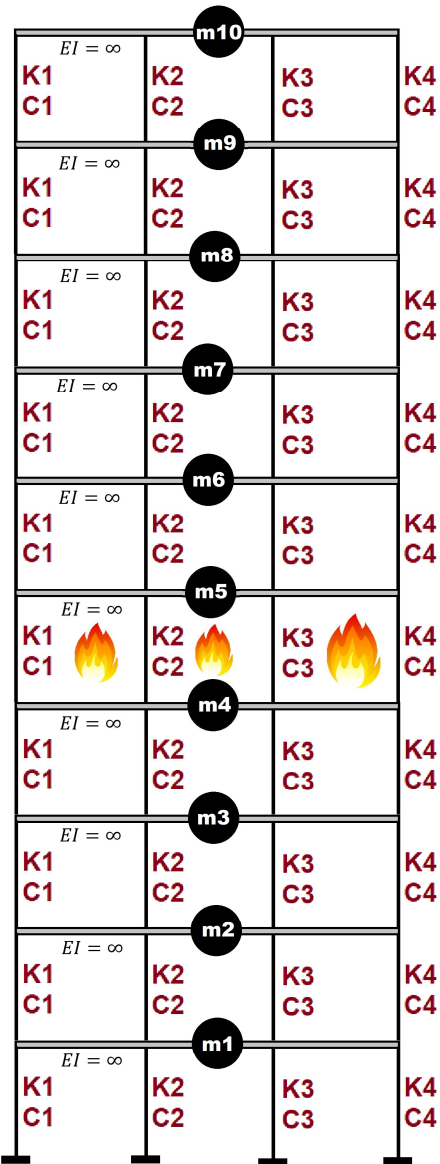
$$m\ddot{u} + c\dot{u} + ku = P(t)$$



ELASTIC MATERIAL



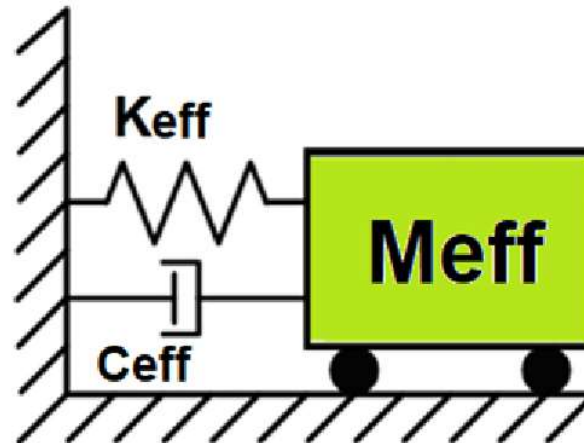
INELASTIC MATERIAL



Displacement-based seismic analysis transformation,
converting a multi-degree-of-freedom (MDOF) system
into an equivalent single-degree-of-freedom (SDOF)
system for seismic assessment

$$\Delta_d = \frac{\sum_{i=1}^n m_i \Delta_i^2}{\sum_{i=1}^n m_i \Delta_i} \quad m_{eff} = \frac{\sum_{i=1}^n m_i \Delta_i}{\Delta_d}$$

$$k_{eff} = \frac{\sum_{i=1}^n k_i \Delta_i}{\Delta_d} \quad T_{eff} = \frac{2\pi}{\sqrt{\frac{k_{eff}}{m_{eff}}}}$$



C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...IRE+EXAMPLE_04_MDOF\P(t)_POST-FIRE_MDOF_8_ANA.py

P(t)_POST-FIRE_MDOF_8_ANA.py X

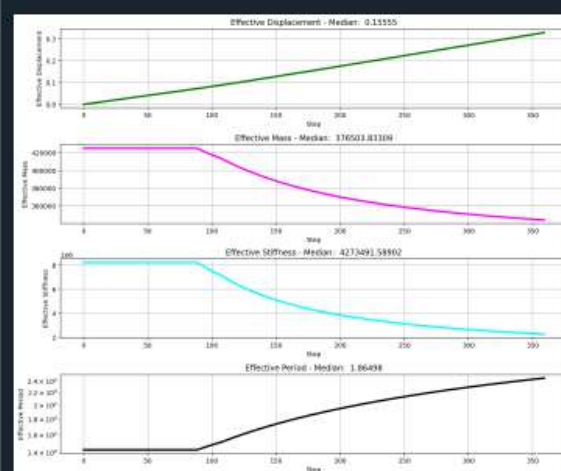
```

1 #####
2 # >> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<
3 # COMPREHENSIVE NONLINEAR SEISMIC ASSESSMENT OF A POST-FIRE MULTI-DEGREE-FREEDOM WITH FOUR
4 # ELEMENTS STRUCTURE : AN OPENSEES FRAMEWORK FOR STATIC PUSHOVER, CYCLIC DEGRADATION,
5 # STATIC TIME-HISTORY AND DYNAMIC TIME-HISTORY ANALYSIS
6 #
7 # EQUIVALENT VISCOUS DAMPING RATIO:  $\xi_{eq} = 100 * E_d / (4 * \pi * E_s)$ 
8 #
9 # ENERGY DISSIPATION CAPACITY INDEX =  $100 * E_d(\text{earthquake}) / E_d(\text{cyclic displacement})$ 
10 #
11 # EQUIVALENT SDOF SYSTEM DERIVATION VIA DISPLACEMENT-BASED SEISMIC DESIGN PROCEDURE WITH PUSHOVER ANAL
12 #
13 #  $P(t) = P_0 \sin(\omega t)$ 
14 #  $P(t) = P_0 \exp(-0.05\omega t) \sin(\omega t)$ 
15 #
16 # ASSESSMENT OF DUCTILITY DAMAGE INDICES FOR STRUCTURAL ELEMENTS AND SYSTEMS AND EVALUATION
17 # OF ENERGY DISSIPATION CAPACITY INDEX
18 #
19 # MARKOV CHAIN FOR EVALUATIONS OF PROBABILITY OF FAILURE
20 #
21 # THIS PYTHON SCRIPT IS WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)
22 # EMAIL: salar.d.ghashghaei@gmail.com
23 #####
24 """
25 Nonlinear Seismic Performance Assessment of a MDOF:
26 An OpenSeesPy Framework for Material and Geometric Nonlinearity Under Static, Cyclic, and Earthquake Lo
27
28 This OpenSeesPy script performs rigorous nonlinear static and dynamic analysis of a MDOF
29 for performance-based earthquake engineering. The 2D model incorporates both material nonlinearity
30 (elastic-perfectly plastic or hysteretic steel with strain hardening)
31 and geometric nonlinearity (corotational truss formulation) to capture P-delta effects
32 and large displacements-essential for collapse assessment.
33
34 Eight analysis protocols are implemented:

```

C:\Users\Dell

21 %



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C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...IRE+\EXAMPLE_04_MDOF\P(t)_POST-FIRE_MDOF_8_ANA.py

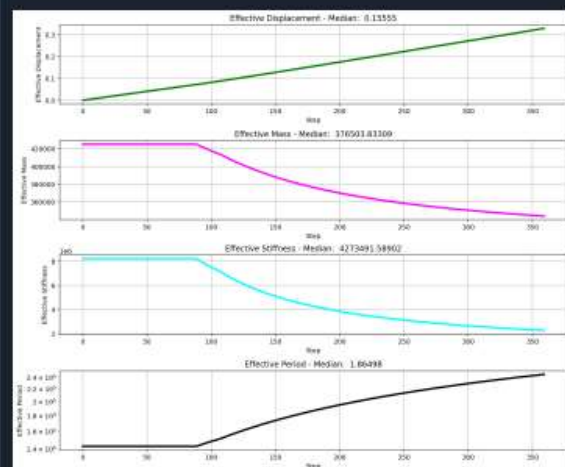
P(t)_POST-FIRE_MDOF_8_ANA.py X

STEEL FIRE PROPERTIES

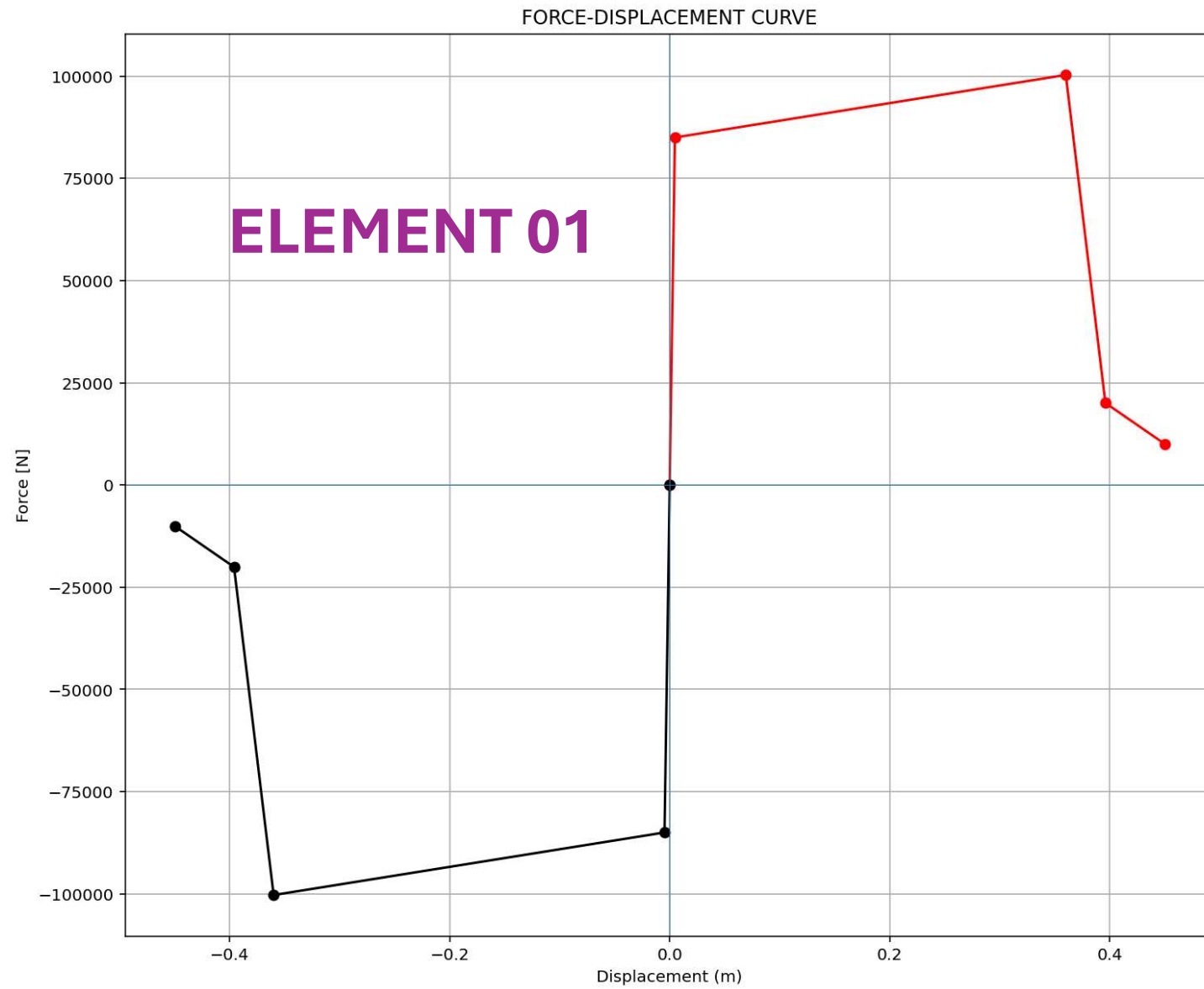
```
149
150 def STEEL_FIRE_PROPERTIES(T):
151     # THERMAL EFFECTS OF STEEL MATERIAL STRESS-STRAIN
152     # WRITTEN BY SALAR DELAVAR GHASHGHAIEI (QASHQAI)
153     # PAPER: Mechanical properties of structural steel at elevated temperatures and after cooling c
154     # https://www.researchgate.net/publication/227780727\_Mechanical\_properties\_of\_structural\_steel\_c
155     if T <= 20:
156         kE = 1.00
157         kFy = 1.00
158         kFu = 1.00
159         esu = 0.25
160
161     elif 20 < T <= 100:
162         kE = 0.90
163         kFy = 0.90
164         kFu = 0.90
165         esu = 0.277
166
167     elif 100 < T <= 200:
168         kE = 0.80
169         kFy = 0.80
170         kFu = 0.80
171         esu = 0.3125
172
173     elif 200 < T <= 300:
174         kE = 0.70
175         kFy = 0.70
176         kFu = 0.70
177         esu = 0.3571
178
179     elif 300 < T <= 400:
180         kE = 0.60
181         kFy = 0.60
182         kFu = 0.60
```

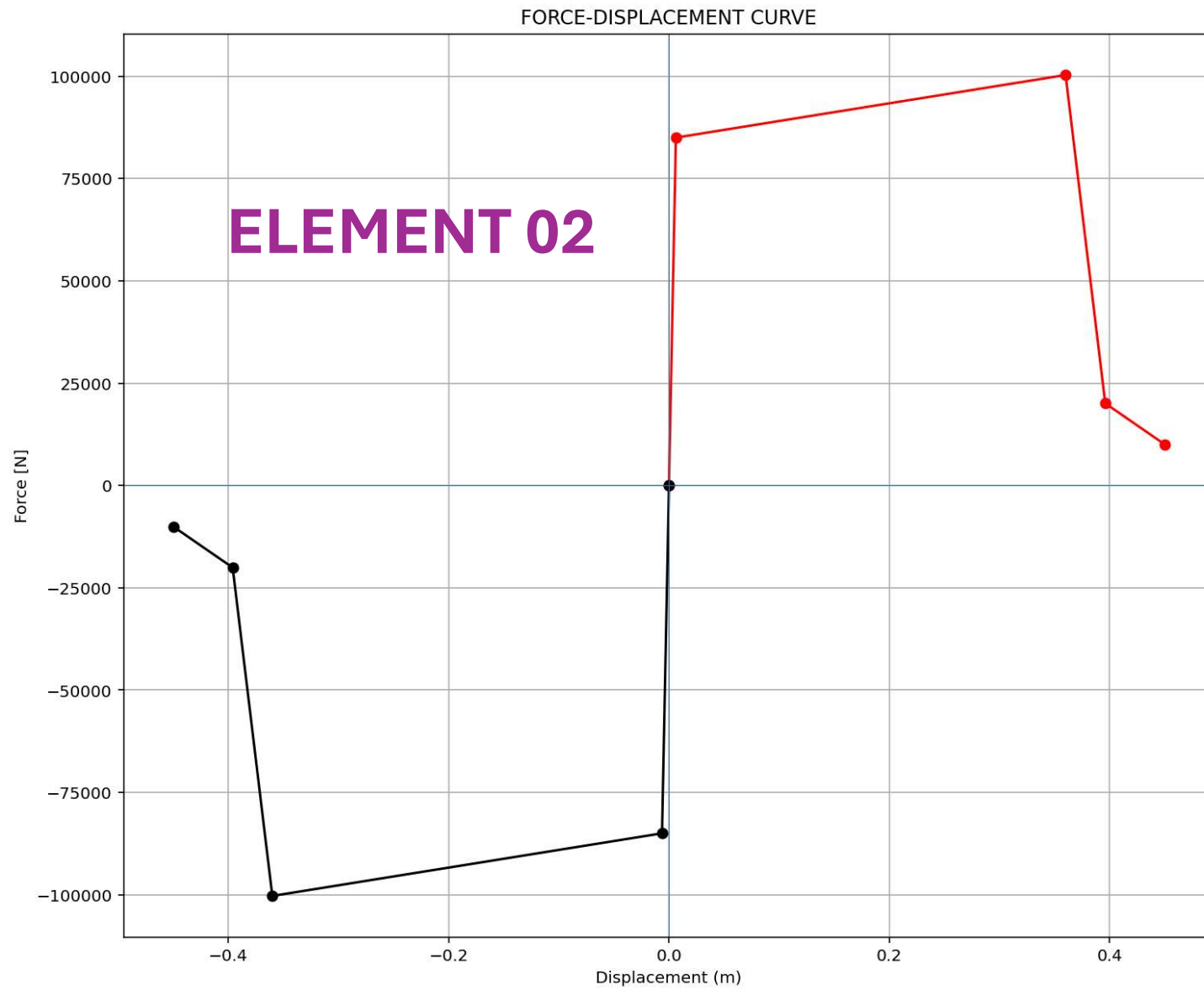
...GY DISSIPATION_CAPACITY_INDEX\57_POST_FIRE+\EXAMPLE_04_MDOF

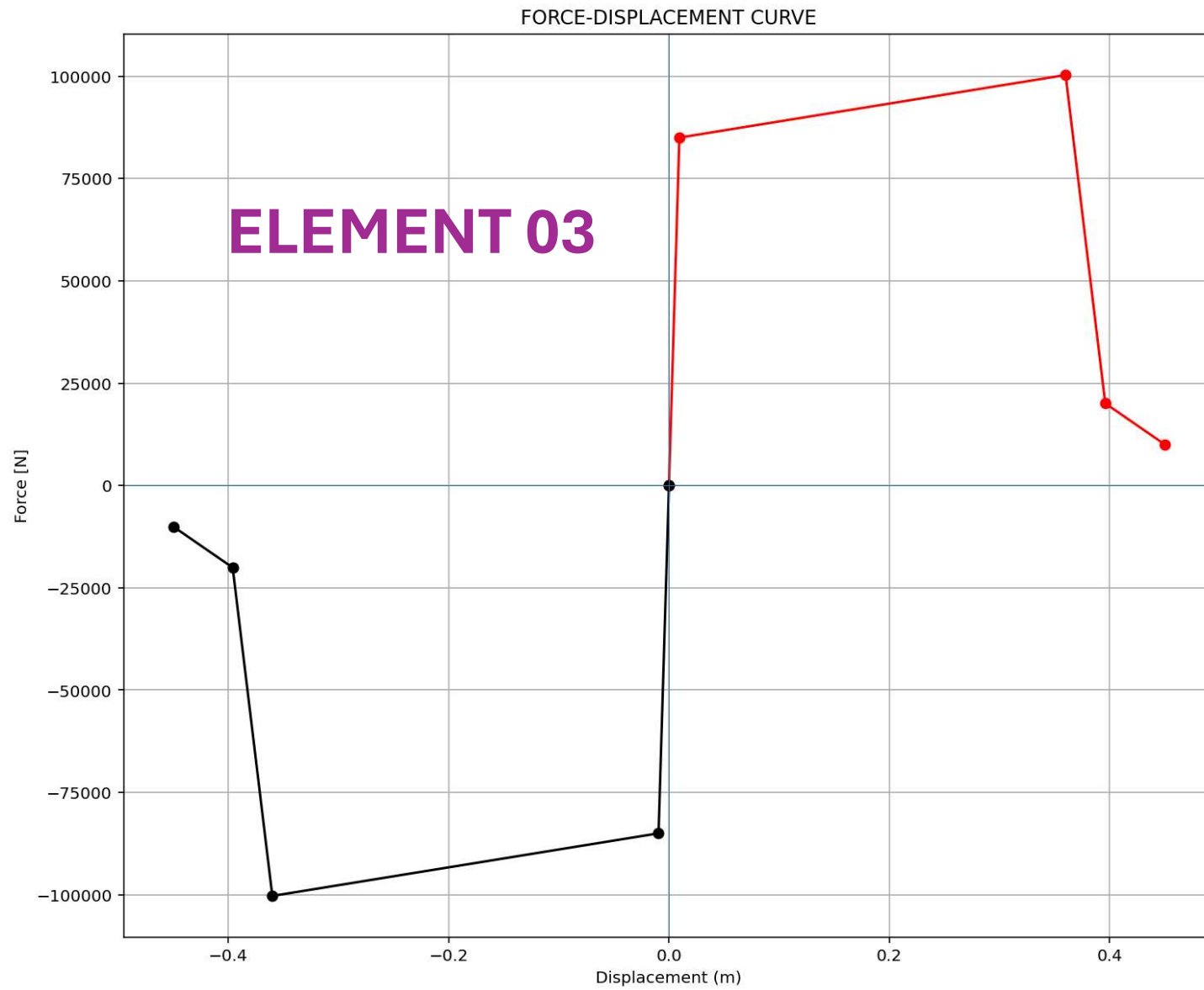
21 %

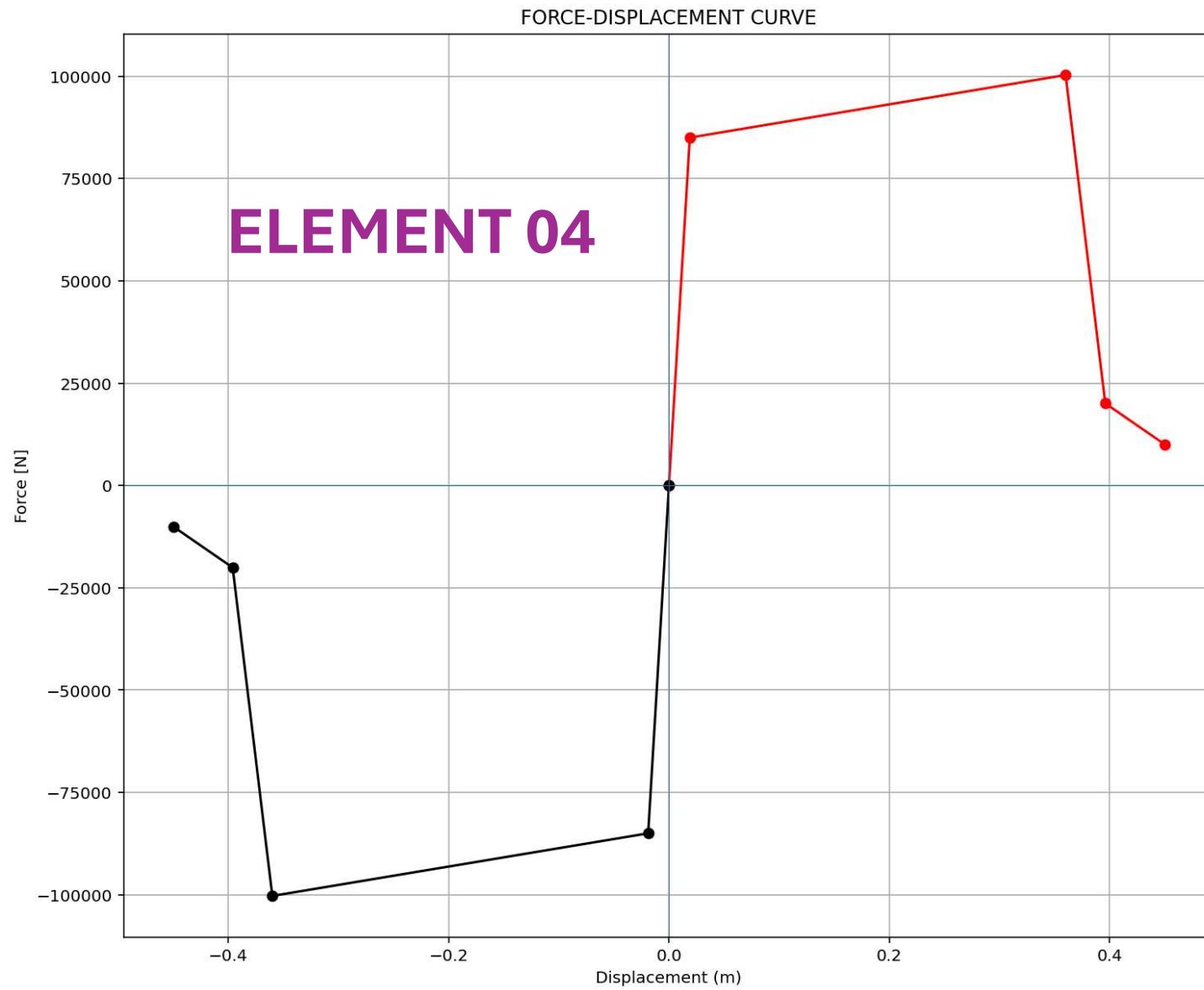


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STATIC ANALYSIS

Spyder (Python 3.12)

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C:\Users\De\l\Desktop\OPENSEES_FILES\+EIGHT_ANALY...IRE+\EXAMPLE_04_MDOF\P(t)_POST-FIRE_MDOF_8_ANA.py

P(t)_POST-FIRE_MDOF_8_ANA.py X

```
1114 # STATIC ANALYSIS
1115 MAT_TYPE = 'INELASTIC' # 'ELASTIC' OR 'INELASTIC'
1116 ANAL_TYPE = 'STATIC'
1117
1118 DATA = MDOF(MAT_TYPE, TOTAL_MASS, ANAL_TYPE)
1119 (reaction, disp, DI) = DATA
1120
1121 S01.PLOT 1D SPRING(deformed scale=1.0, virtual spring length=1.0, virtual spring angle=0.0)
1122 %%-----
1123 # PUSHOVER ANALYSIS (STATIC TIME-HISTORY ANALYSIS)
1124 MAT_TYPE = 'INELASTIC' # 'ELASTIC' OR 'INELASTIC'
1125 ANAL_TYPE = 'PUSHOVER'
1126
1127 DATA = MDOF(MAT_TYPE, TOTAL_MASS, ANAL_TYPE)
1128 (reaction_PUSH, disp_PUSH, DI_PUSH,
1129  ele_force_PUSH, node_displacements_PUSH,
1130  PERIOD_MIN_PUSH, PERIOD_MAX_PUSH) = DATA
1131
1132
1133 XDATA = disp_PUSH
1134 YDATA = reaction_PUSH
1135 XLABEL = 'Displacement [m]'
1136 YLABEL = 'Base Reaction [N]'
1137 TITLE = 'Base Reaction and Displacement of Structure During Pushover Analysis'
1138 COLOR = 'black'
1139 SEMIOLOGY = False
1140 PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMIOLOGY)
1141
1142 DATA = S07.BILINEAR_CURVE(np.abs(disp_PUSH), np.abs(reaction_PUSH), SLOPE_NODE=10)
1143 (X_PUSH, Y_PUSH, Elastic_ST, Plastic_ST, Tangent_ST, Ductility_Rito, Over_Strength_Factor) = DATA
1144
1145 # PLOT STRUCTURAL PERIOD DURING THE ANALYSIS
1146 plt.figure(0, figsize=(12, 8))
1147 plt.plot(disp_PUSH, PERIOD_MIN_PUSH, linewidth=3)
```

...GY DISSIPATION_CAPACITY_INDEX\57_POST_FIRE+\EXAMPLE_04_MDOF

12 %



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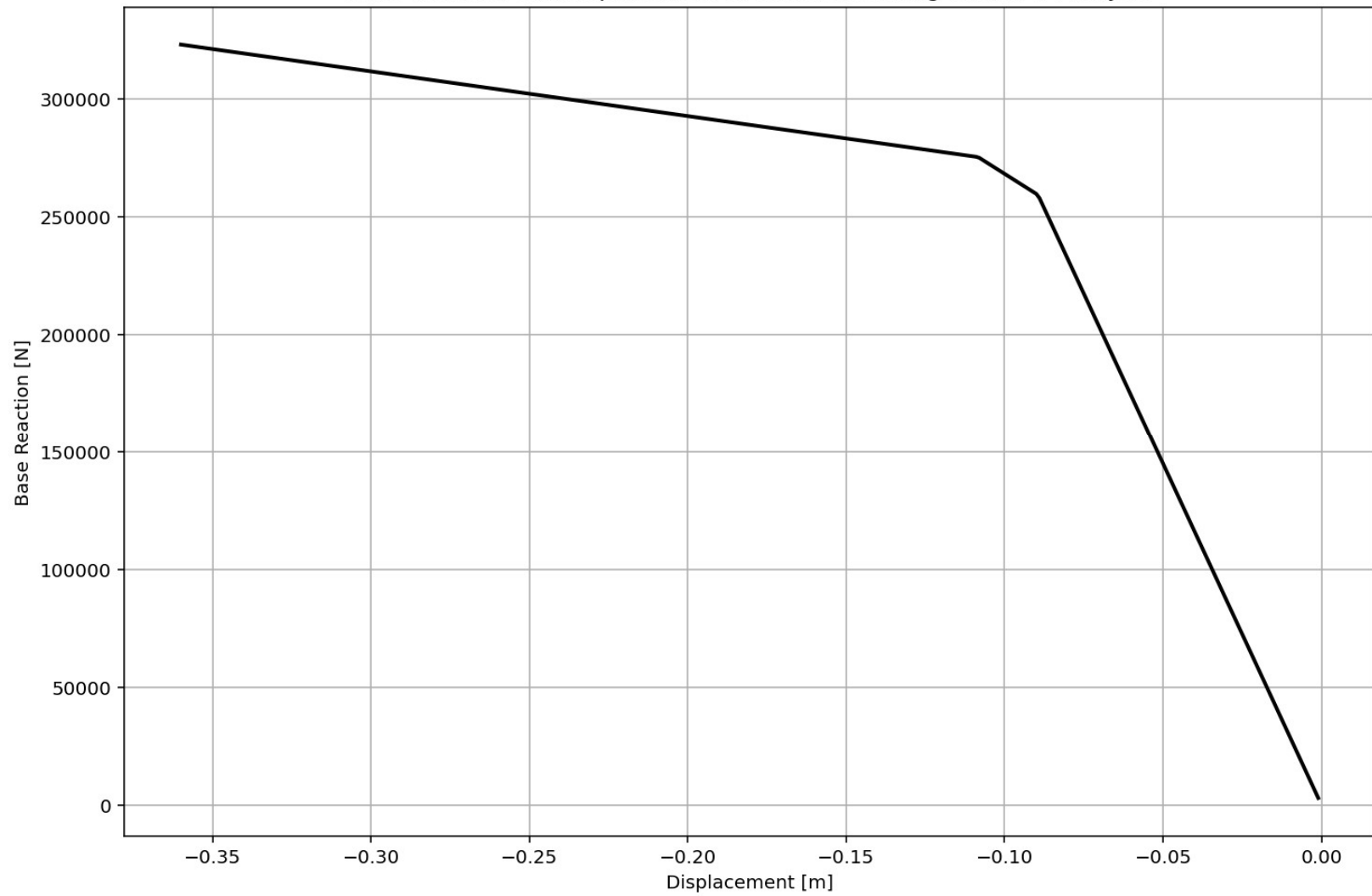
File Tools Conda: anaconda3 (Python 3.12.7) USB: Python Line 127 Col 1 UTF-8 CRLF RW Mem 51%

PUSHOVER ANALYSIS

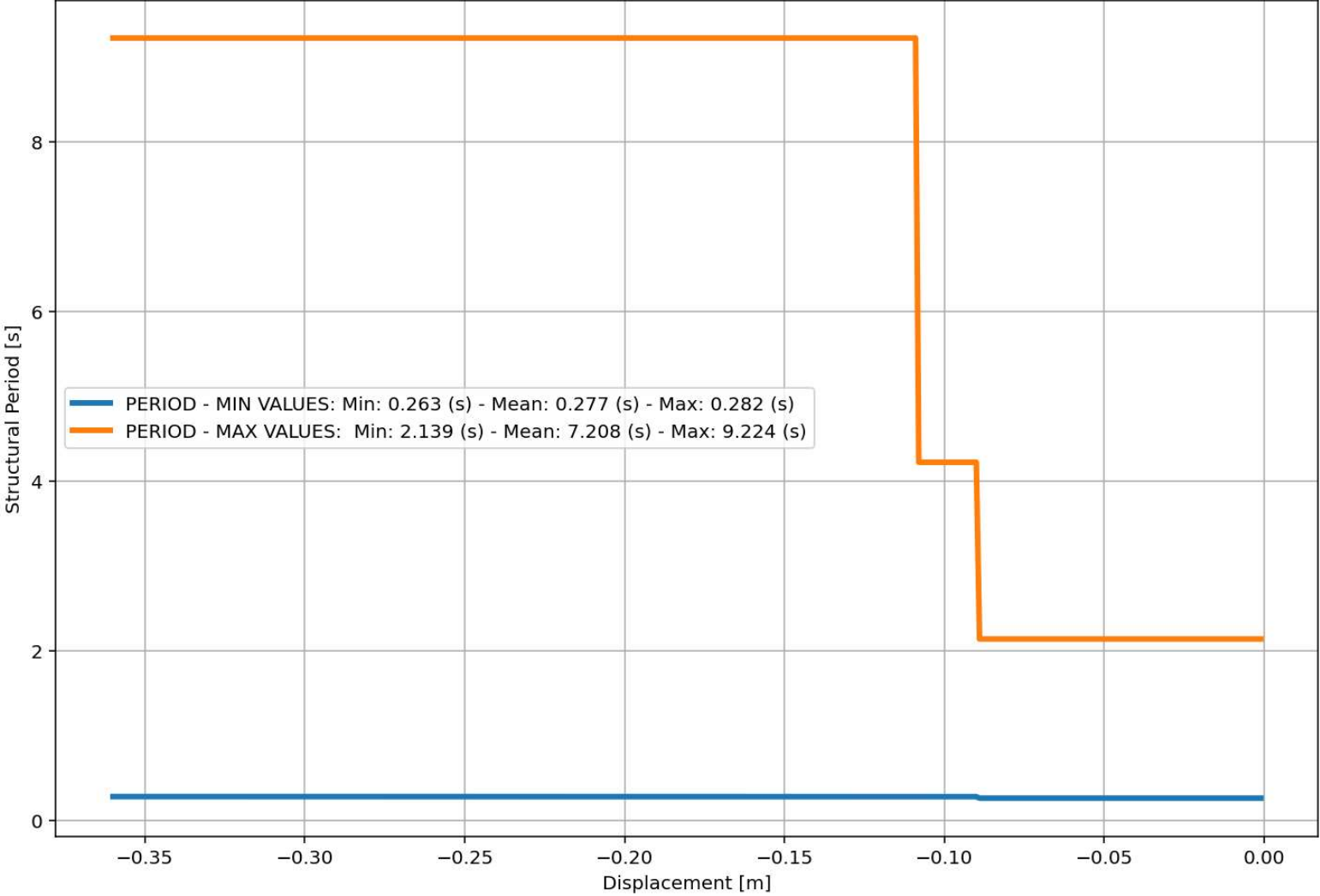
...GY DISSIPATION_CAPACITY_INDEX\57_POST_FIRE+\EXAMPLE_04_MDOF



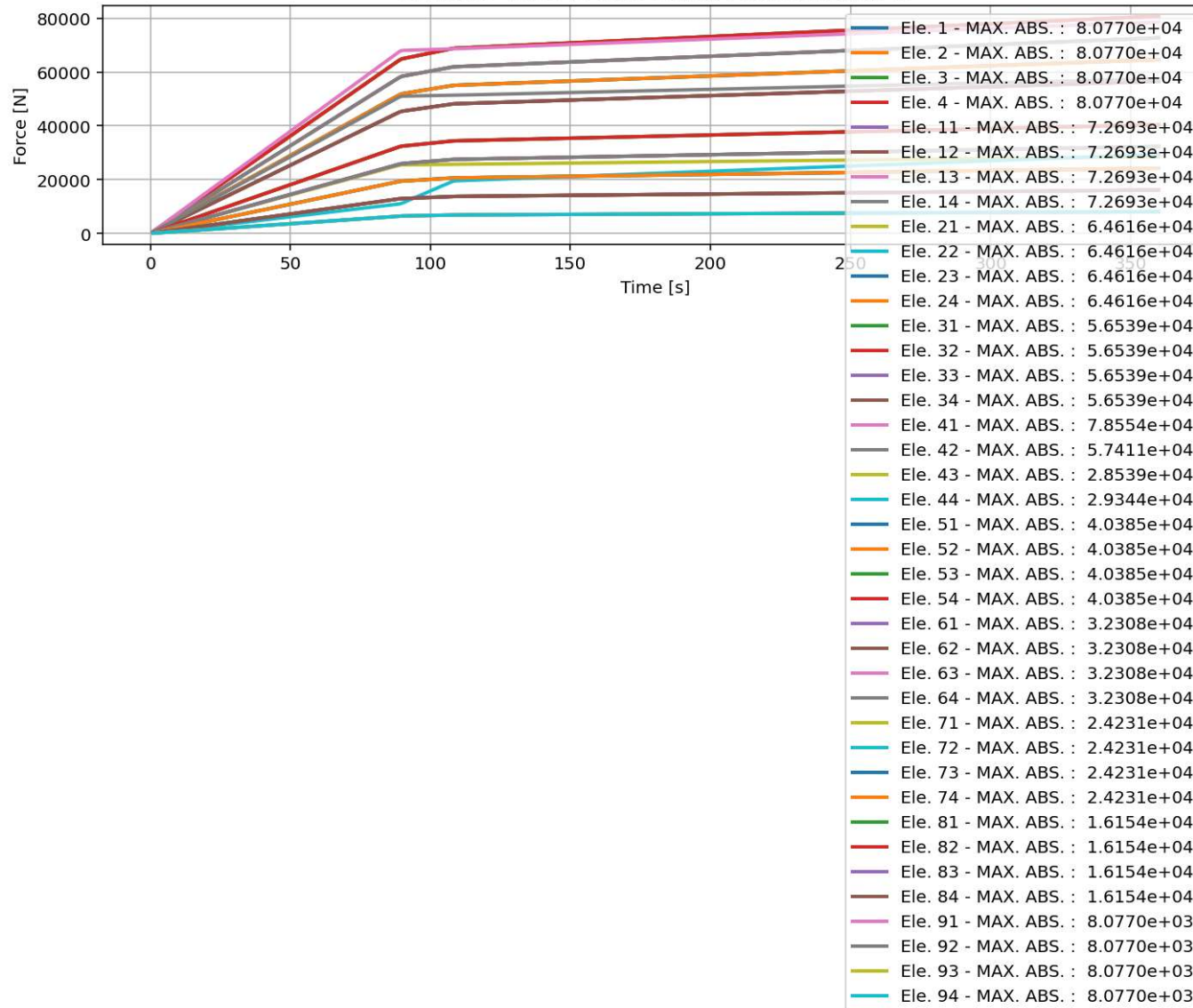
Base Reaction and Displacement of Structure During Pushover Analysis



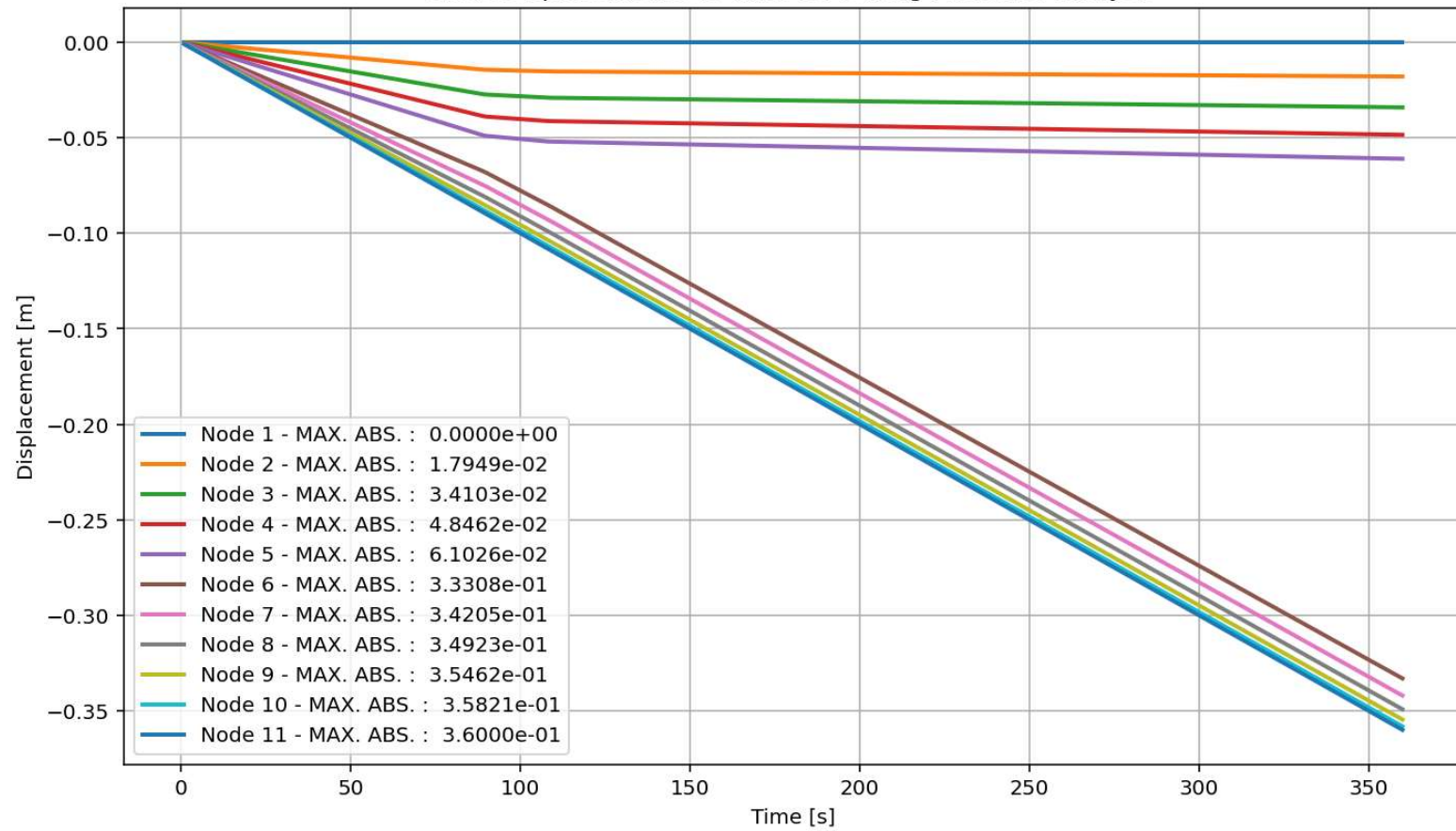
Period of Structure During Pushover Analysis

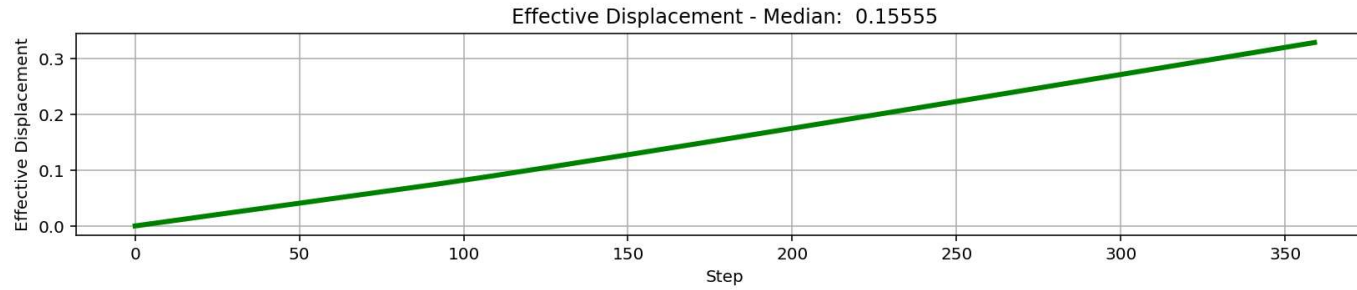


elements force vs Time During Pushover Analysis

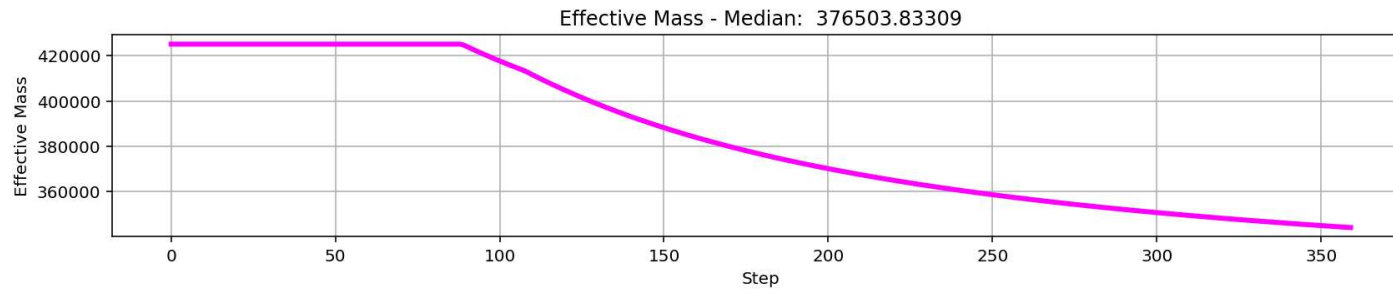


Node Displacements vs Time for During Pushover Analysis

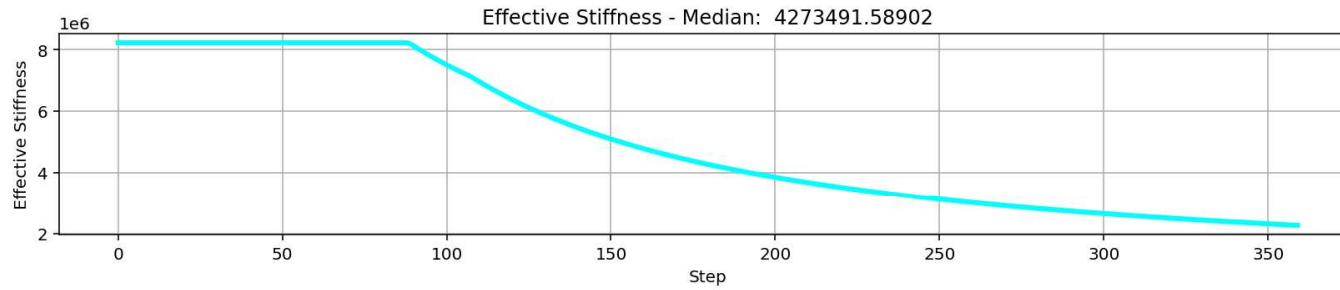




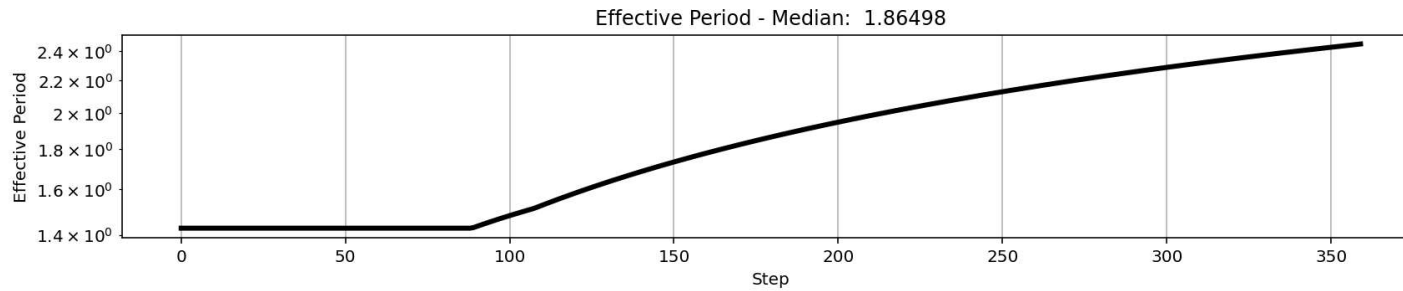
$$\Delta_d = \frac{\sum_{i=1}^n m_i \Delta_i^2}{\sum_{i=1}^n m_i \Delta_i}$$



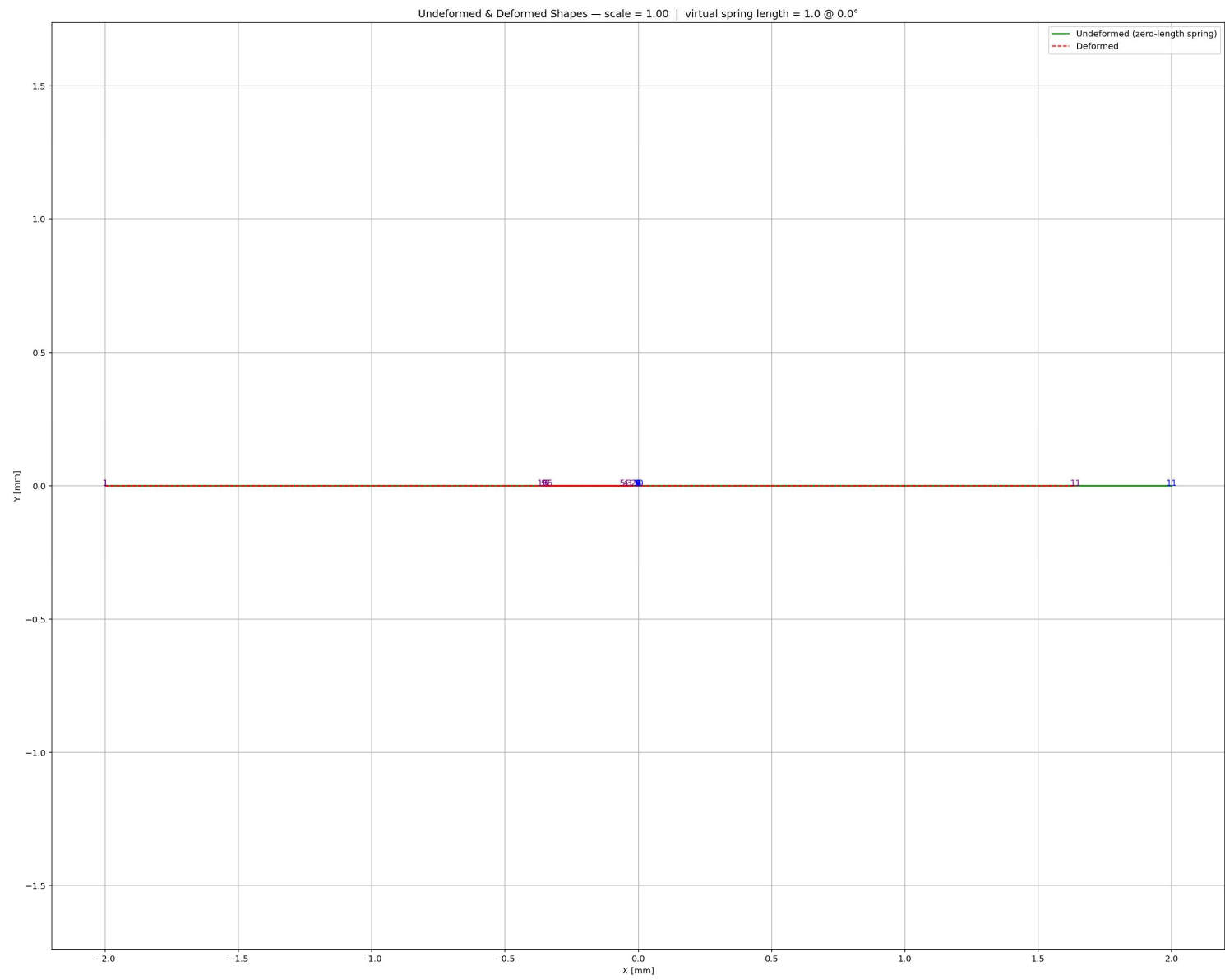
$$m_{eff} = \frac{\sum_{i=1}^n m_i \Delta_i}{\Delta_d}$$



$$k_{eff} = \frac{\sum_{i=1}^n k_i \Delta_i}{\Delta_d}$$



$$T_{eff} = \frac{2\pi}{\sqrt{\frac{k_{eff}}{m_{eff}}}}$$



CYCLIC DISPLACEMENT ANALYSIS

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P(t)_POST-FIRE_MDOF_8_ANA.py X

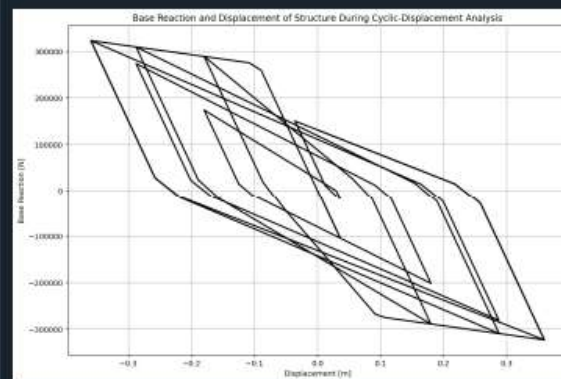
```

1183 # CYCLIC DISPLACEMENT ANALYSIS (STATIC TIME-HISTORY ANALYSIS)
1184 MAT_TYPE = 'INELASTIC' # 'ELASTIC' OR 'INELASTIC'
1185 ANAL_TYPE = 'CYCLIC_DISPLACEMENT'
1186
1187 DATA = MDOF(MAT_TYPE, TOTAL_MASS, ANAL_TYPE)
1188 (reaction_CP, disp_CP, DI_CP,
1189  ele_force_CP, node_displacements_CP,
1190  PERIOD_MIN_CP, PERIOD_MAX_CP) = DATA
1191
1192
1193 XDATA = disp_CP
1194 YDATA = reaction_CP
1195 XLABEL = 'Displacement [m]'
1196 YLABEL = 'Base Reaction [N]'
1197 TITLE = 'Base Reaction and Displacement of Structure During Cyclic-Displacement Analysis'
1198 COLOR = 'black'
1199 SEMIOLOGY = False
1200 PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMIOLOGY)
1201
1202 # PLOT STRUCTURAL PERIOD DURING THE ANALYSIS
1203 plt.figure(0, figsize=(12, 8))
1204 plt.plot(disp_CP, PERIOD_MIN_CP, linewidth=3)
1205 plt.plot(disp_CP, PERIOD_MAX_CP, linewidth=3)
1206 plt.title('Period of Structure During Cyclic Displacement Analysis')
1207 plt.ylabel('Structural Period [s]')
1208 plt.xlabel('Displacement [m]')
1209 #plt.semilogy()
1210 plt.grid()
1211 plt.legend([f'PERIOD - MIN VALUES: Min: {np.min(PERIOD_MIN_CP):.3f} (s) - Mean: {np.mean(PERIOD_MIN_CP,
1212  f'PERIOD - MAX VALUES: Min: {np.min(PERIOD_MAX_CP):.3f} (s) - Mean: {np.mean(PERIOD_MAX_CP,
1213  ])
1214 plt.show()
1215
1216

```

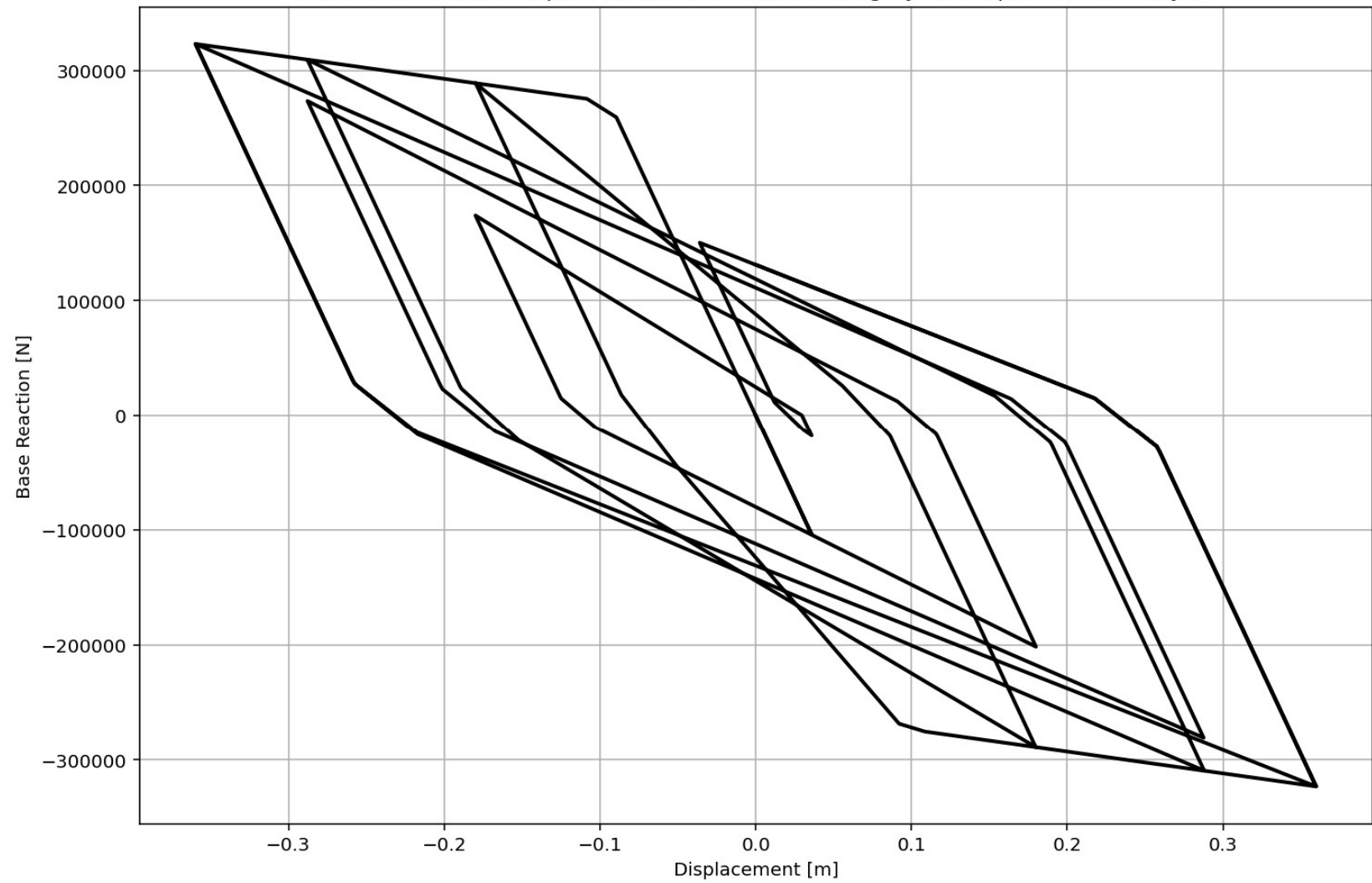
...GY DISSIPATION_CAPACITY_INDEX\57_POST_FIRE+\EXAMPLE_04_MDOF

24 %

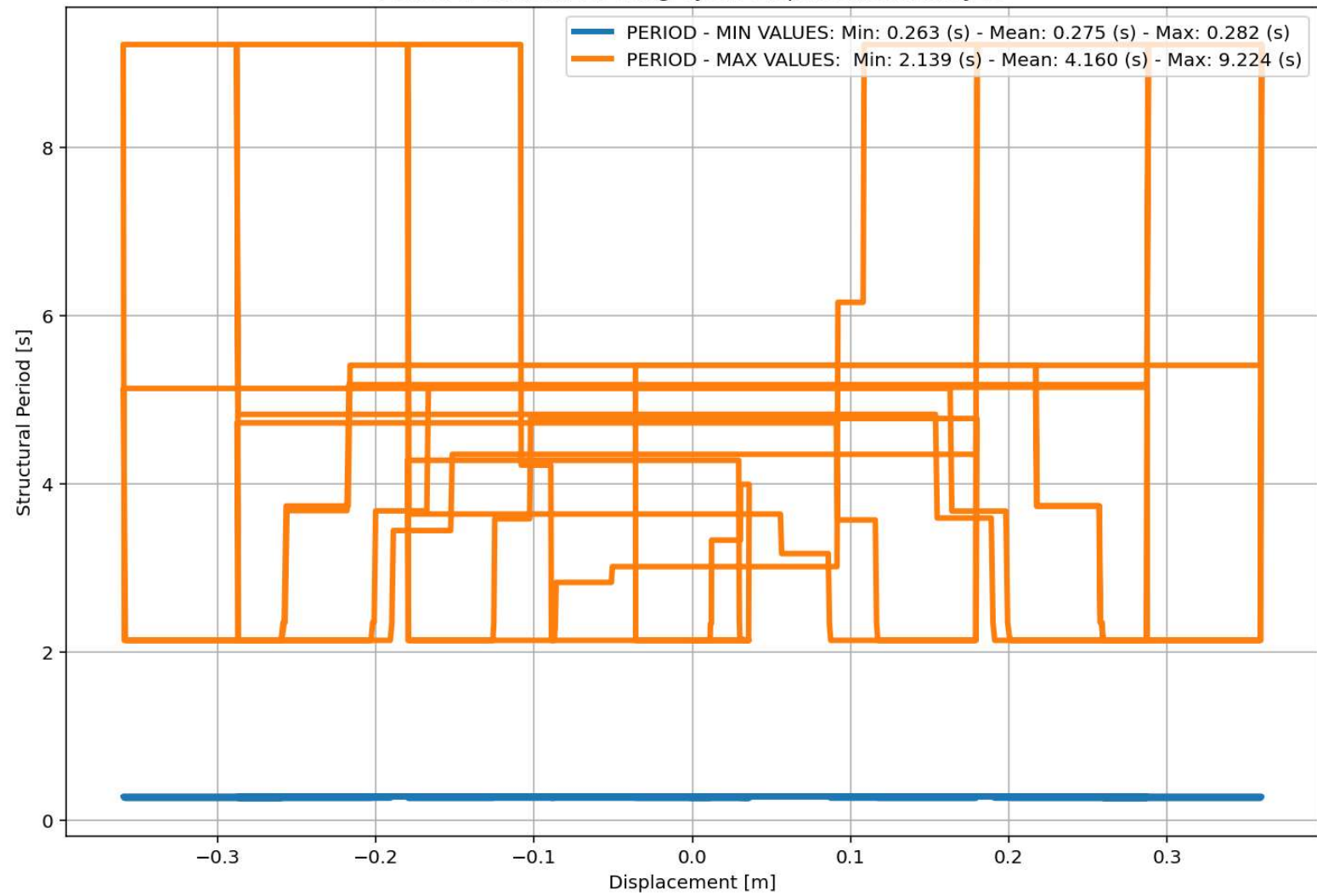


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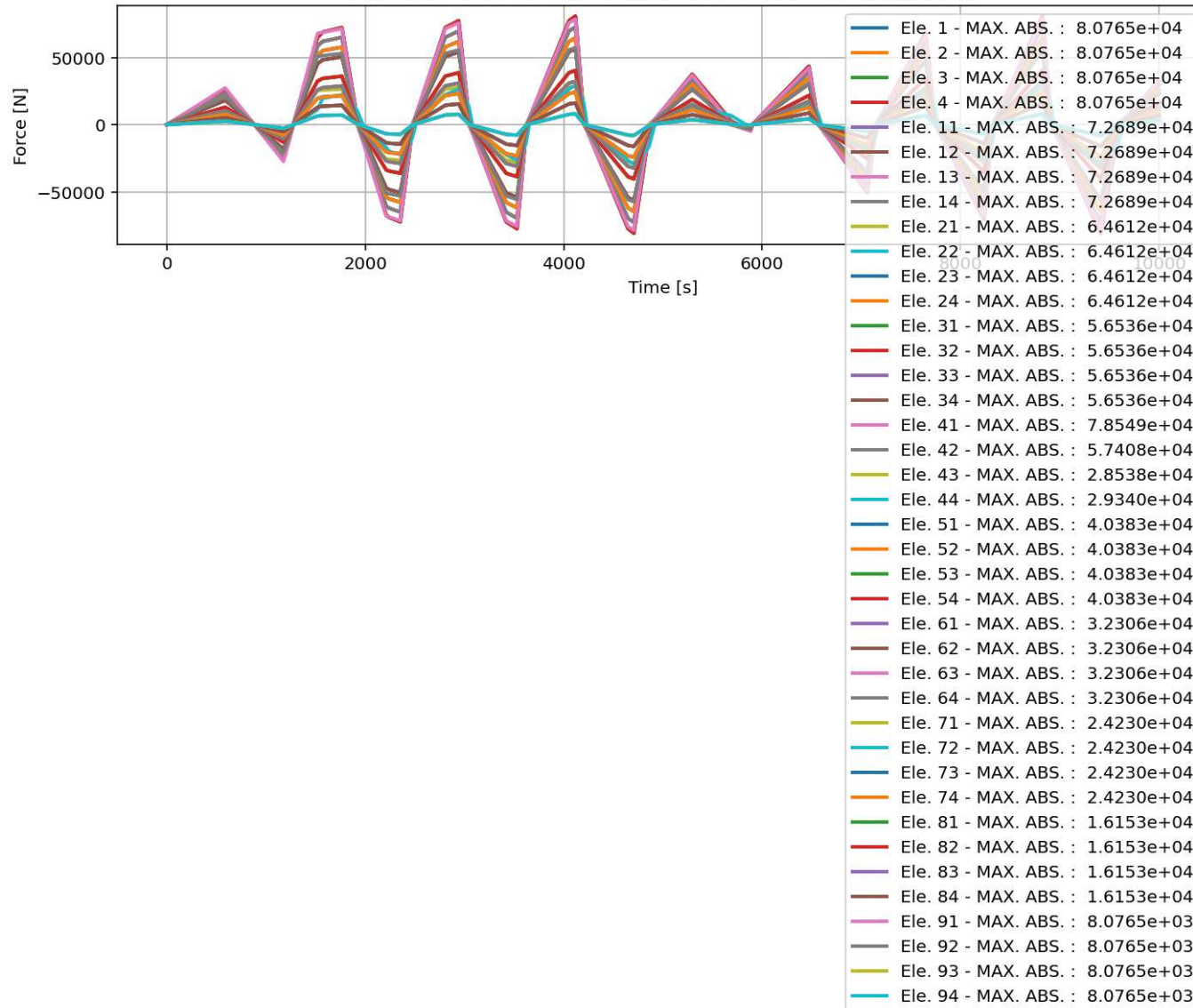
Base Reaction and Displacement of Structure During Cyclic-Displacement Analysis



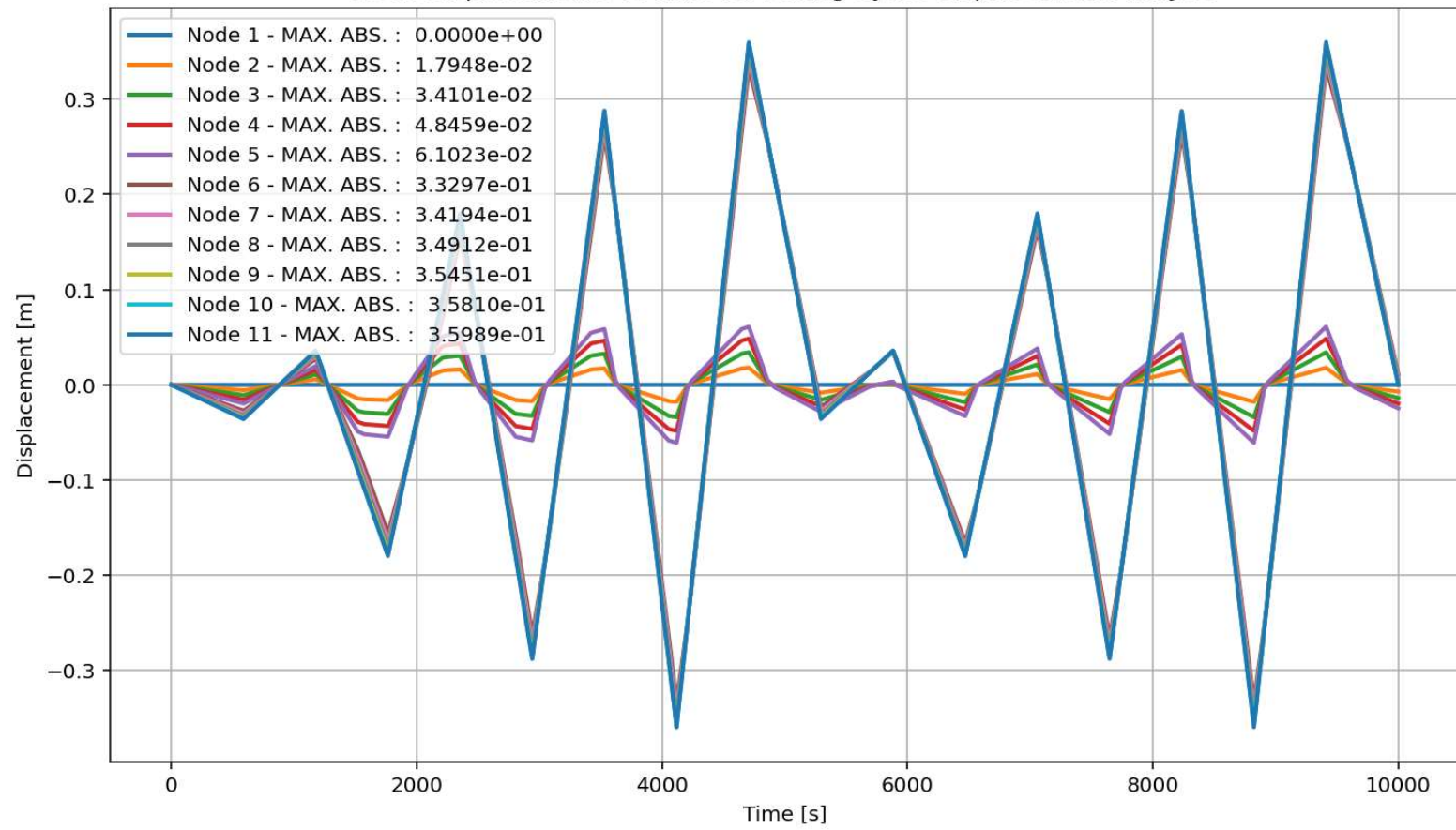
Period of Structure During Cyclic Displacement Analysis

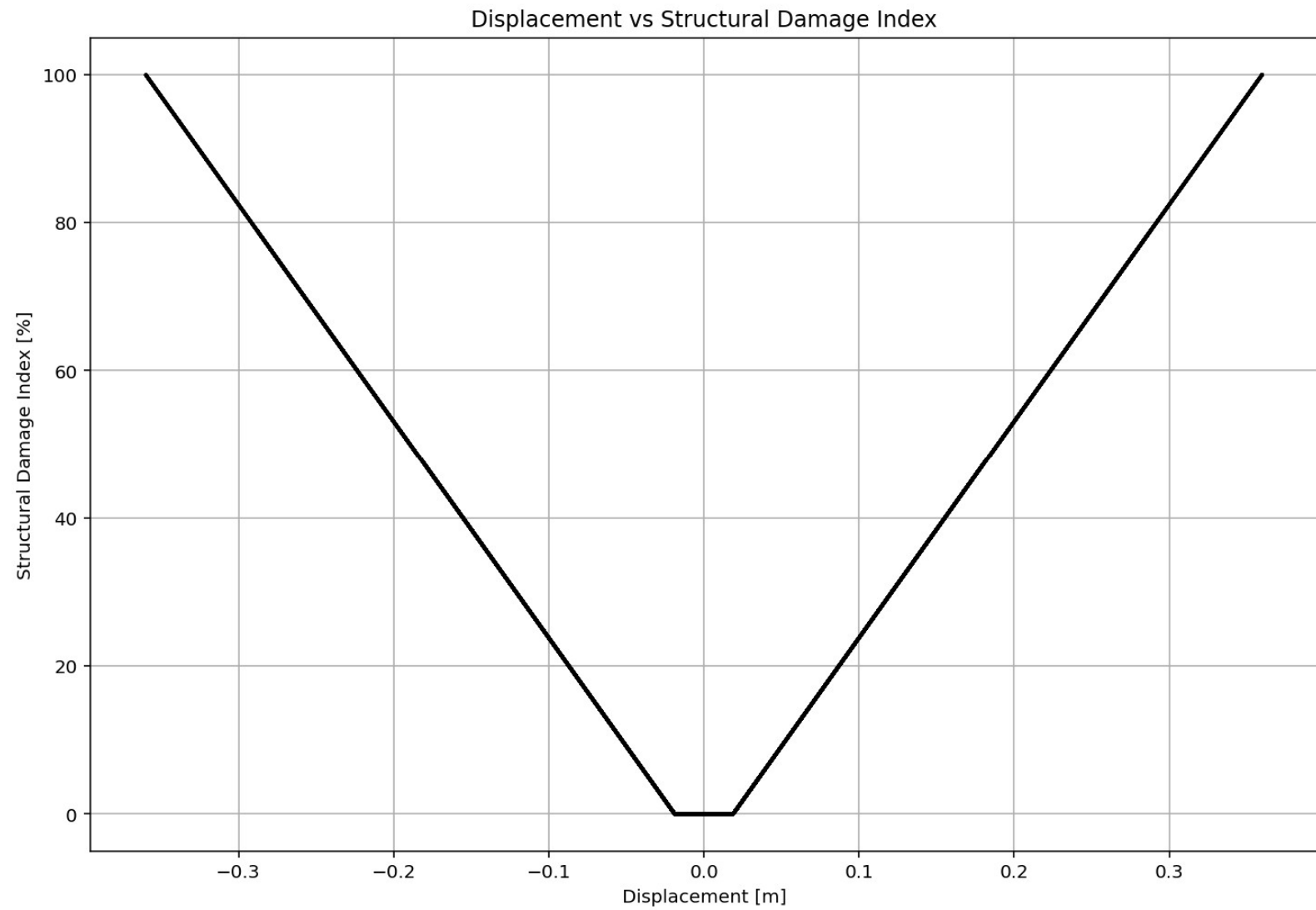


elements force vs Time During Cyclic Displacement Analysis

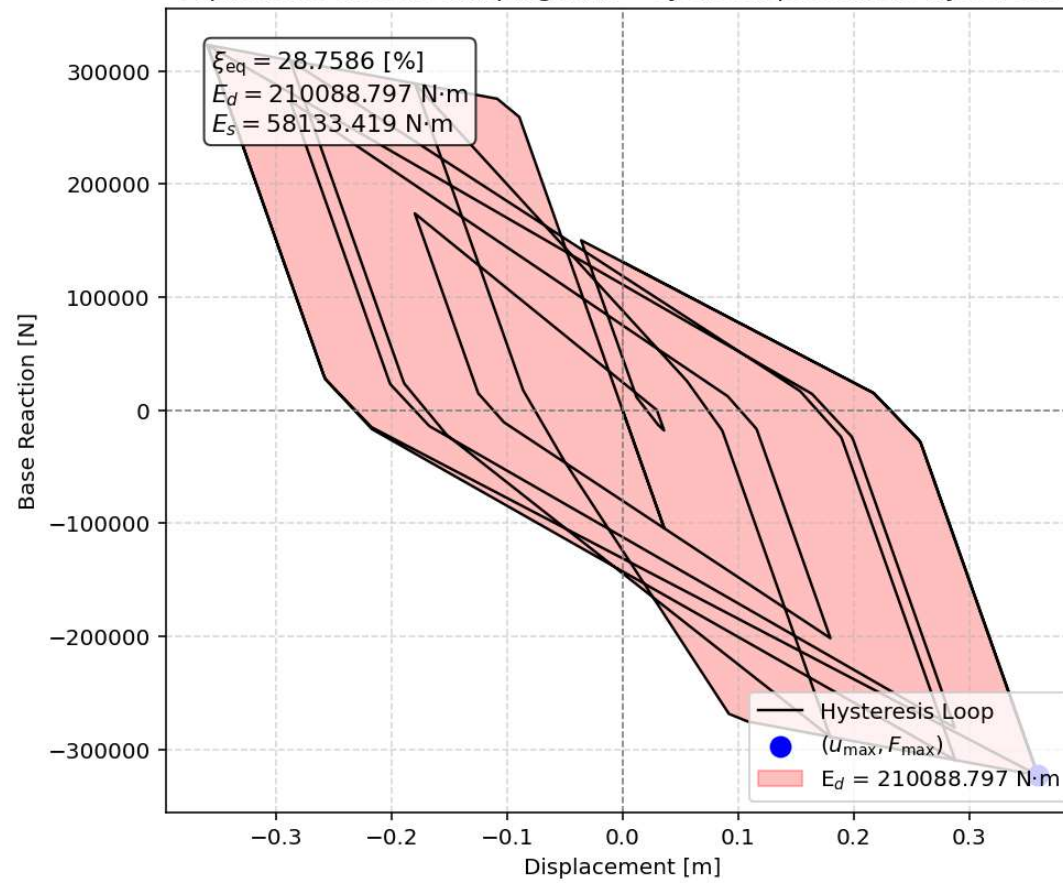


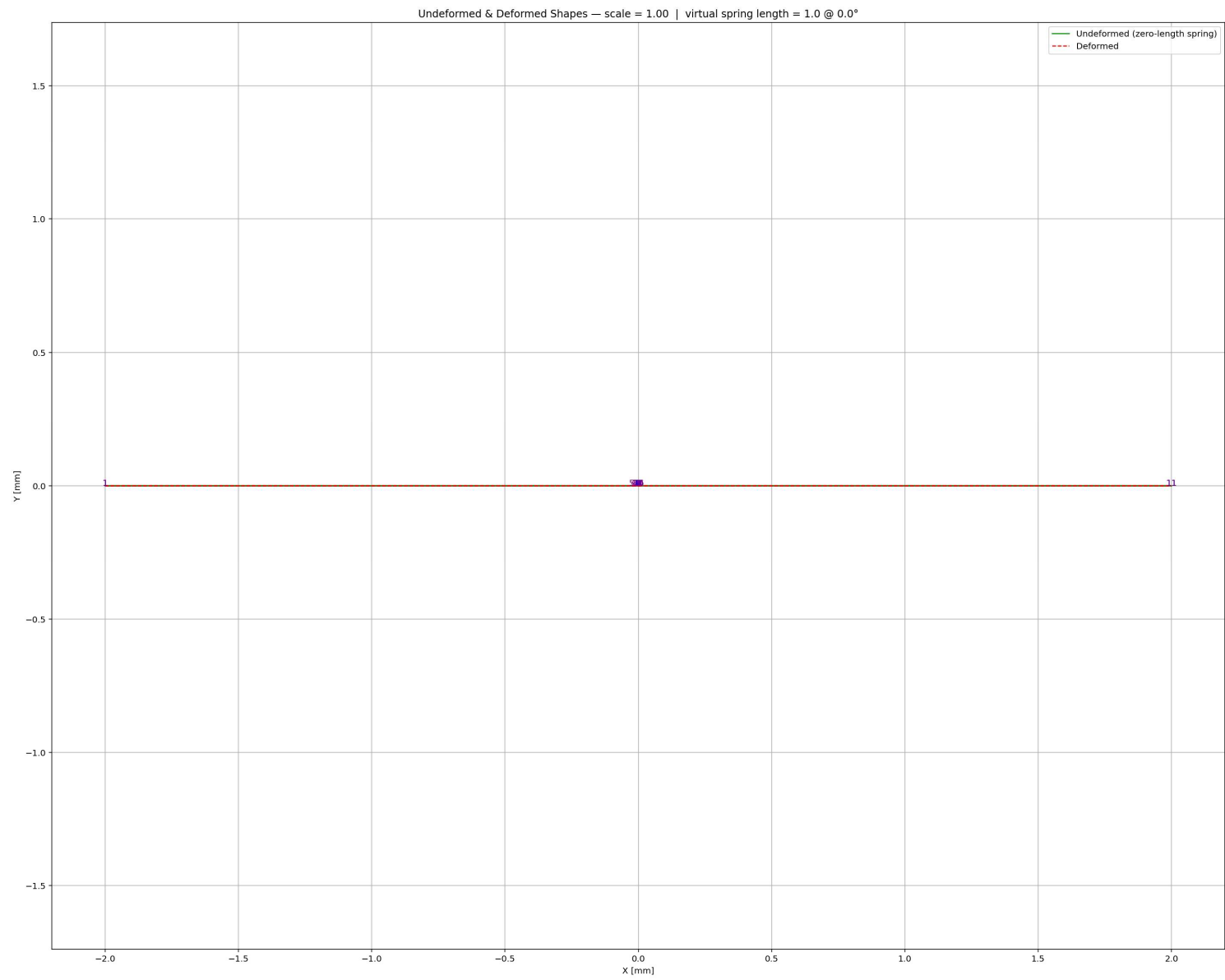
Node Displacements vs Time for During Cyclic Displacement Analysis





Equivalent viscous damping ratio - Cyclic Displacement Hysteresis



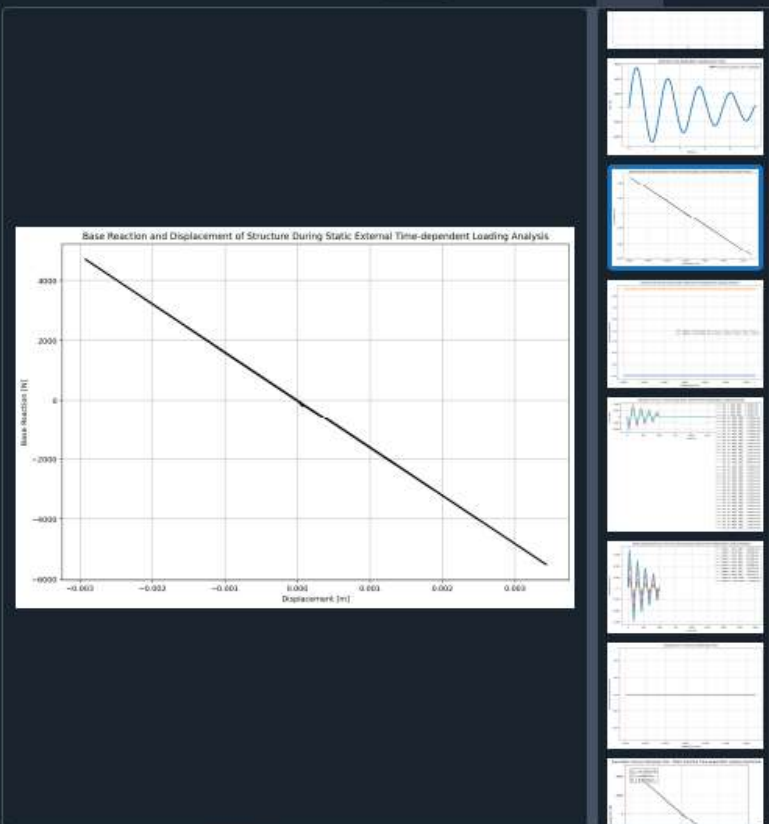


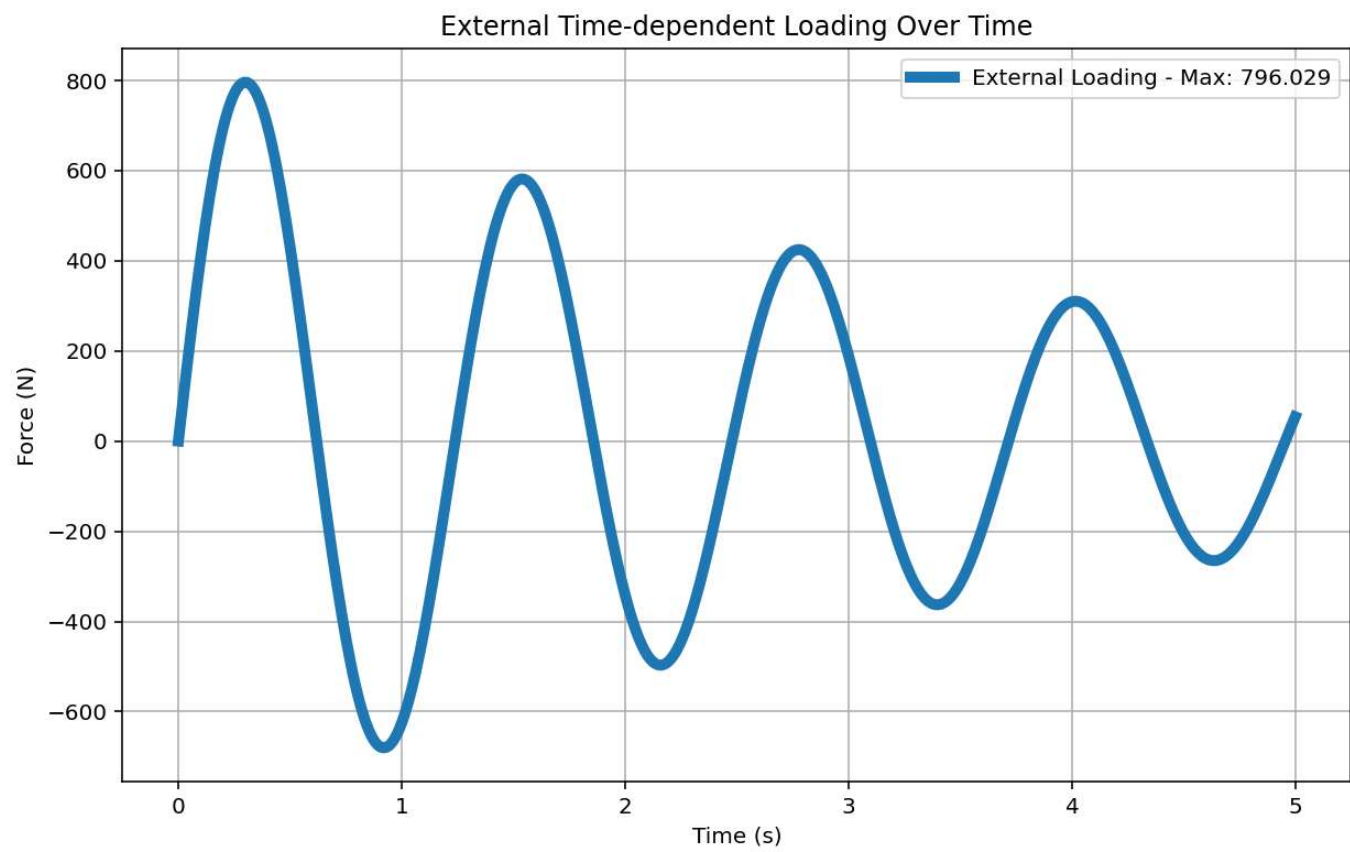
STATIC TIME-HISTORY WITH EXTERNAL TIME-DEPENDENT LOADING ANALYSIS

```

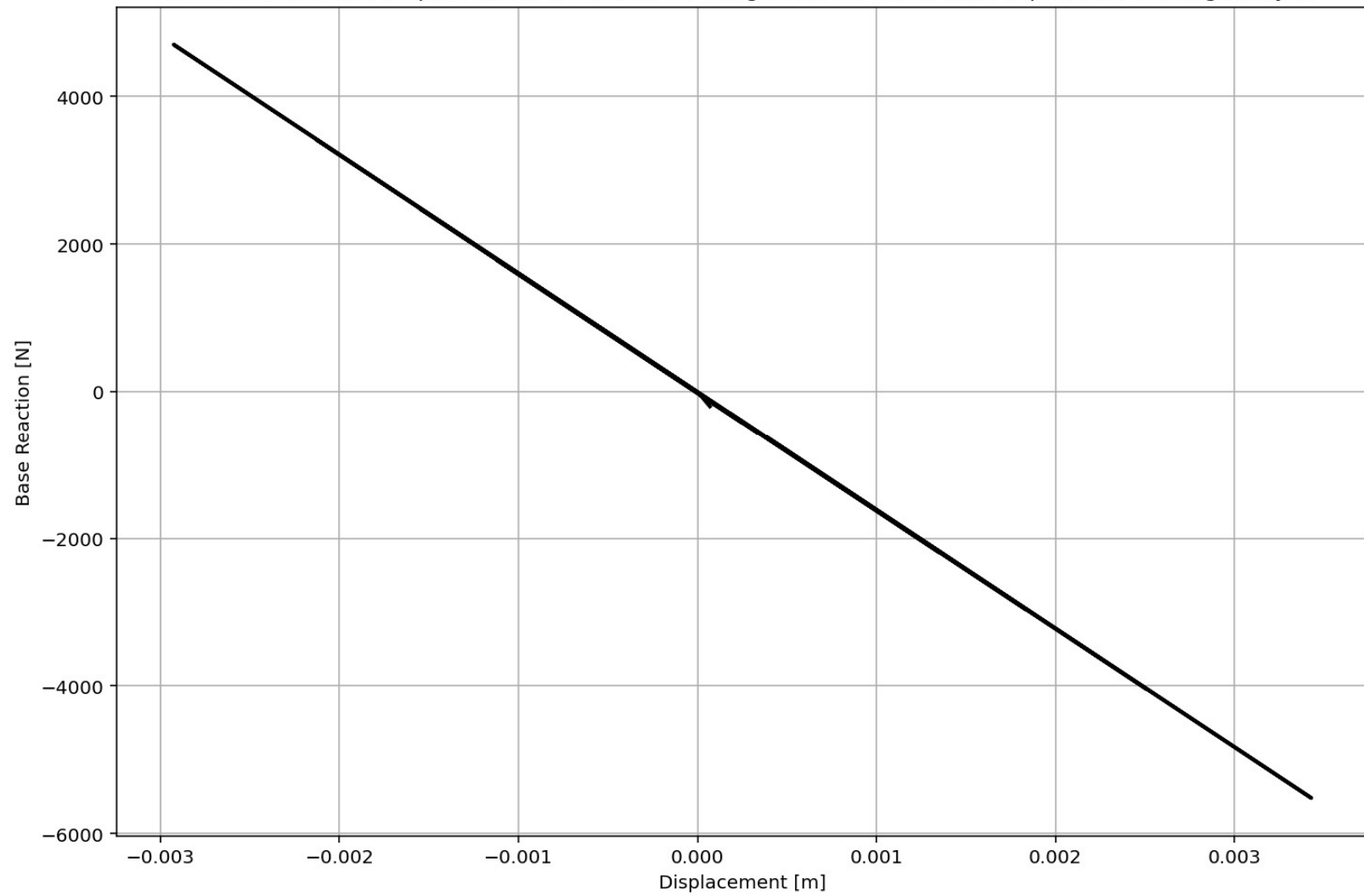
1249 # EXTERNAL TIME-DEPENDENT LOADING ANALYSIS (STATIC TIME-HISTORY ANALYSIS)
1250 MAT_TYPE = 'INELASTIC' # 'ELASTIC' OR 'INELASTIC'
1251 ANAL_TYPE = 'STATIC_EXTERNAL_TIME-DEPENDENT_LOADING'
1252
1253 DATA = MDOF(MAT_TYPE, TOTAL_MASS, ANAL_TYPE)
1254
1255 (reaction_ETDLS, disp_ETDLS, DI_ETDLS,
1256  ele_force_ETDLS, node_displacements_ETDLS,
1257  PERIOD_MIN_ETDLS, PERIOD_MAX_ETDLS) = DATA
1258
1259
1260 XDATA = disp_ETDLS
1261 YDATA = reaction_ETDLS
1262 XLABEL = 'Displacement [m]'
1263 YLABEL = 'Base Reaction [N]'
1264 TITLE = 'Base Reaction and Displacement of Structure During Static External Time-dependent Loading Anal
1265 COLOR = 'black'
1266 SEMIOLOGY = False
1267 PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMIOLOGY)
1268
1269 # PLOT STRUCTURAL PERIOD DURING THE ANALYSIS
1270 plt.figure(0, figsize=(12, 8))
1271 plt.plot(disp_ETDLS, PERIOD_MIN_ETDLS, linewidth=3)
1272 plt.plot(disp_ETDLS, PERIOD_MAX_ETDLS, linewidth=3)
1273 plt.title('Period of Structure During Static External Time-dependent Loading Analysis')
1274 plt.ylabel('Structural Period [s]')
1275 plt.xlabel('Displacement [m]')
1276 #plt.semilogy()
1277 plt.grid()
1278 plt.legend([f'PERIOD - MIN VALUES: Min: {np.min(PERIOD_MIN_ETDLS):.3f} (s) - Mean: {np.mean(PERIOD_MIN_
1279             f'PERIOD - MAX VALUES: Min: {np.min(PERIOD_MAX_ETDLS):.3f} (s) - Mean: {np.mean(PERIOD_MA
1280             ])
1281 plt.show()
1282

```

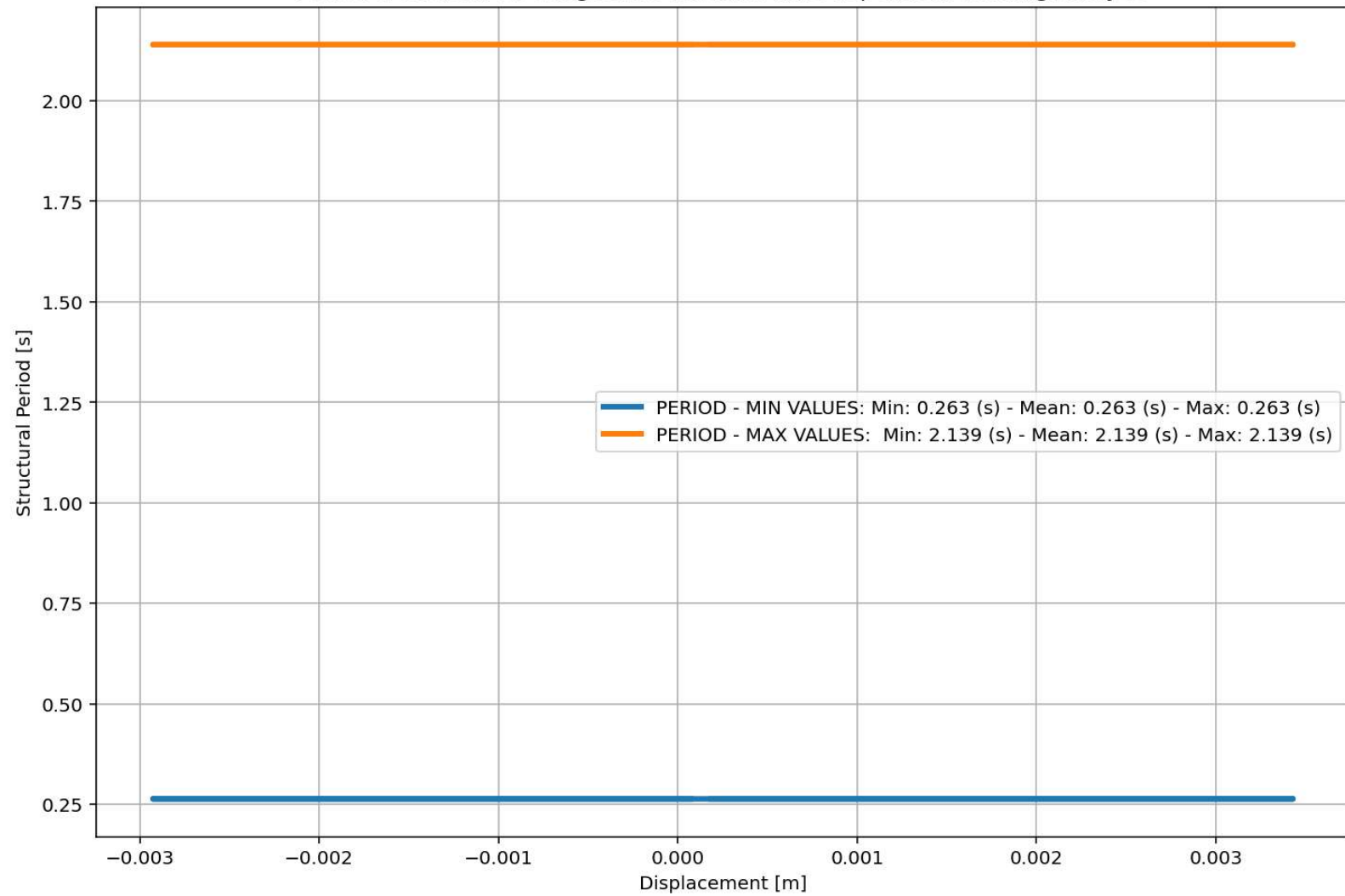


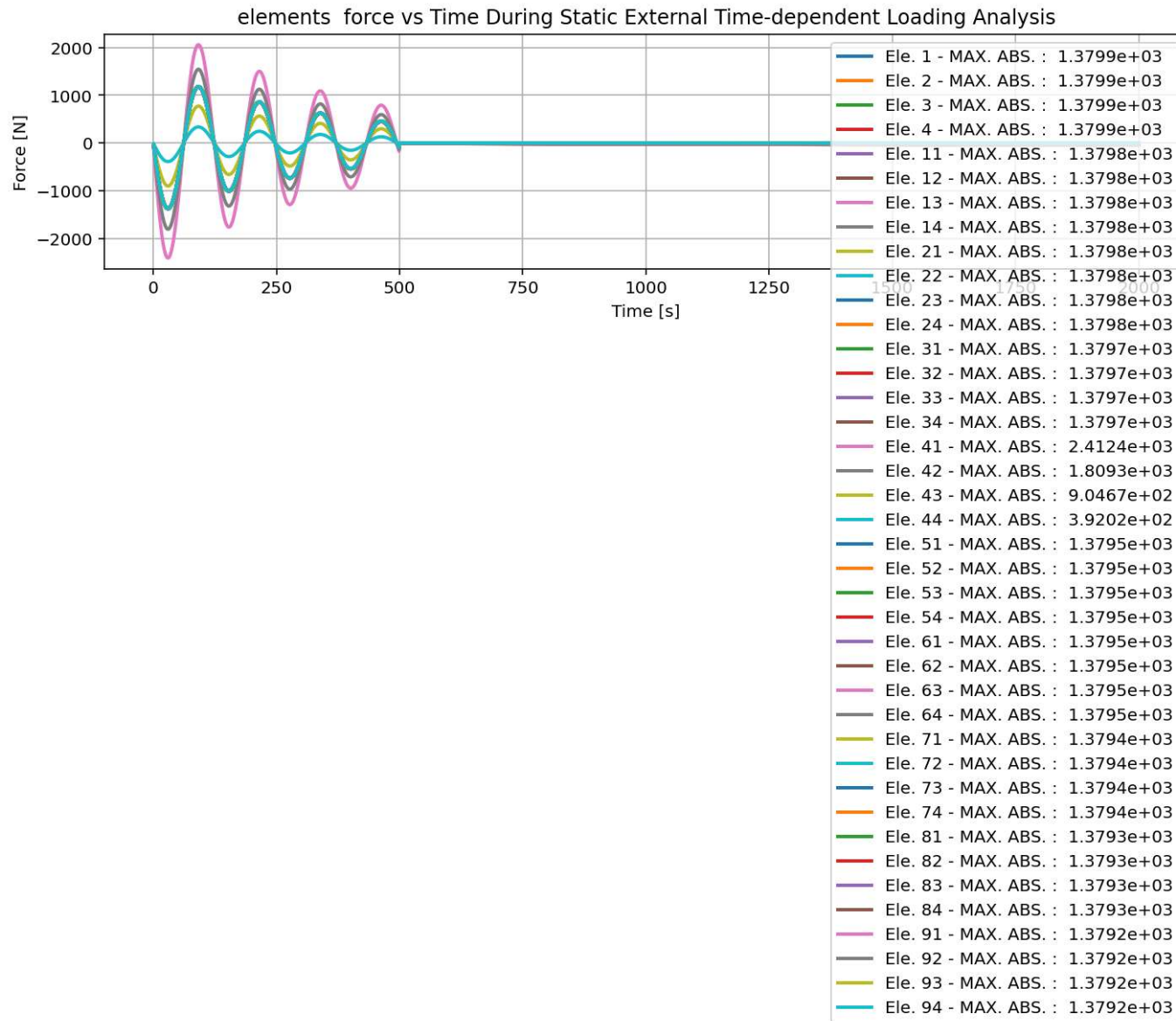


Base Reaction and Displacement of Structure During Static External Time-dependent Loading Analysis

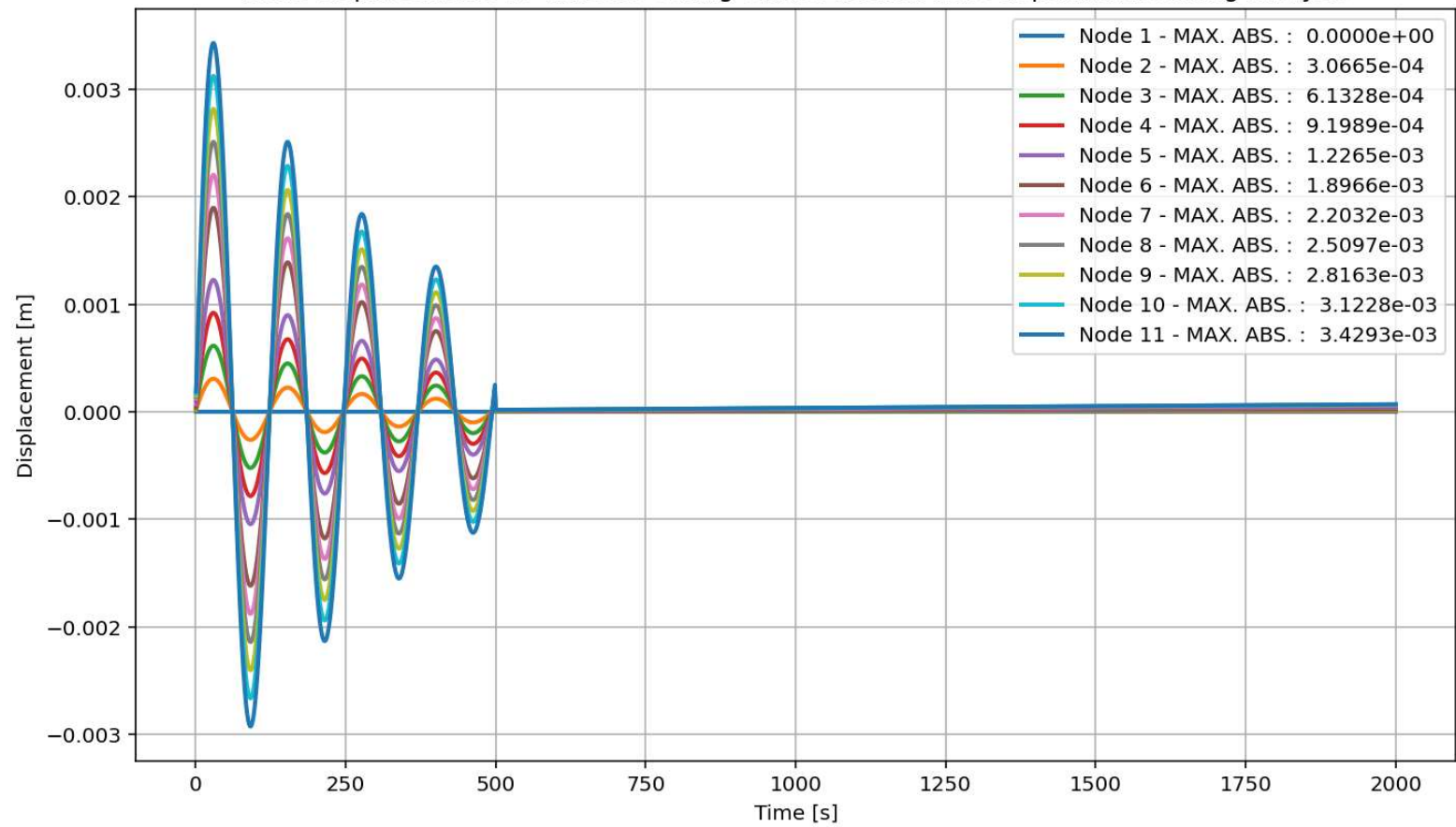


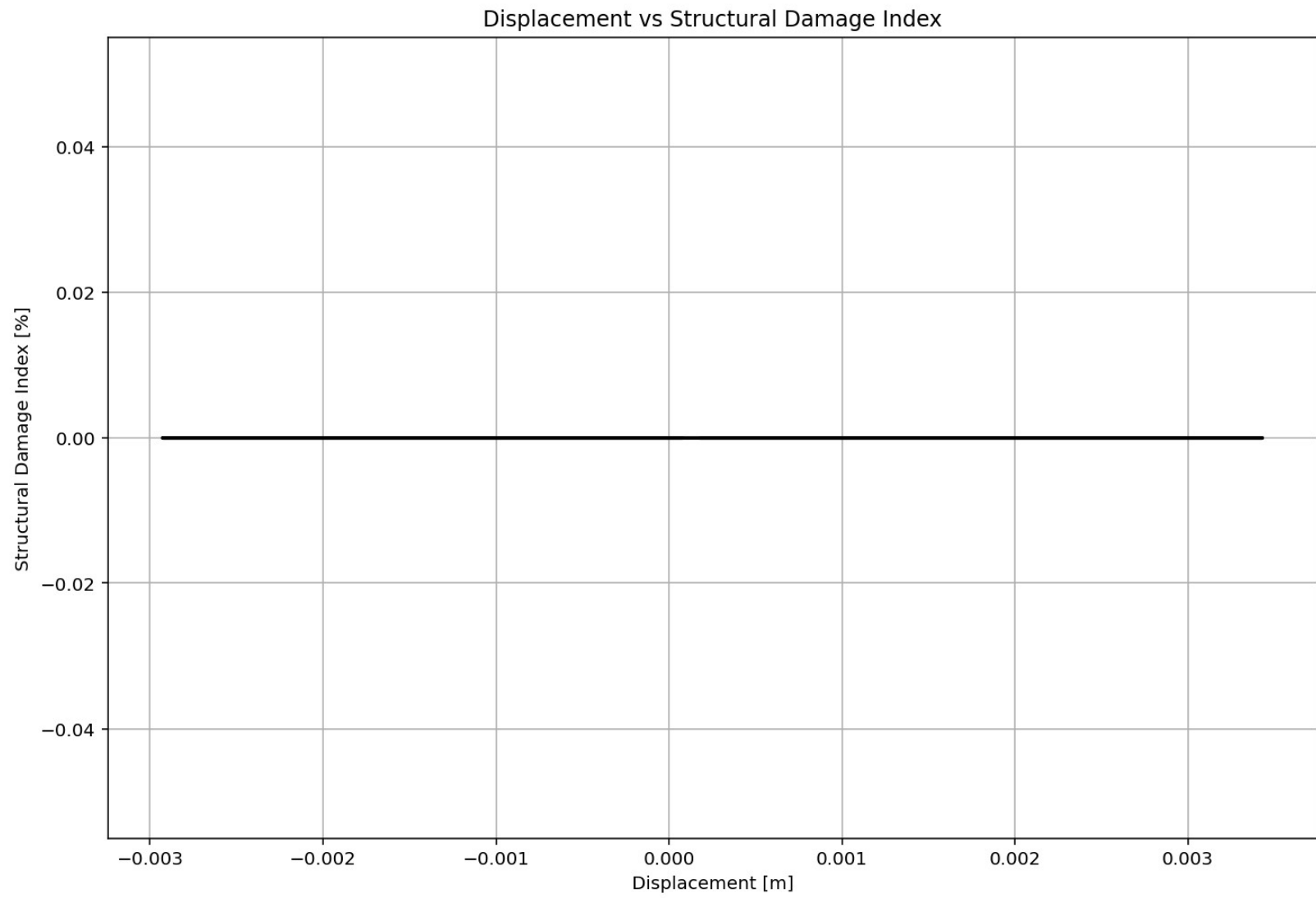
Period of Structure During Static External Time-dependent Loading Analysis



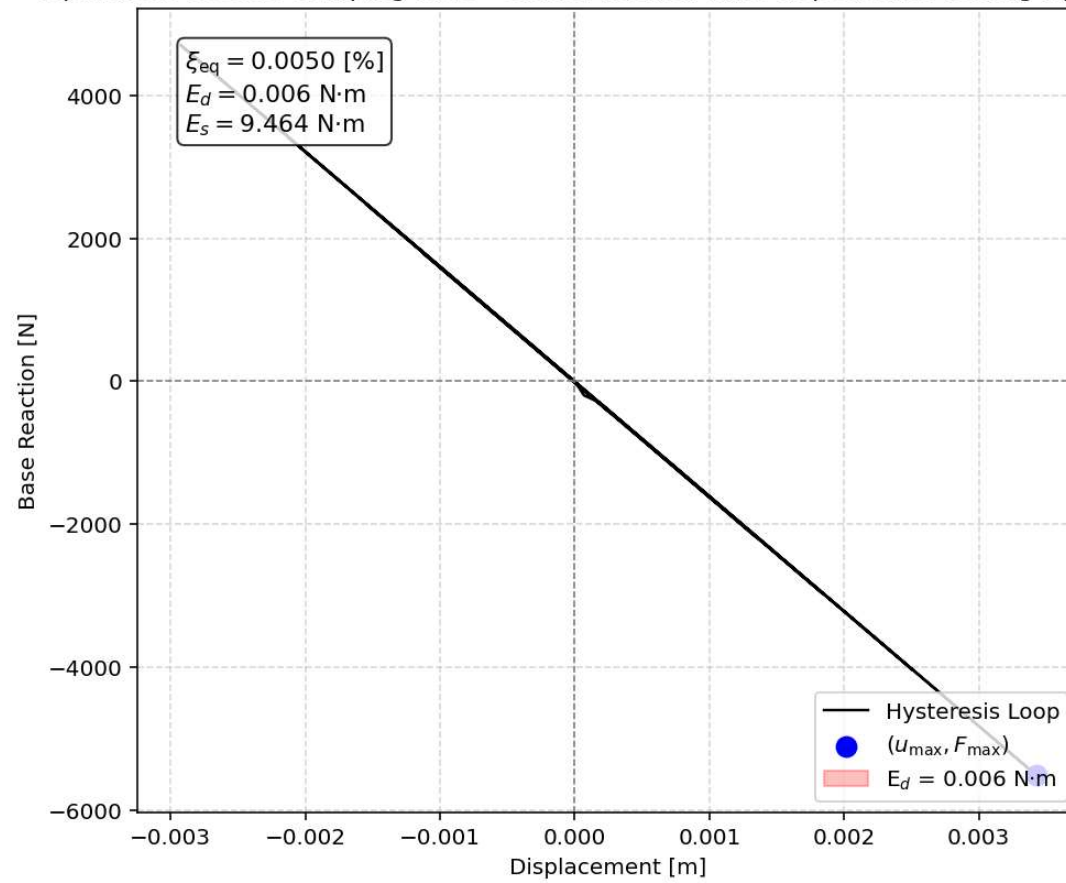


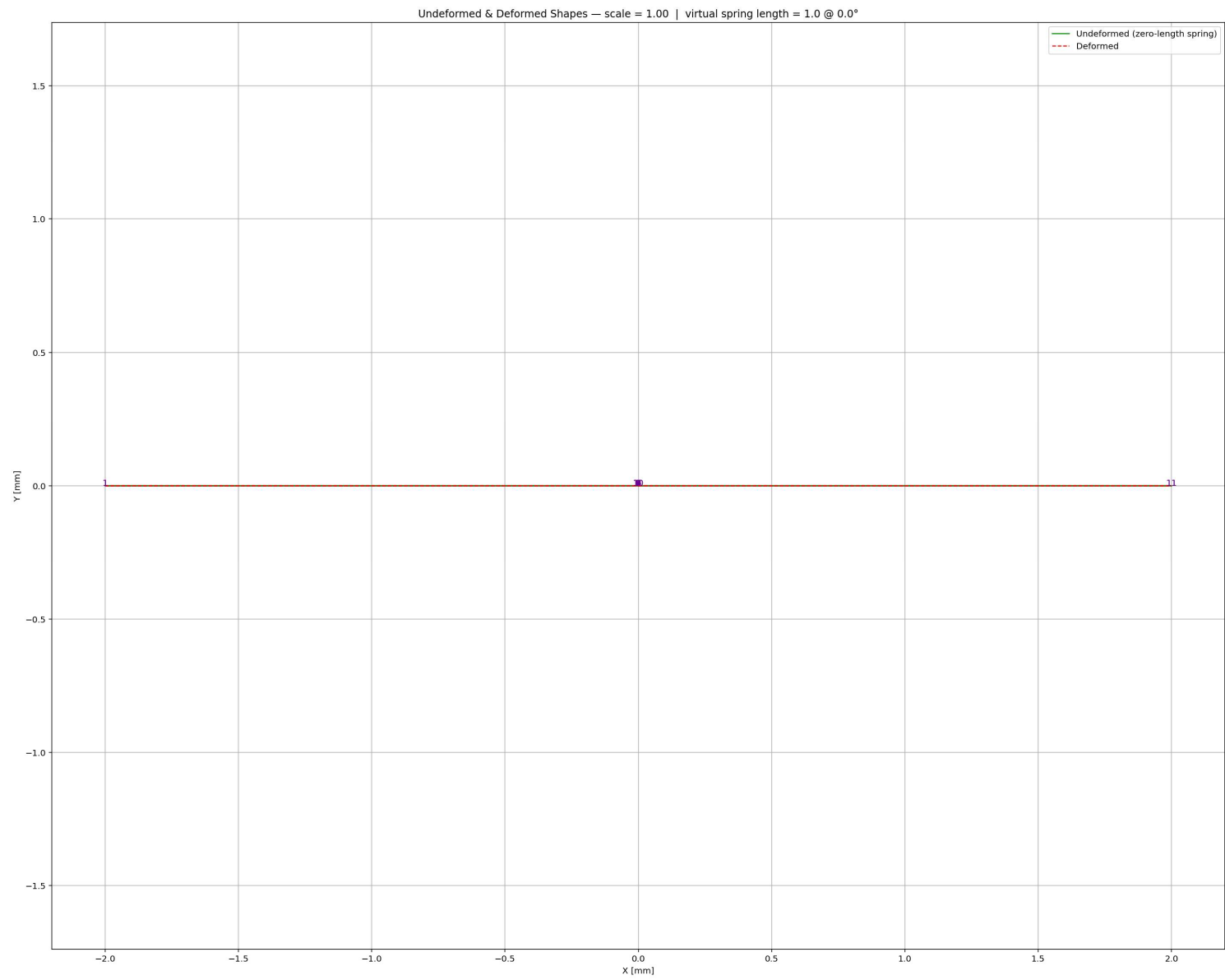
Node Displacements vs Time for During Static External Time-dependent Loading Analysis





Equivalent viscous damping ratio - Static External Time-dependent Loading Hysteresis





DYNAMIC TIME-HISTORY WITH EXTERNAL TIME-DEPENDENT LOADING ANALYSIS

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P(t)_POST-FIRE_MDOF_8_ANA.py X

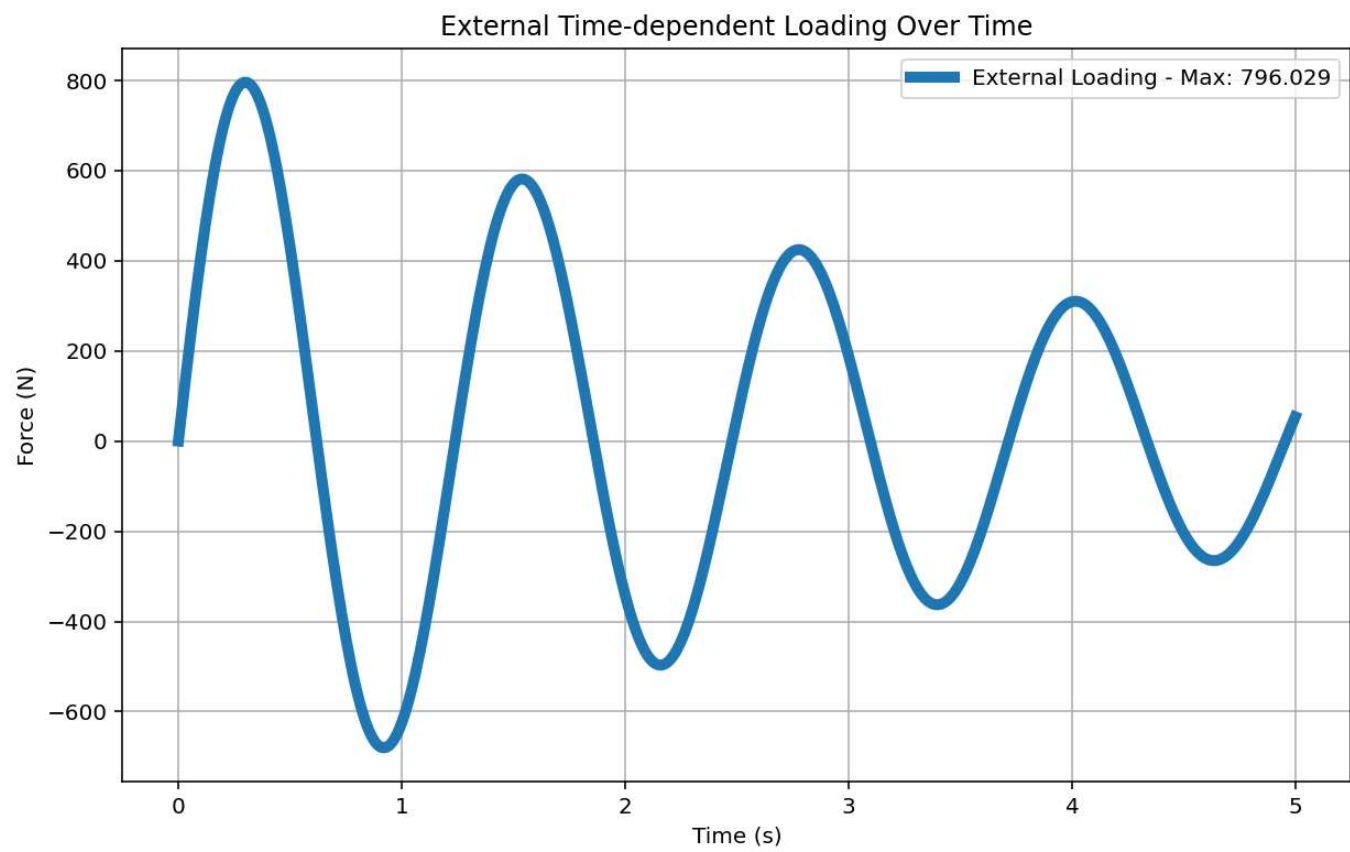
```
1316 # EXTERNAL TIME-DEPENDENT LOADING ANALYSIS (DYNAMIC TIME-HISTORY ANALYSIS)
1317 MAT_TYPE = 'INELASTIC' # 'ELASTIC' OR 'INELASTIC'
1318 ANAL_TYPE = 'DYNAMIC_EXTERNAL_TIME-DEPENDENT_LOADING'
1319
1320 DATA = MDOF(MAT_TYPE, TOTAL_MASS, ANAL_TYPE)
1321
1322 (time_ETDLD, reaction_ETDLD, disp_ETDLD, velo_ETDLD, acc_ETDLD, DI_ETDLD,
1323  ele_force_ETDLD, node_displacements_ETDLD,
1324  stiffness, PERIOD, damping_ratio,
1325  PERIOD_MIN_ETDLD, PERIOD_MAX_ETDLD) = DATA
1326
1327
1328
1329 XDATA = disp_ETDLD
1330 YDATA = reaction_ETDLD
1331 XLABEL = 'Displacement [m]'
1332 YLABEL = 'Base Reaction [N]'
1333 TITLE = 'Base Reaction and Displacement of Structure During Dynamic External Time-dependent Loading Analysis'
1334 COLOR = 'black'
1335 SEMILOGY = False
1336 PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMILOGY)
1337
1338 PLOT_TIME_HISTORY(time_ETDLD, reaction_ETDLD, disp_ETDLD, velo_ETDLD, acc_ETDLD)
1339
1340 # PLOT STRUCTURAL PERIOD DURING THE ANALYSIS
1341 plt.figure(0, figsize=(12, 8))
1342 plt.plot(disp_ETDLD, PERIOD_MIN_ETDLD, linewidth=3)
1343 plt.plot(disp_ETDLD, PERIOD_MAX_ETDLD, linewidth=3)
1344 plt.title('Period of Structure During Dynamic External Time-dependent Loading Analysis')
1345 plt.ylabel('Structural Period [s]')
1346 plt.xlabel('Displacement [m]')
1347 #plt.semilogy()
1348 plt.grid()
1349 plt.legend([f'PERIOD - MIN VALUES: Min: {np.min(PERIOD_MIN_ETDLD):.3f} (s) - Mean: {np.mean(PERIOD_MIN
```

...GY DISSIPATION_CAPACITY_INDEX\57_POST_FIRE+\EXAMPLE_04_MDOF

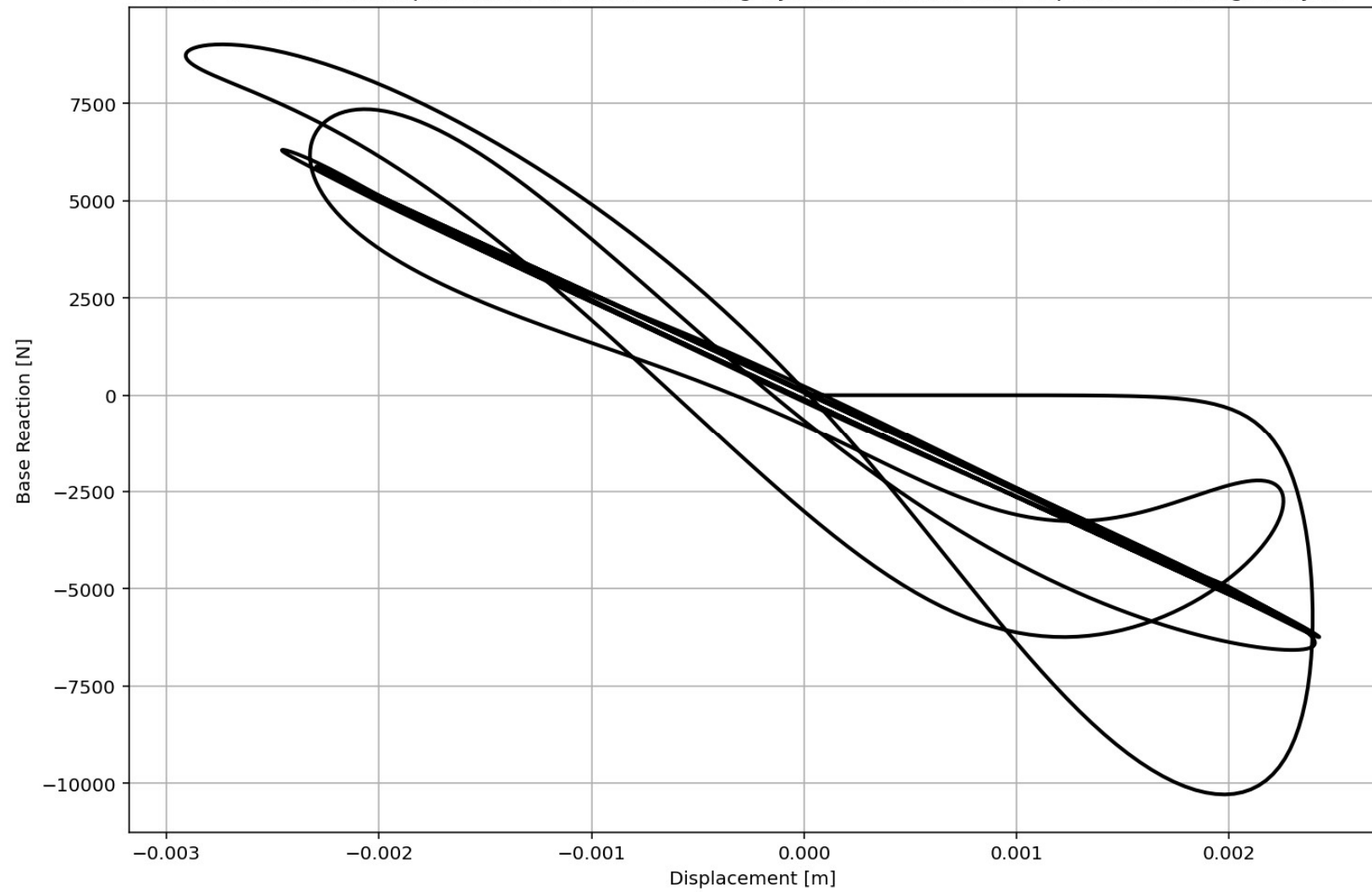
24 %



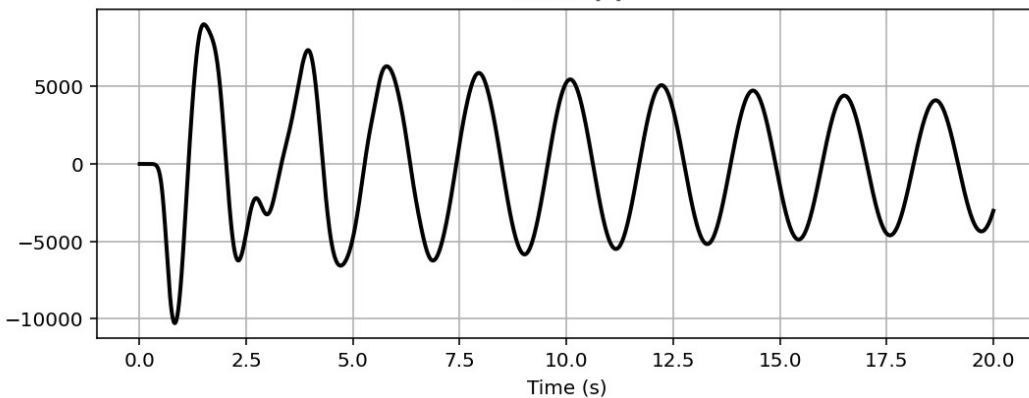
IPython Console Plots Variable Explorer Help Debugger History Files



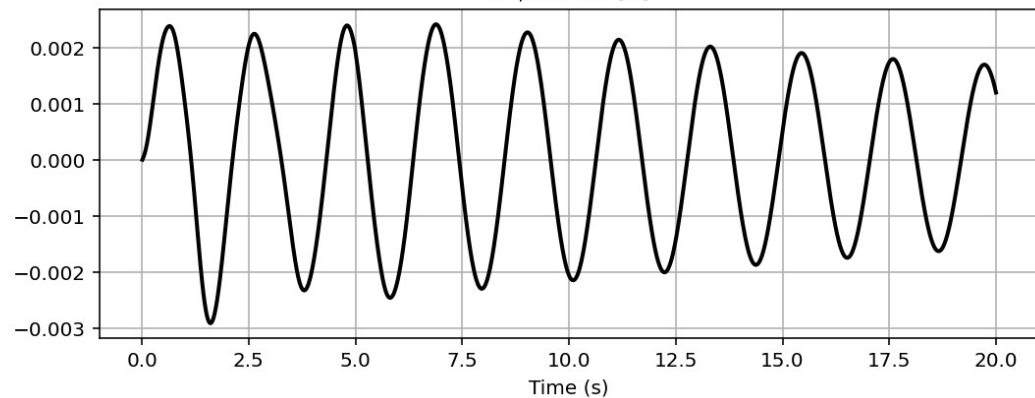
Base Reaction and Displacement of Structure During Dynamic External Time-dependent Loading Analysis



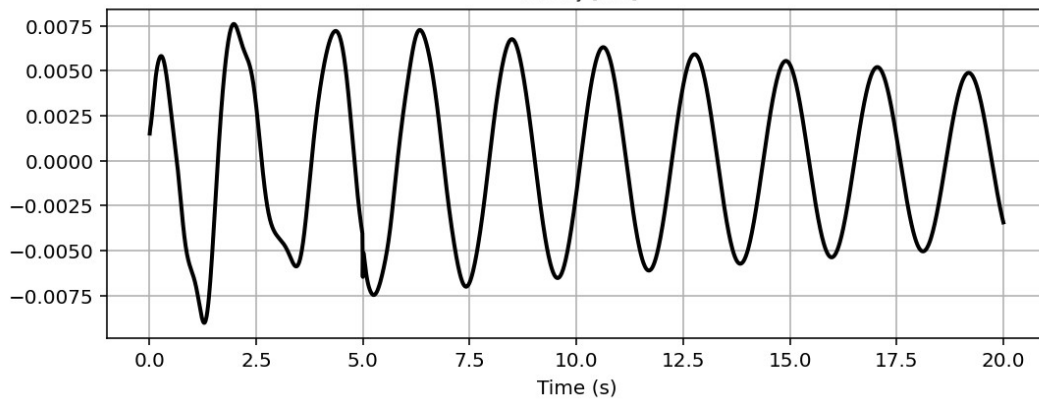
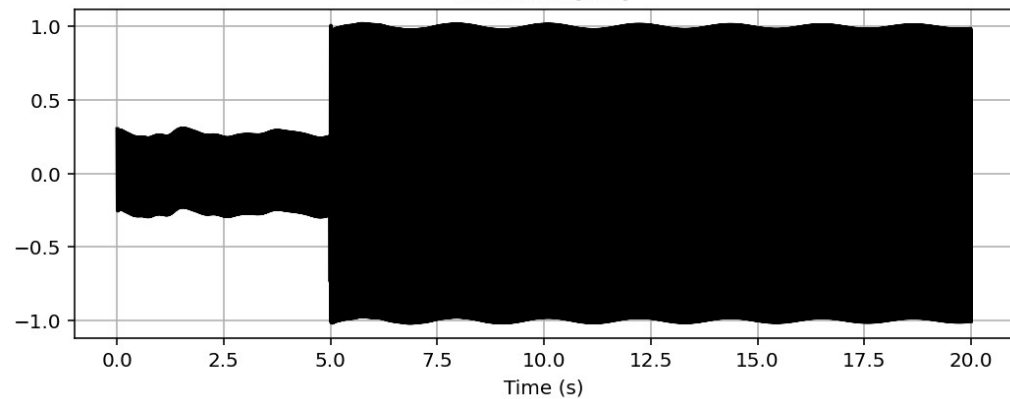
Reaction [N]



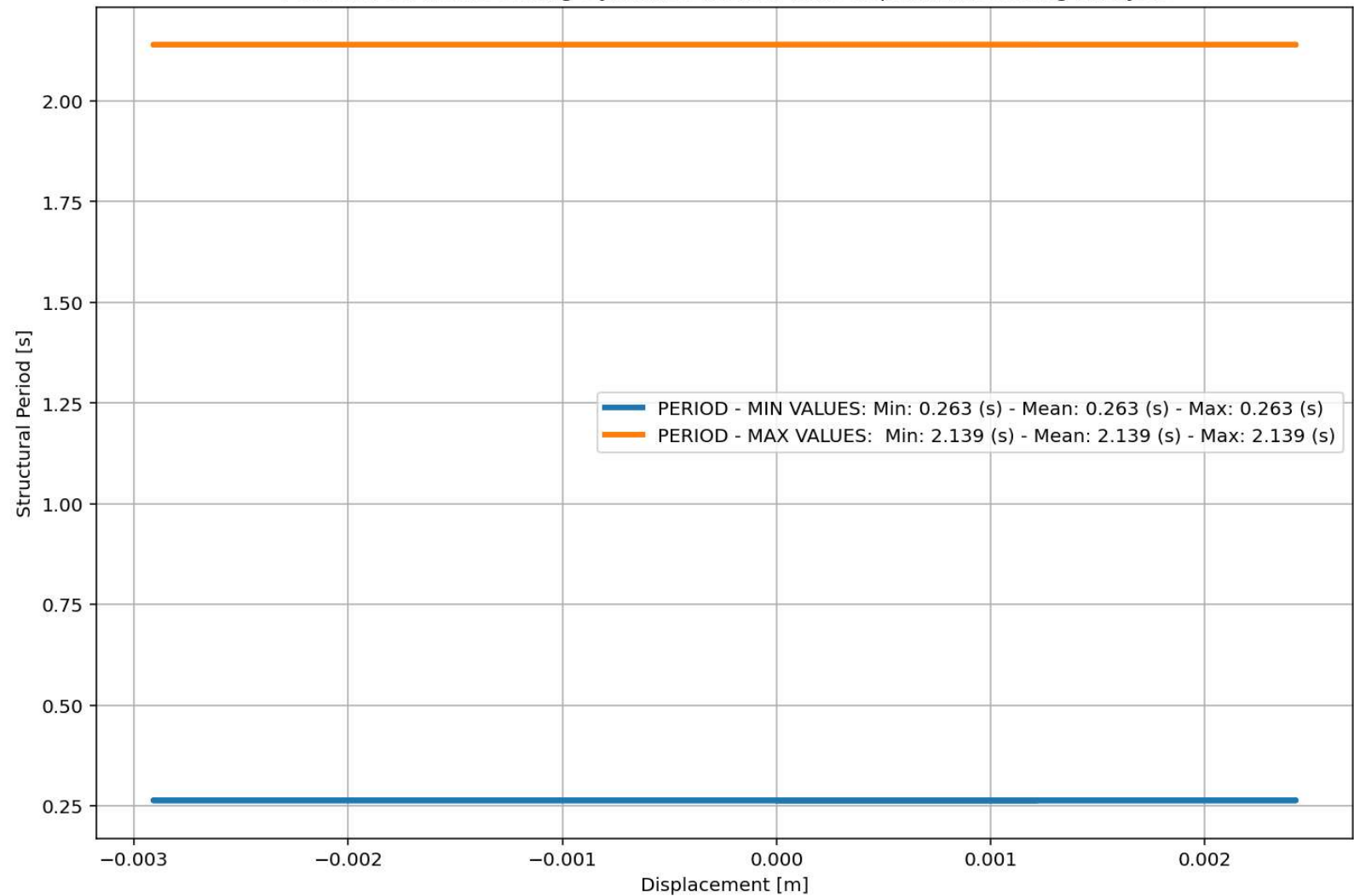
Displacement [m]

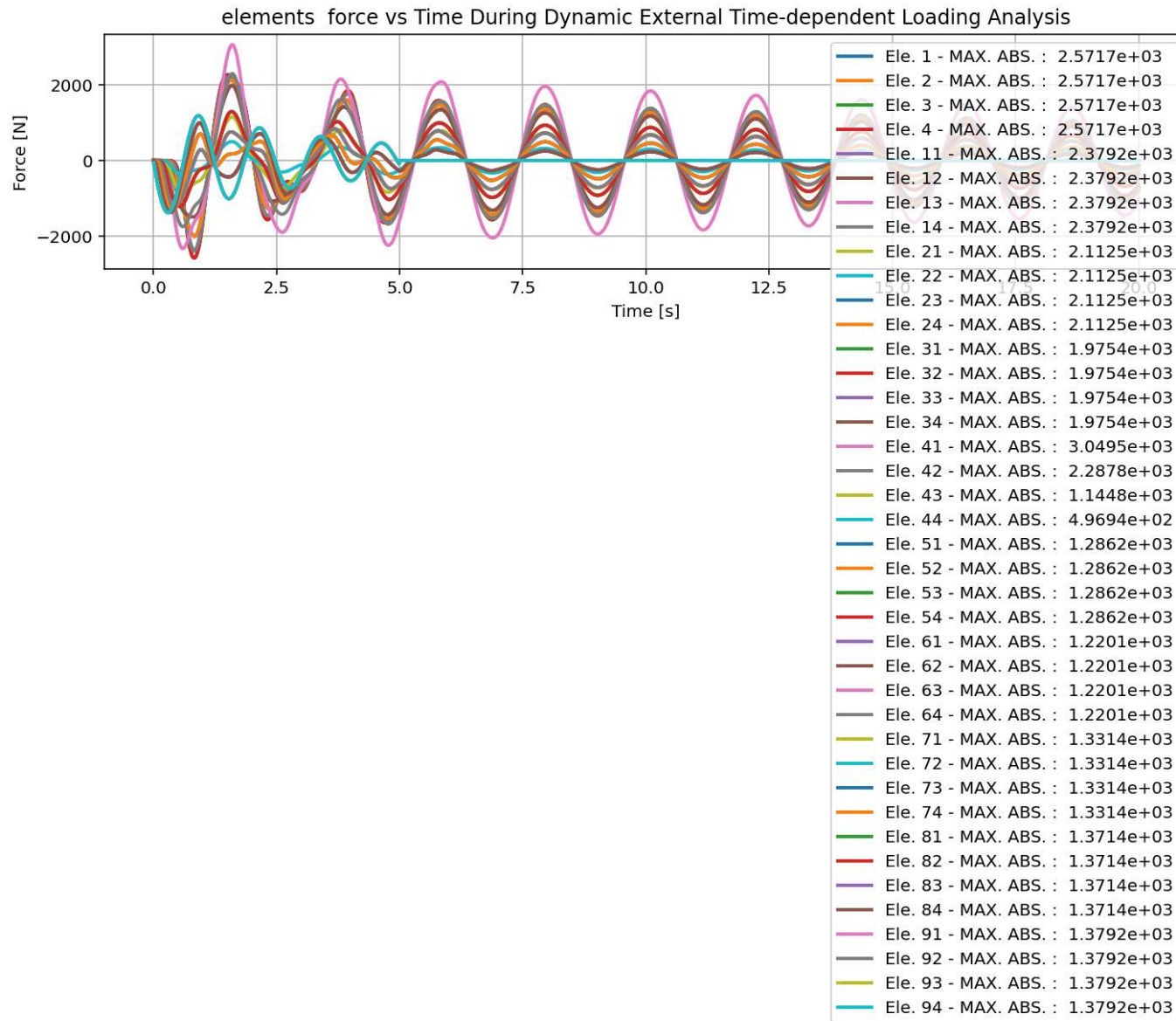


Velocity [m/s]

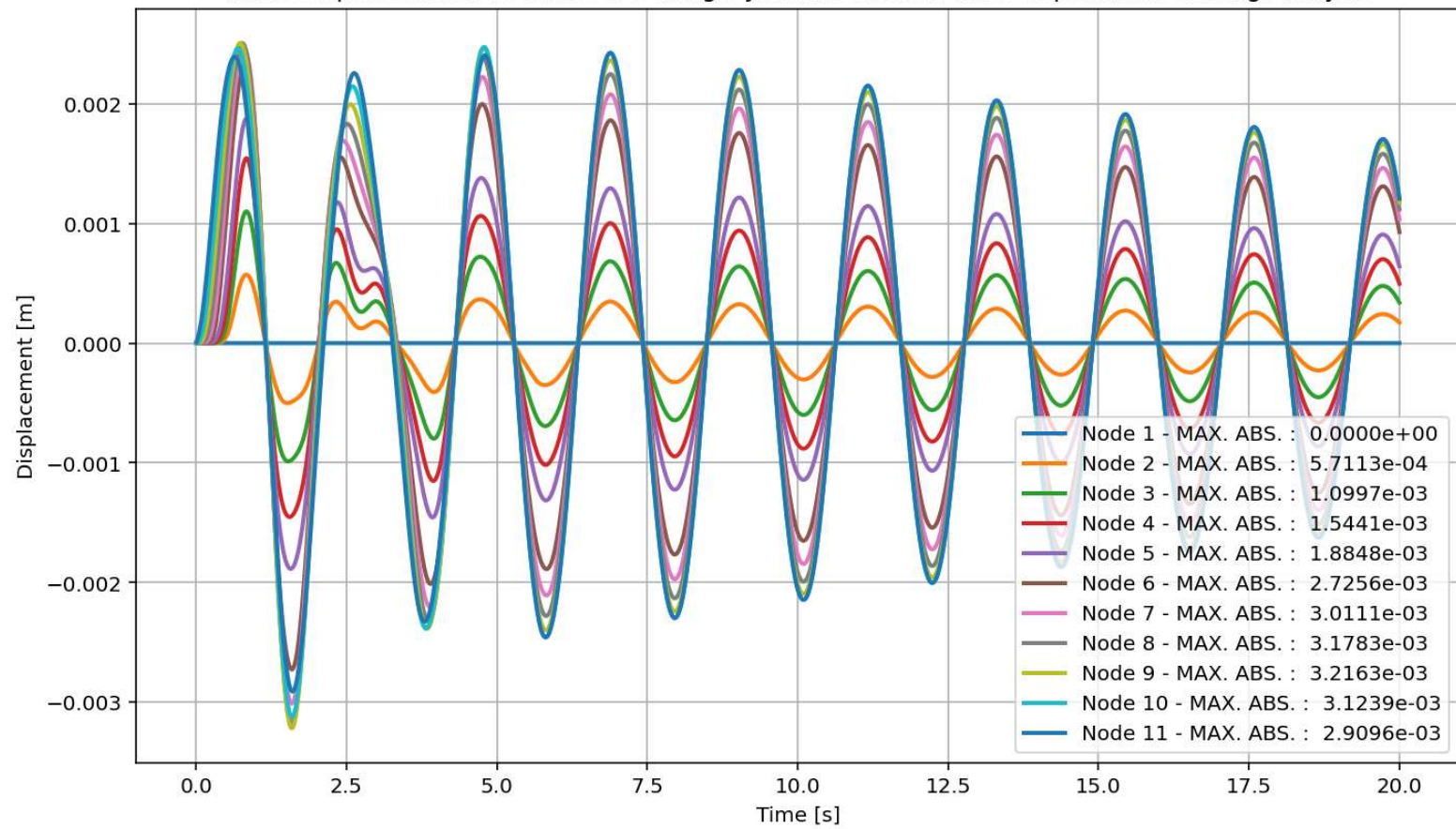
Acceleration [m/s²]

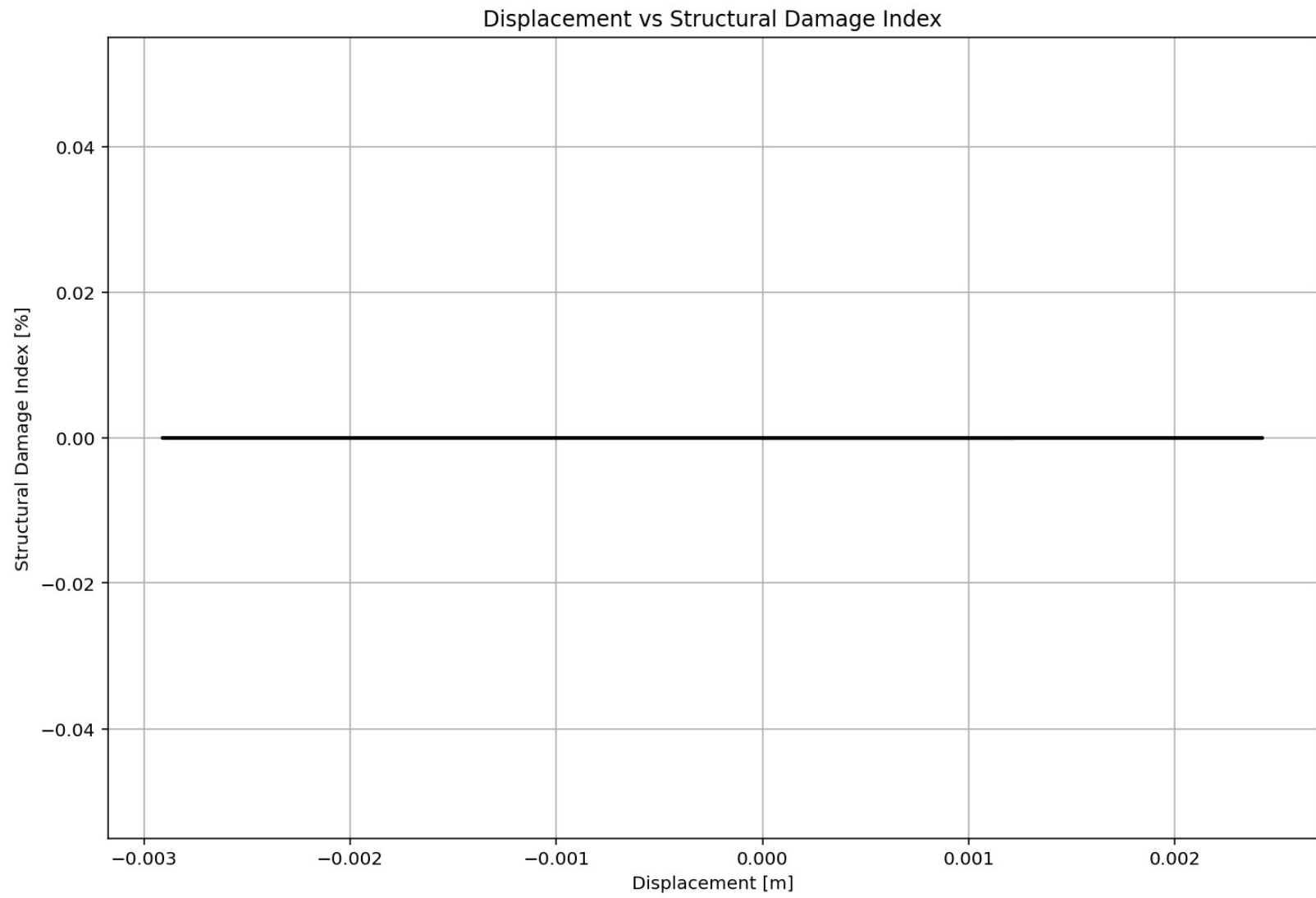
Period of Structure During Dynamic External Time-dependent Loading Analysis



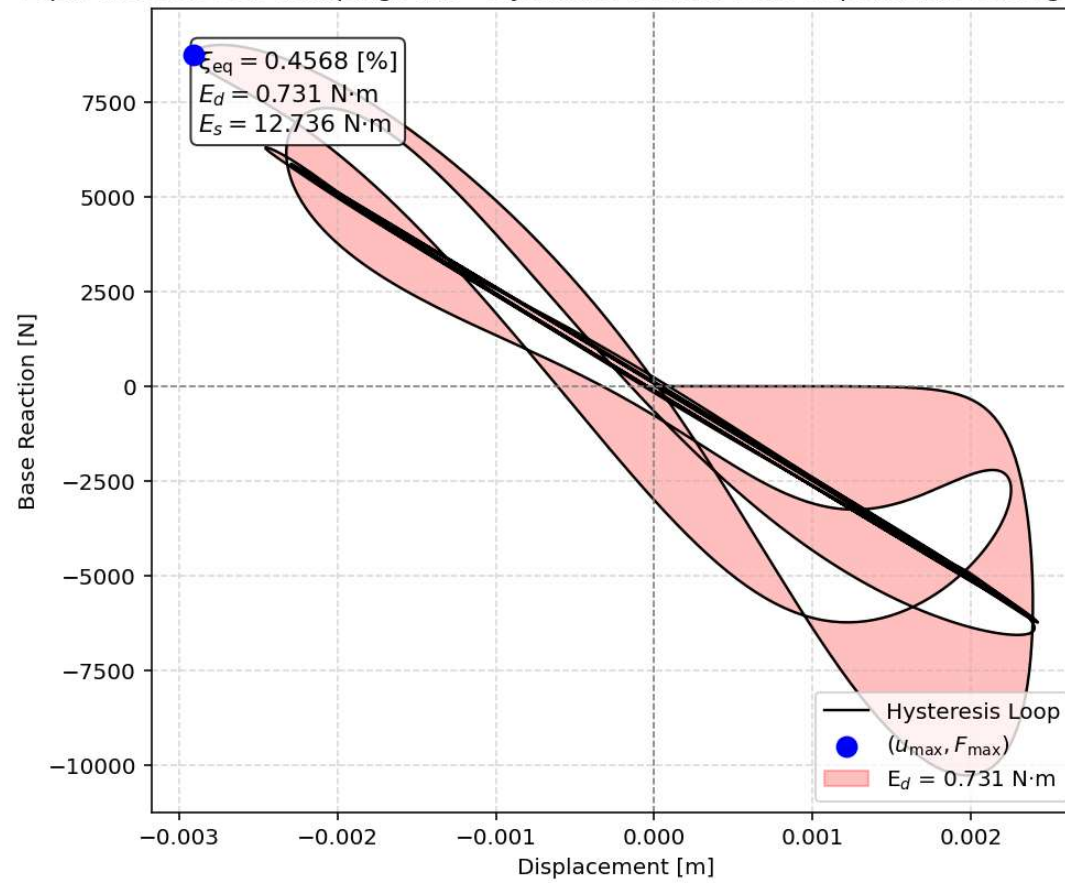


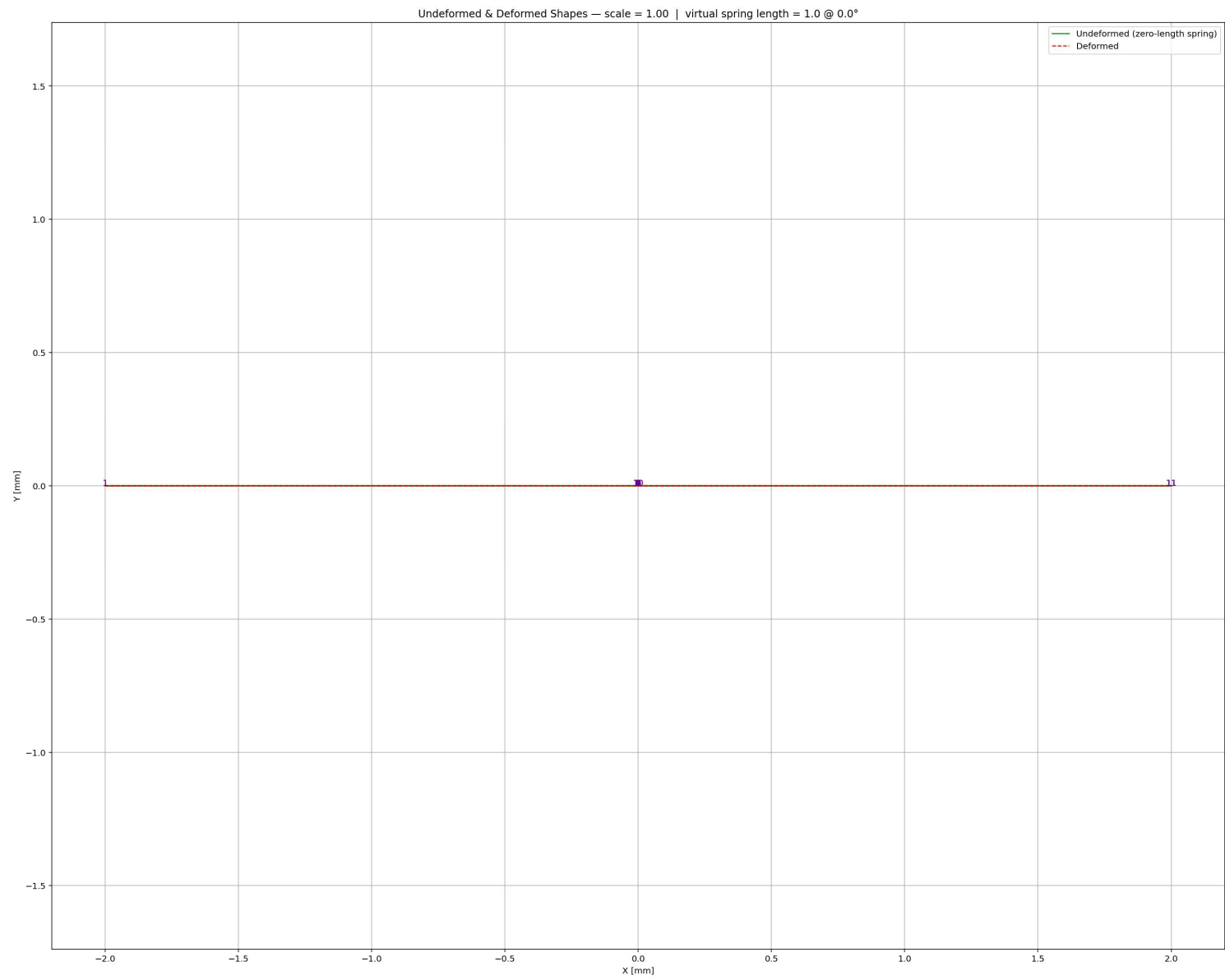
Node Displacements vs Time for During Dynamic External Time-dependent Loading Analysis





Equivalent viscous damping ratio - Dynamic External Time-dependent Loading Hysteresis





FREE-VIBRATION ANALYSIS

C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...IRE+\EXAMPLE_04_MDOF\P(t)_POST-FIRE_MDOF_8_ANA.py

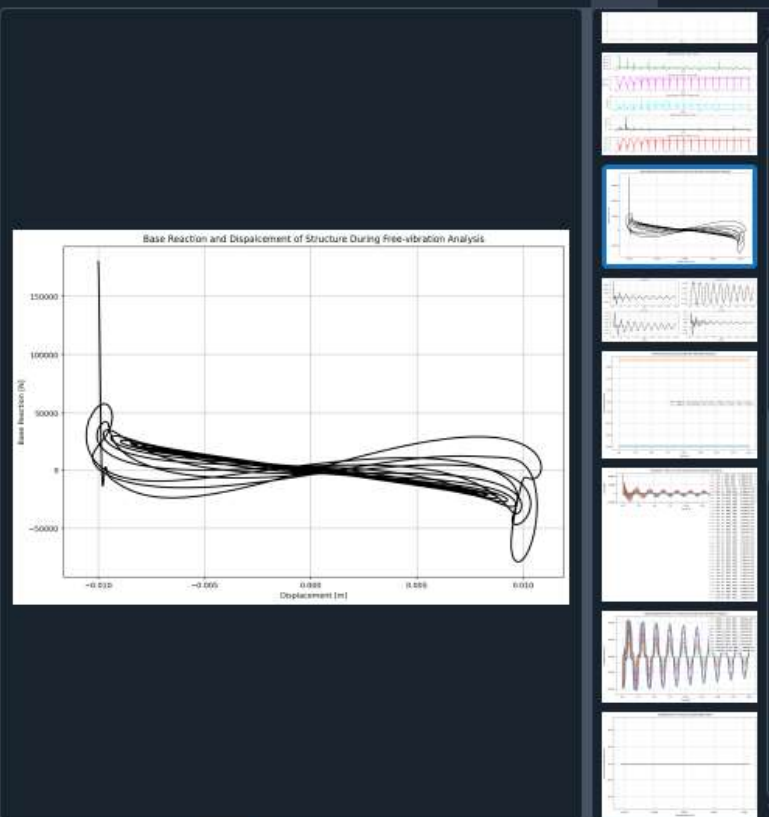
P(t)_POST-FIRE_MDOF_8_ANA.py X

```

1379 # FREE-VIBRATION ANALYSIS (DYNAMIC TIME-HISTORY ANALYSIS)
1380 MAT_TYPE = 'INELASTIC' # 'ELASTIC' OR 'INELASTIC'
1381 ANAL_TYPE = 'FREE-VIBRATION'
1382 DATA = MDOF(MAT_TYPE, TOTAL_MASS, ANAL_TYPE)
1383
1384 (time_FV, reaction_FV, disp_FV, velo_FV, acc_FV, DI_FV,
1385  ele_force_FV, node_displacements_FV,
1386  stiffness_FV, PERIOD_FV, damping_ratio_FV,
1387  PERIOD_MIN_FV, PERIOD_MAX_FV) = DATA
1388
1389 XDATA = disp_FV
1390 YDATA = reaction_FV
1391 XLABEL = 'Displacement [m]'
1392 YLABEL = 'Base Reaction [N]'
1393 TITLE = 'Base Reaction and Displacment of Structure During Free-vibration Analysis'
1394 COLOR = 'black'
1395 SEMIOLOGY = False
1396 PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMIOLOGY)
1397
1398 PLOT_TIME_HISTORY(time_FV, reaction_FV, disp_FV, velo_FV, acc_FV)
1399
1400 # PLOT STRUCTURAL PERIOD DURING THE ANALYSIS
1401 plt.figure(0, figsize=(12, 8))
1402 plt.plot(time_FV, PERIOD_MIN_FV, linewidth=3)
1403 plt.plot(time_FV, PERIOD_MAX_FV, linewidth=3)
1404 plt.title('Period of Structure During Free-vibration Analysis')
1405 plt.ylabel('Structural Period [s]')
1406 plt.xlabel('Time [s]')
1407 #plt.semilogy()
1408 plt.grid()
1409 plt.legend([f'PERIOD - MIN VALUES: Min: {np.min(PERIOD_MIN_FV):.3f} (s) - Mean: {np.mean(PERIOD_MIN_FV,
1410  f'PERIOD - MAX VALUES: Min: {np.min(PERIOD_MAX_FV):.3f} (s) - Mean: {np.mean(PERIOD_MAX_FV,
1411  ])
1412 plt.show()
    
```

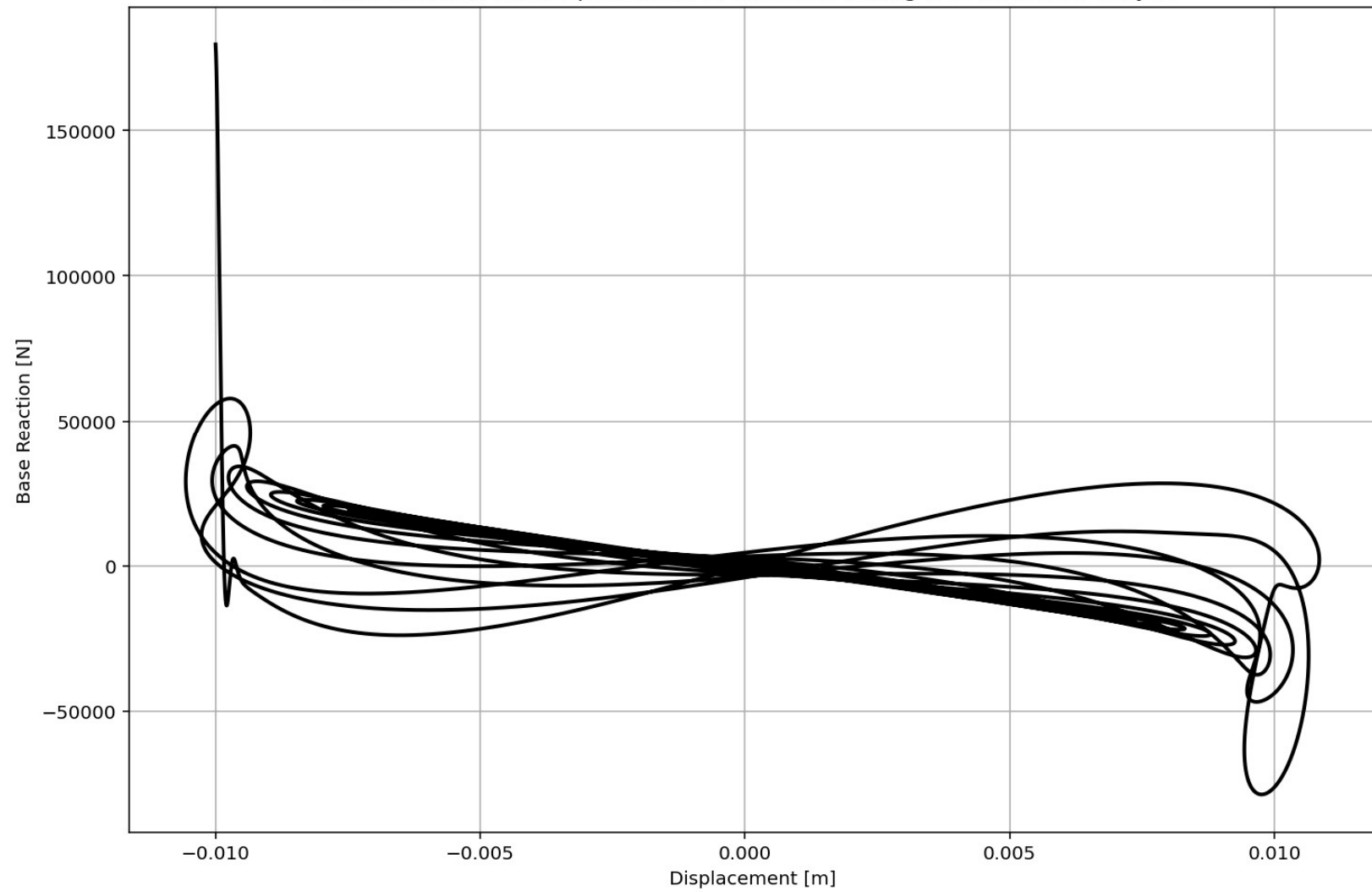
...GY DISSIPATION_CAPACITY_INDEX\57_POST_FIRE+\EXAMPLE_04_MDOF

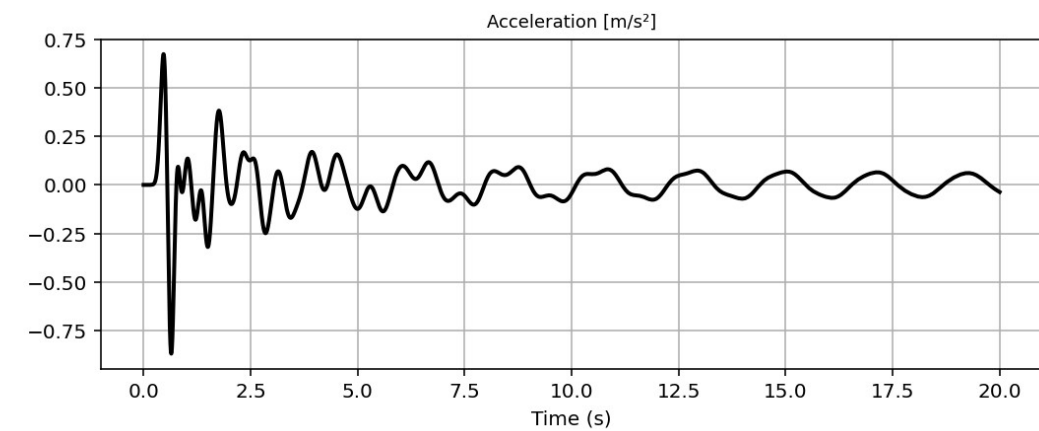
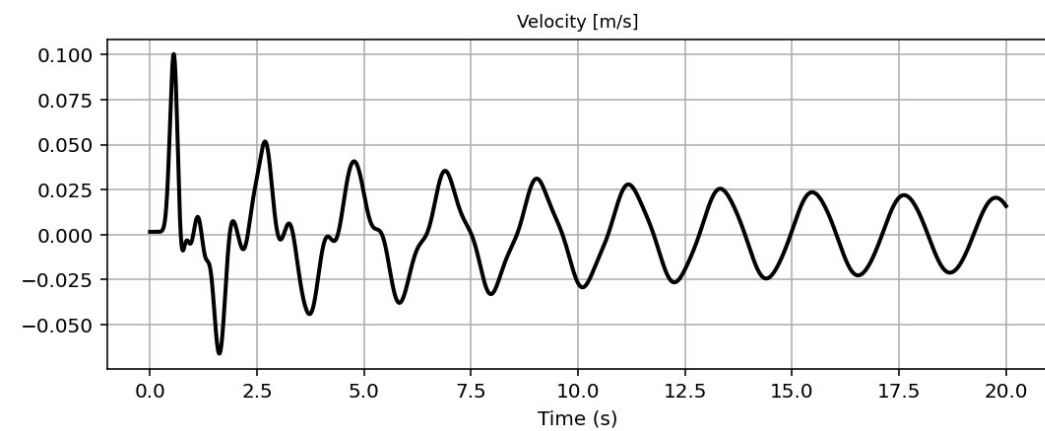
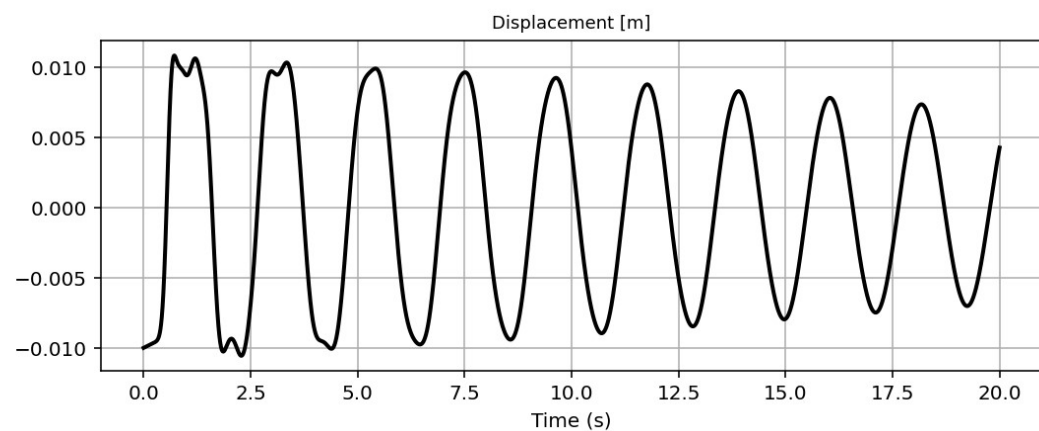
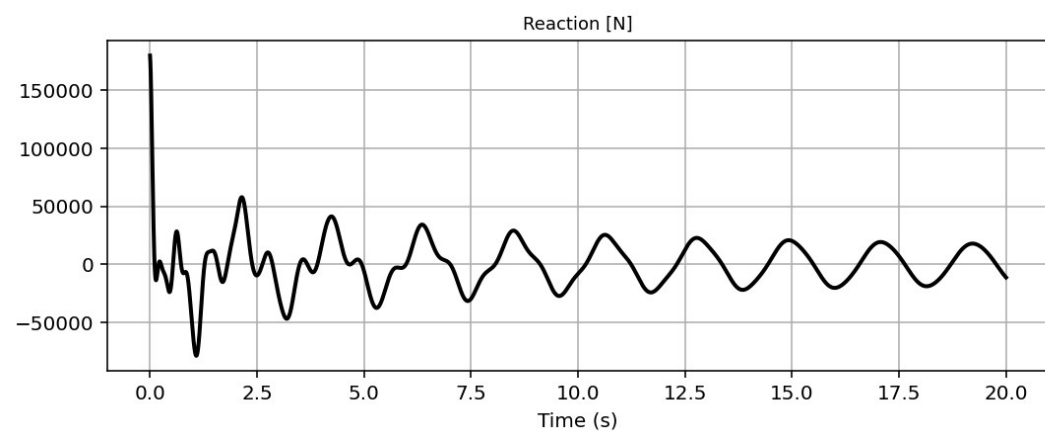
24 %



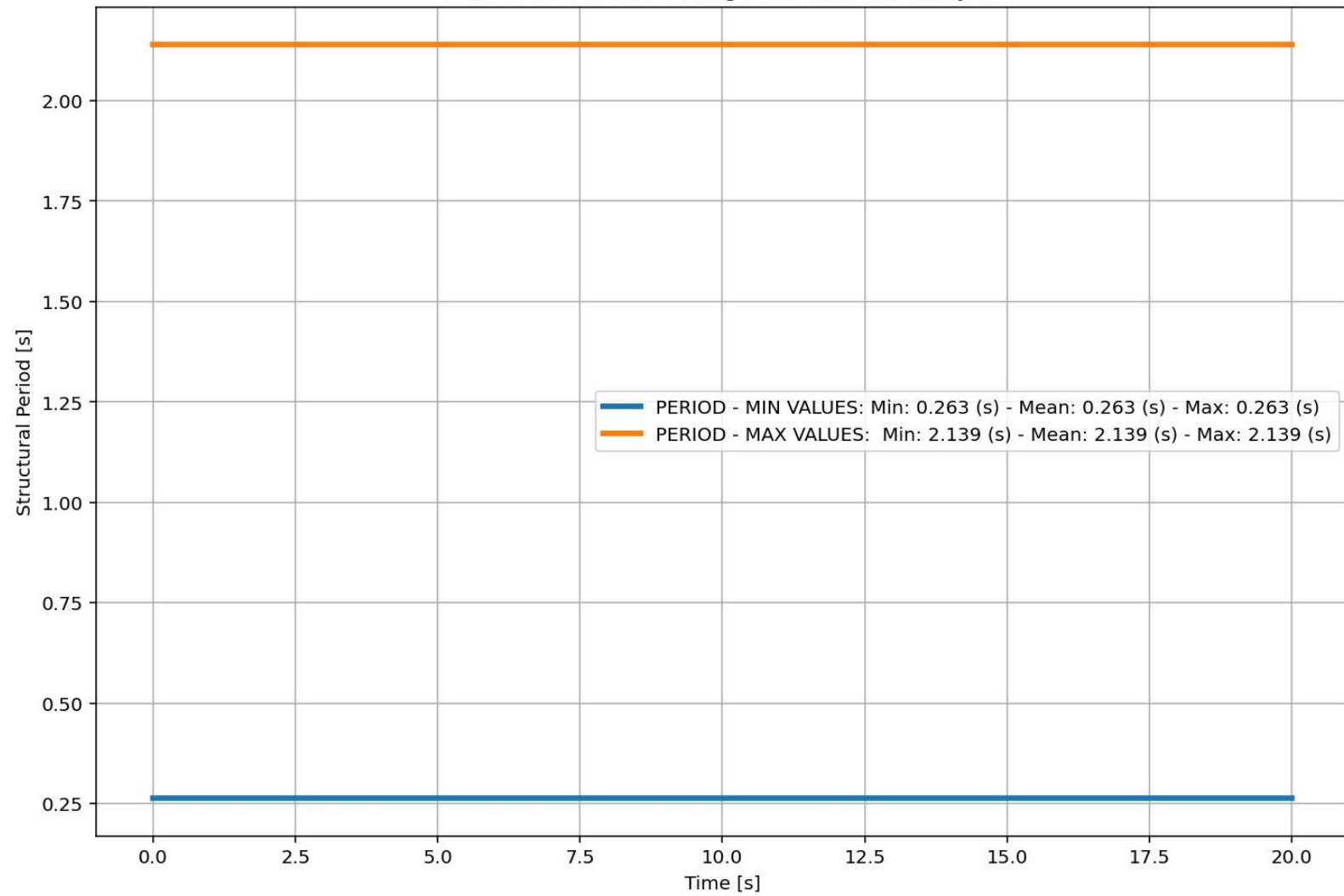
IPython Console Plots Variable Explorer Help Debugger History Files

Base Reaction and Dispalcement of Structure During Free-vibration Analysis

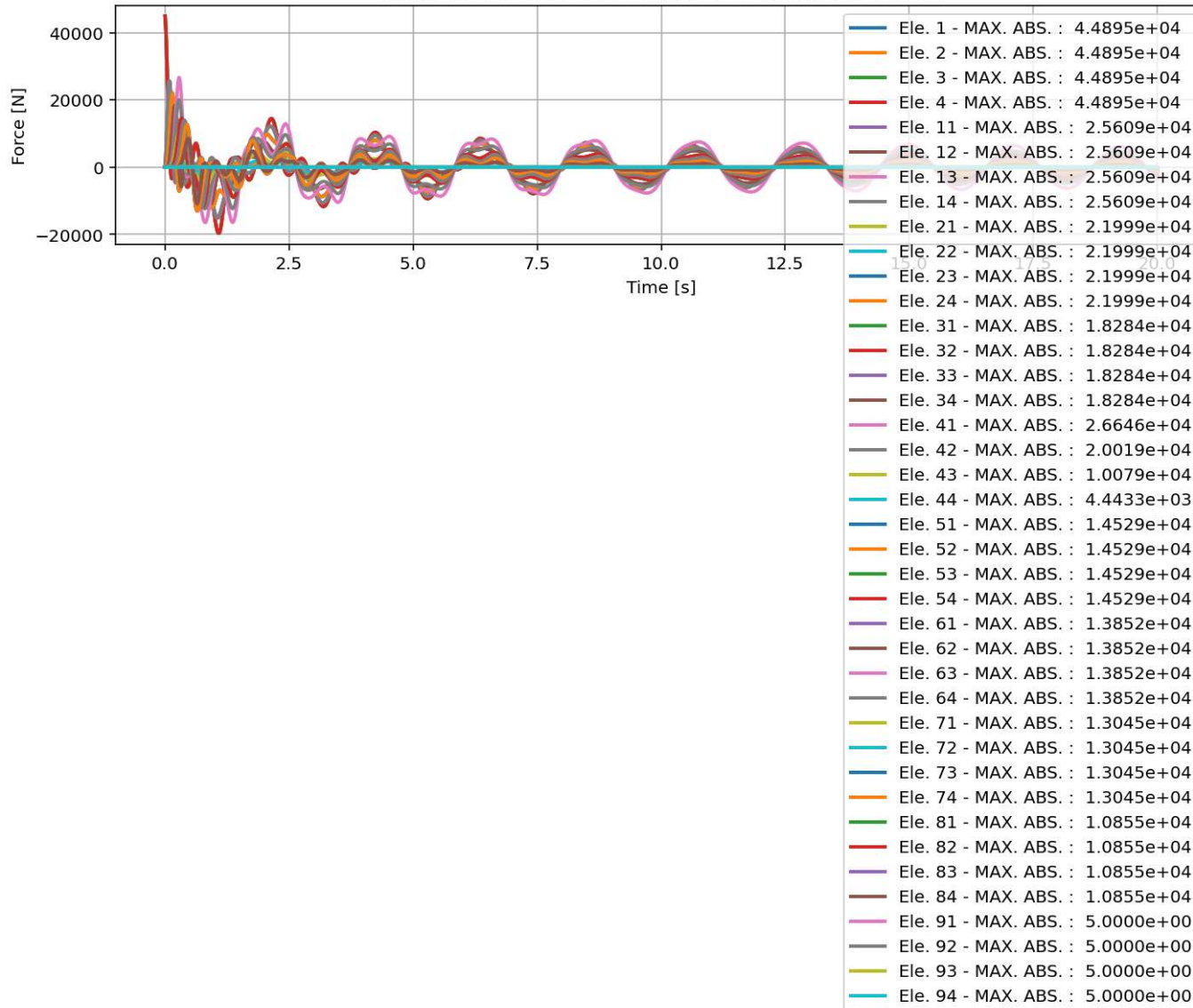




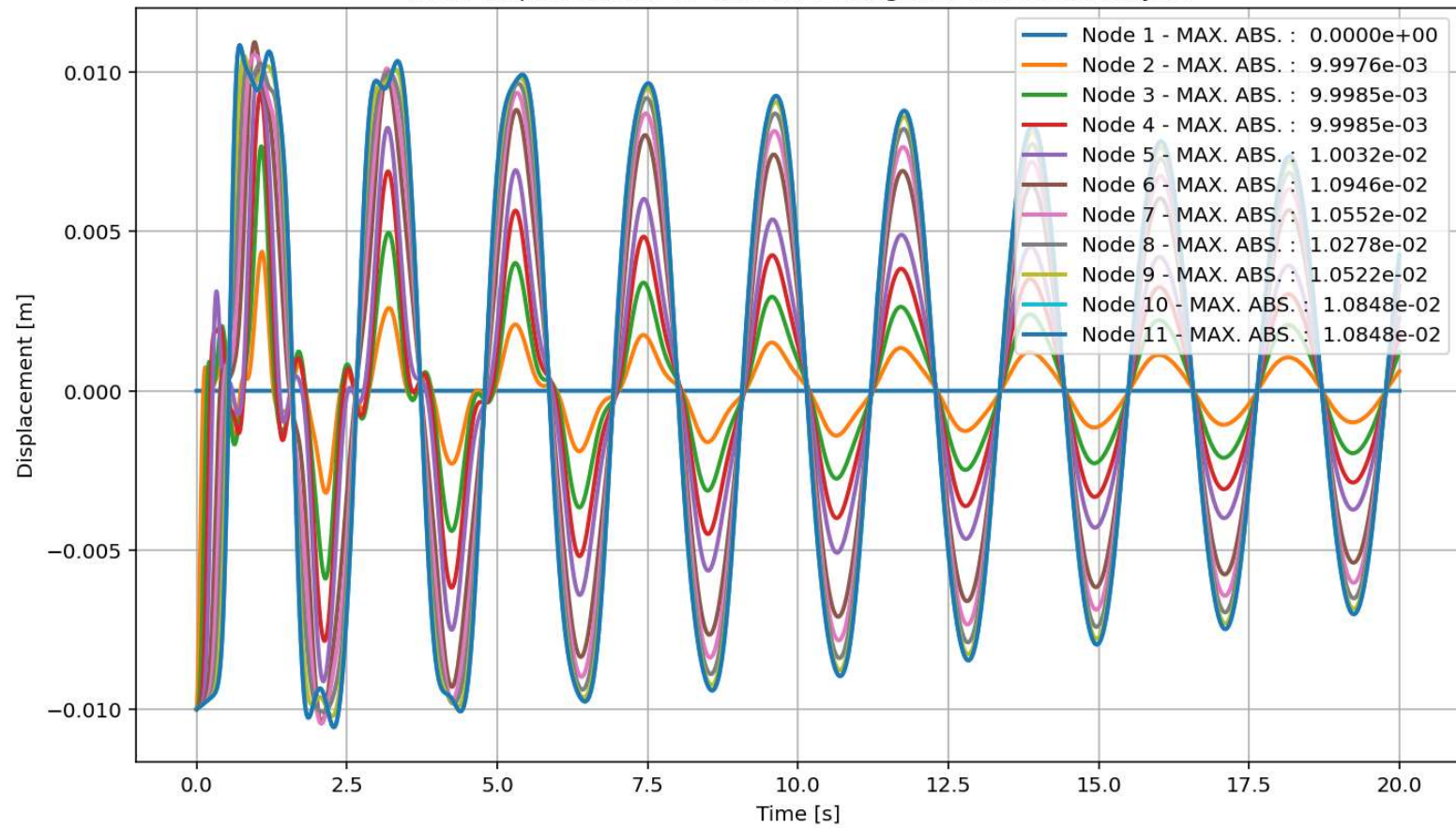
Period of Structure During Free-vibration Analysis

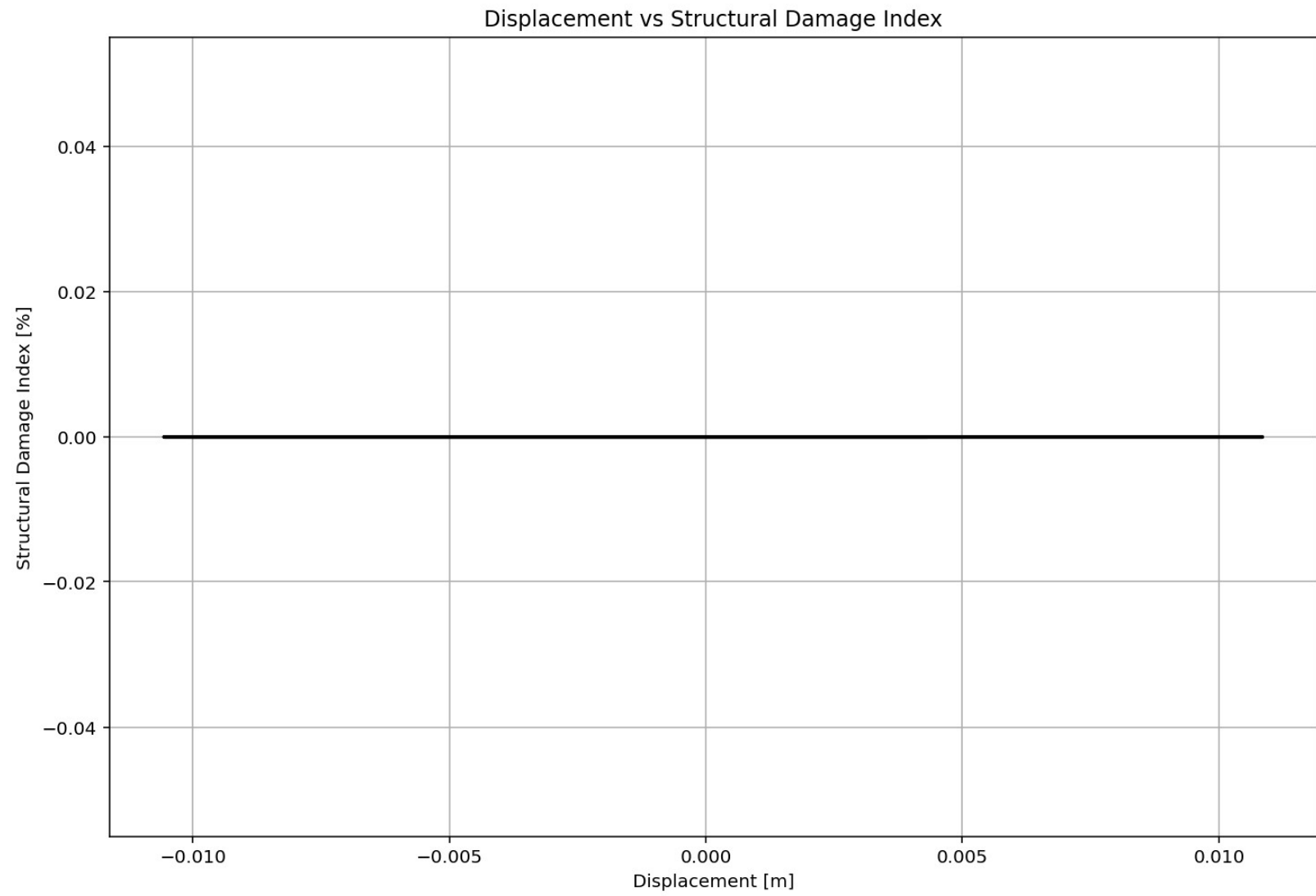


elements force vs Time During Free-vibration Analysis



Node Displacements vs Time for During Free-vibration Analysis





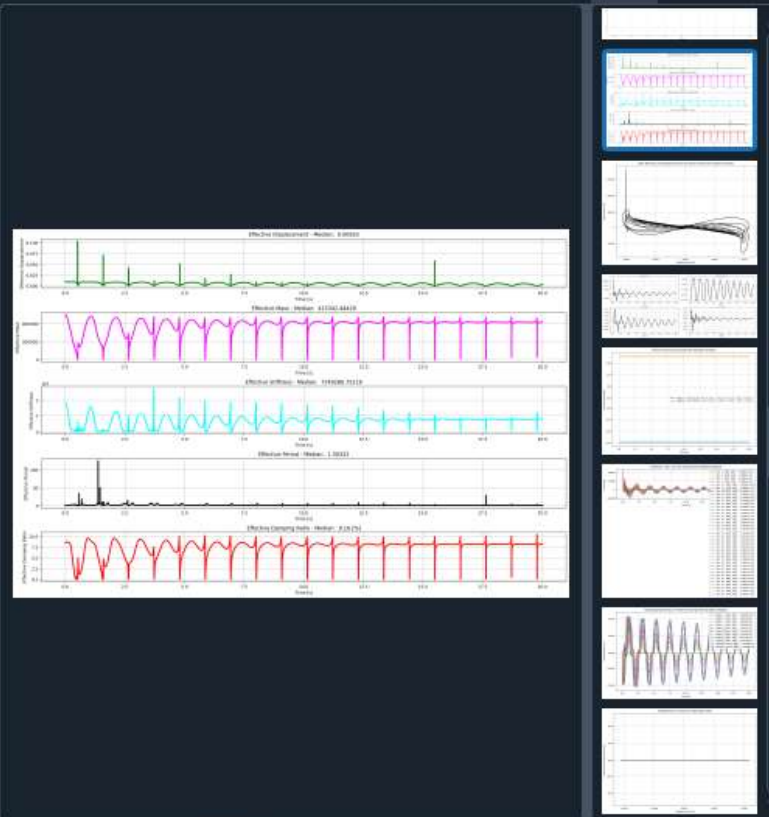
C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...OMPUTE_EFFECTIVE_PROPERTIES_FUN_FREE_VIBRATION.py

P(t)_POST-FIRE_MDOF_8_ANA.py X COMPUTE_EFFECTIVE_..._FREE_VIBRATION.py X

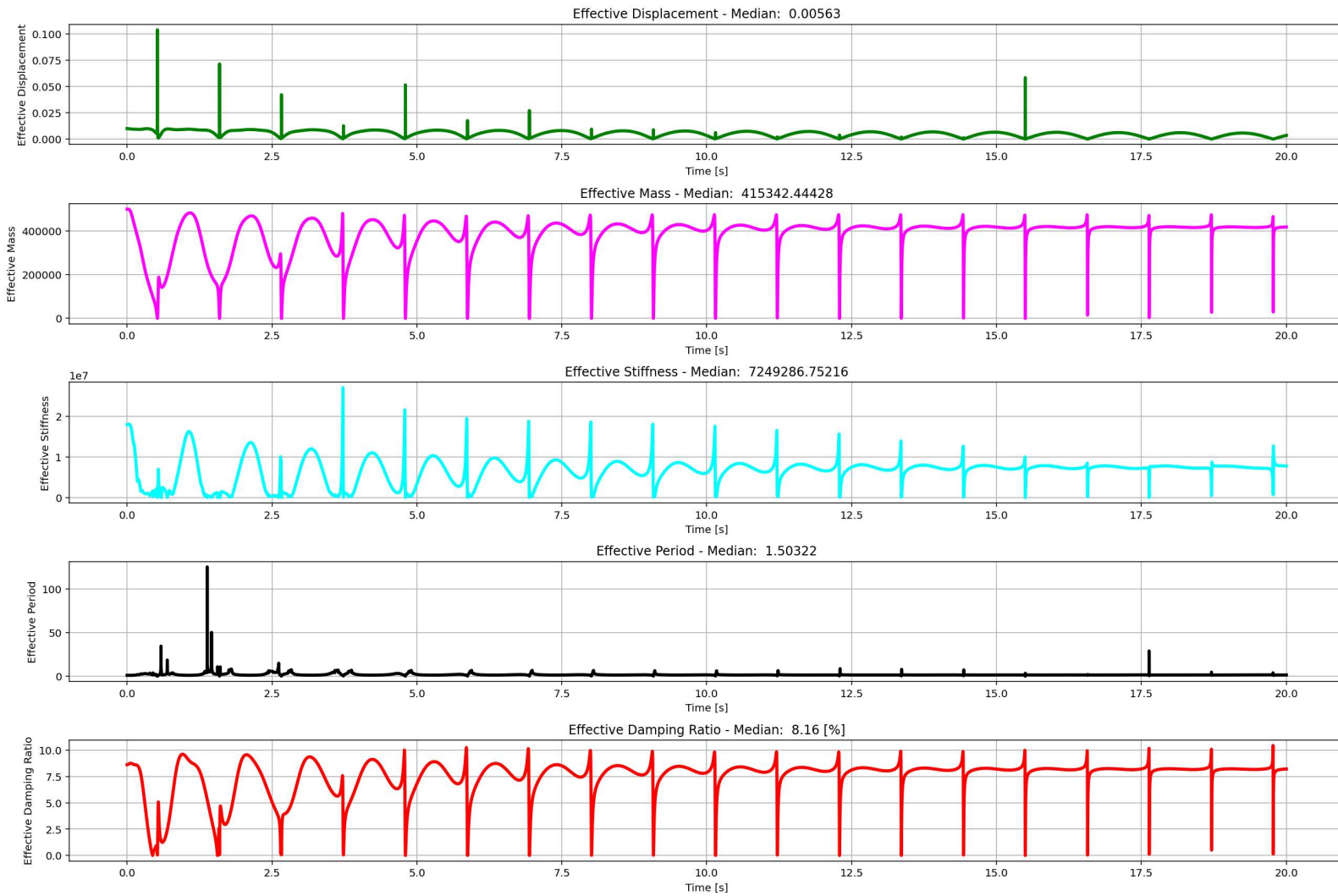
```
1  """ EQUIVALENT SDOF SYSTEM DERIVATION VIA DISPLACEMENT-BASED SEISMIC DESIGN PROCEDURE WITH FREE-VIBRATION
2  # Change MDOF to SDOF System with Displacement Based Design Concept
3  # THIS PYTHON SCRIPT IS WRITTEN BY SALAR DELAVAR GHASHGHAEE (QASHQAI)
4  """
5  This script implements a displacement-based pushover transformation,
6  converting a multi-degree-of-freedom (MDOF) system into an equivalent
7  single-degree-of-freedom (SDOF) system for seismic assessment.
8  It calculates effective modal properties-displacement, mass, and
9  stiffness-by weighting element forces and nodal displacements according
10 to a presumed deformed shape.
11 The derived effective period provides a simplified dynamic characteristic
12 for performance-based engineering. The visualizations effectively track the
13 evolution of these equivalent parameters throughout the nonlinear analysis steps.
14 """
15 #-----
16 import numpy as np
17
18 displacement_X_1 = np.array(list(node_displacements.values())[1]) # DOF 02
19 displacement_X_2 = np.array(list(node_displacements.values())[2]) # DOF 03
20 displacement_X_3 = np.array(list(node_displacements.values())[3]) # DOF 04
21 displacement_X_4 = np.array(list(node_displacements.values())[4]) # DOF 05
22 displacement_X_5 = np.array(list(node_displacements.values())[5]) # DOF 06
23 displacement_X_6 = np.array(list(node_displacements.values())[6]) # DOF 07
24 displacement_X_7 = np.array(list(node_displacements.values())[7]) # DOF 08
25 displacement_X_8 = np.array(list(node_displacements.values())[8]) # DOF 09
26 displacement_X_9 = np.array(list(node_displacements.values())[9]) # DOF 10
27 displacement_X_10 = np.array(list(node_displacements.values())[10]) # DOF 11
28
29 ele_force_01 = np.array(list(ele_force.values())[0]) # ELEMENT 01
30 ele_force_02 = np.array(list(ele_force.values())[1]) # ELEMENT 02
31 ele_force_03 = np.array(list(ele_force.values())[2]) # ELEMENT 03
32 ele_force_04 = np.array(list(ele_force.values())[3]) # ELEMENT 04
33 ele_force_05 = np.array(list(ele_force.values())[4]) # ELEMENT 05
34 ele_force_06 = np.array(list(ele_force.values())[5]) # ELEMENT 06
```

...GY DISSIPATION_CAPACITY_INDEX\57_POST_FIRE+\EXAMPLE_04_MDOF

14 %



IPython Console Plots Variable Explorer Help Debugger History Files



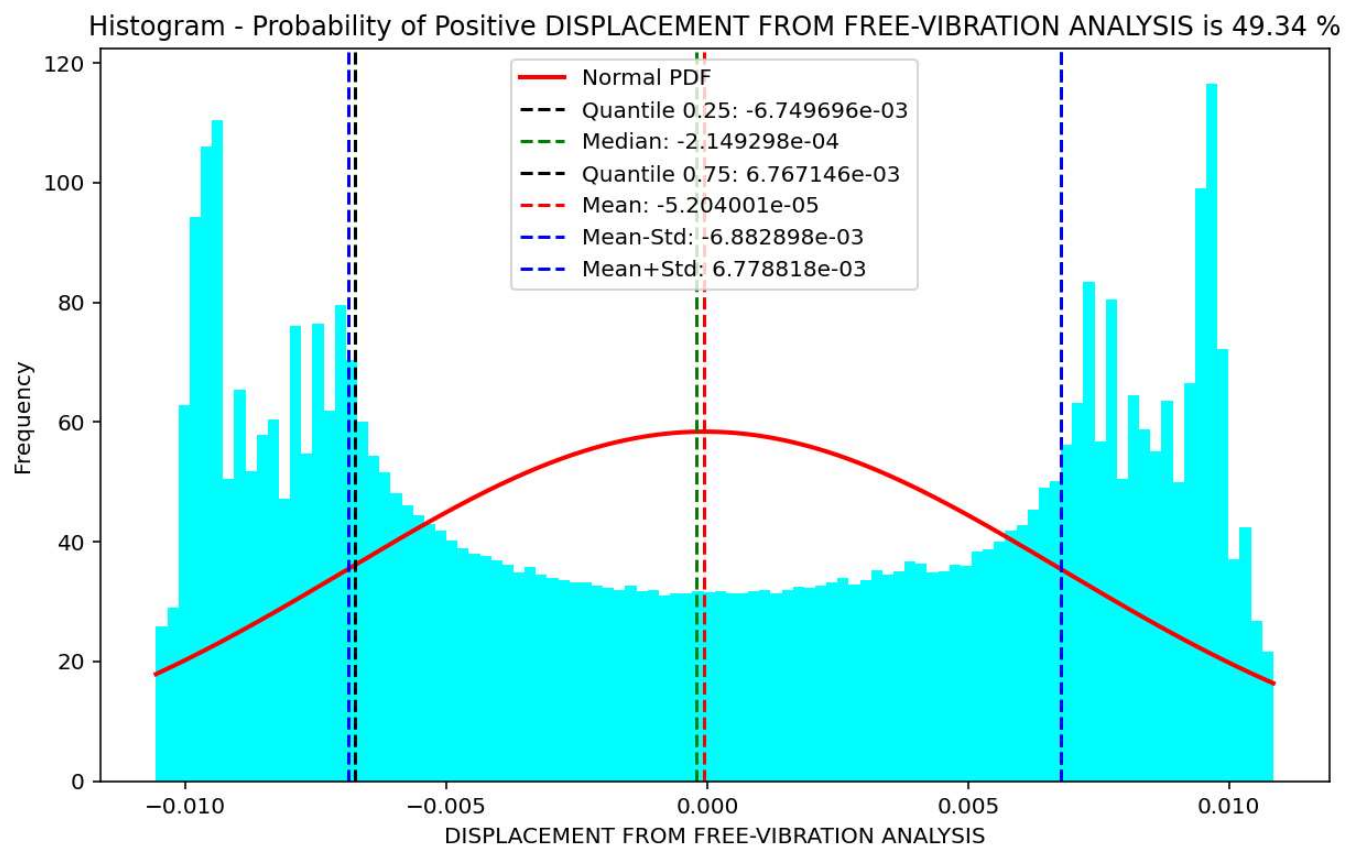
$$\Delta_d = \frac{\sum_{i=1}^n m_i \Delta_i^2}{\sum_{i=1}^n m_i \Delta_i}$$

$$m_{eff} = \frac{\sum_{i=1}^n m_i \Delta_i}{\Delta_d}$$

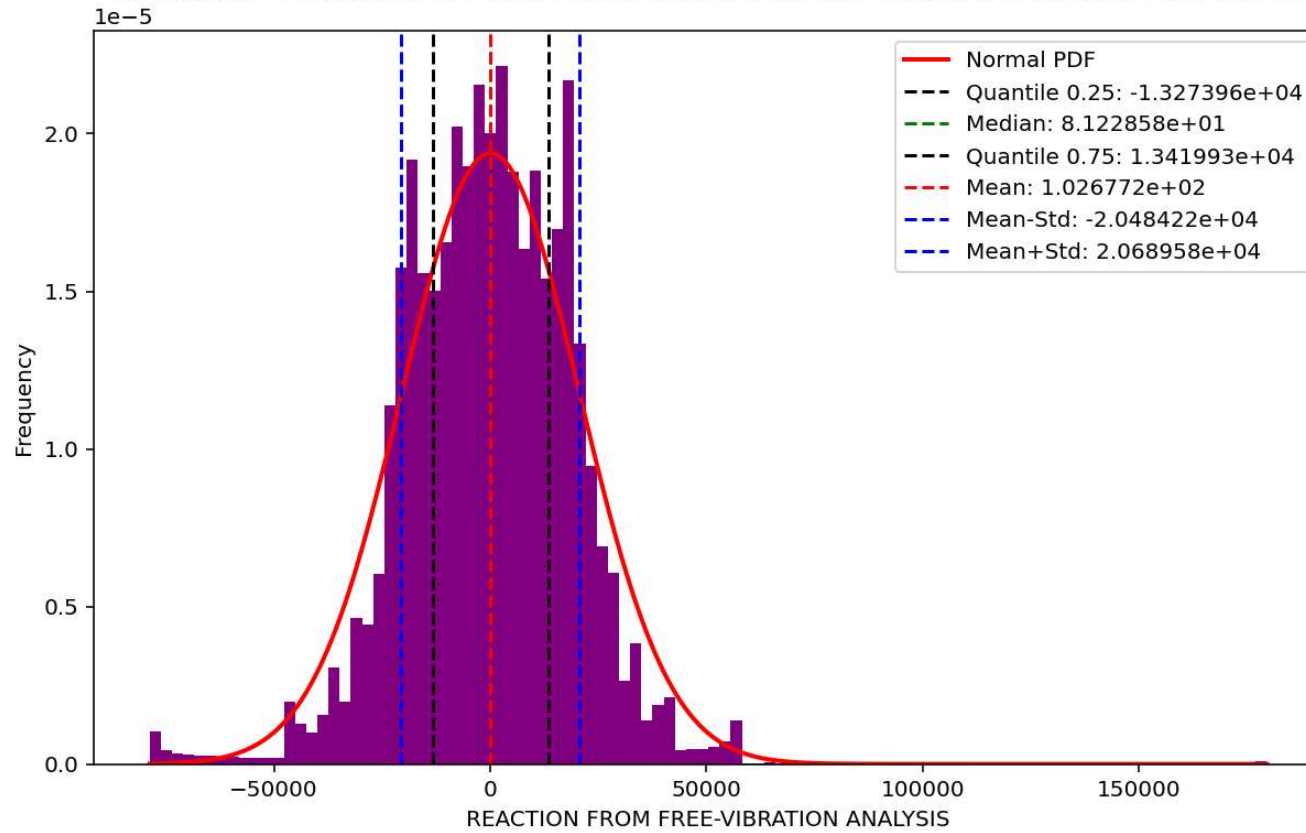
$$k_{eff} = \frac{\sum_{i=1}^n k_i \Delta_i}{\Delta_d}$$

$$T_{eff} = \frac{2\pi}{\sqrt{\frac{k_{eff}}{m_{eff}}}}$$

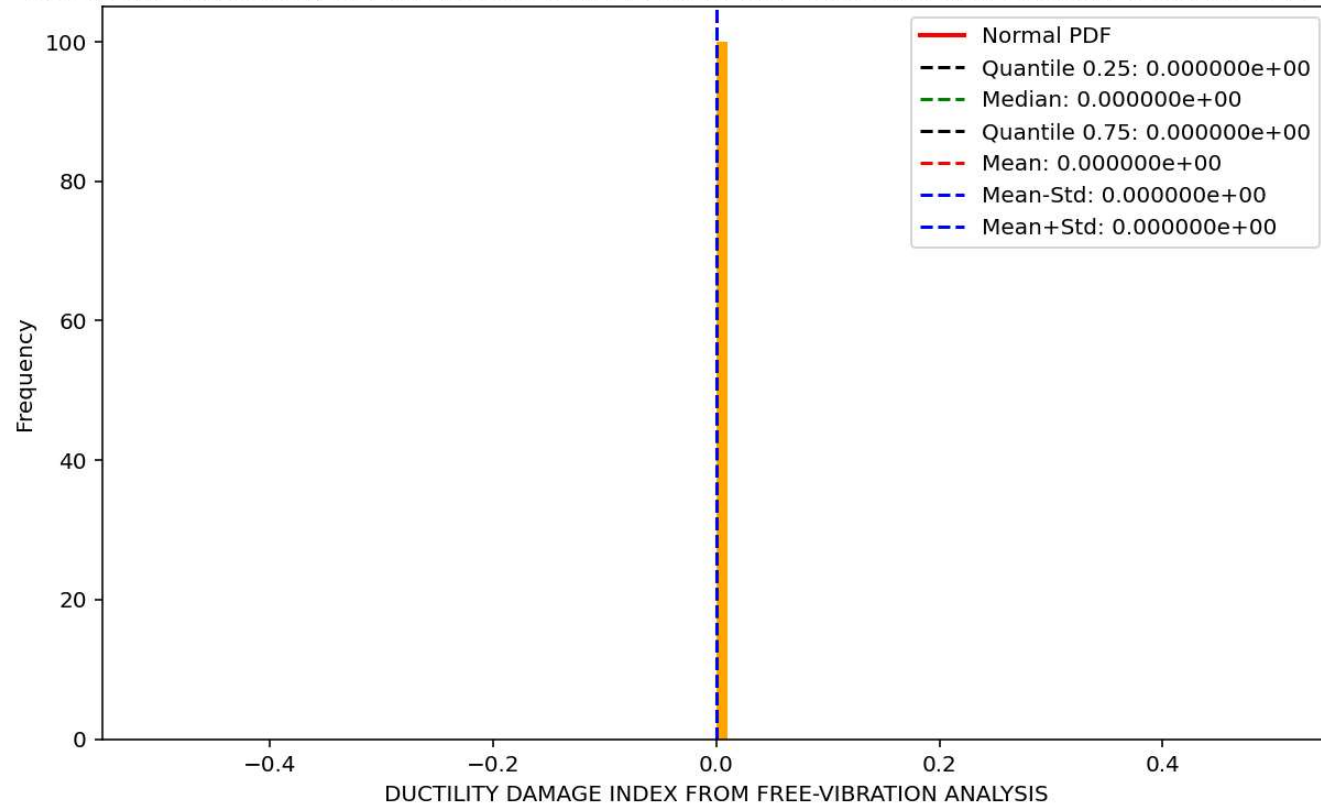
$$\xi_{eff} = \frac{\sum_{i=1}^n \xi_i \Delta_i}{\Delta_d}$$



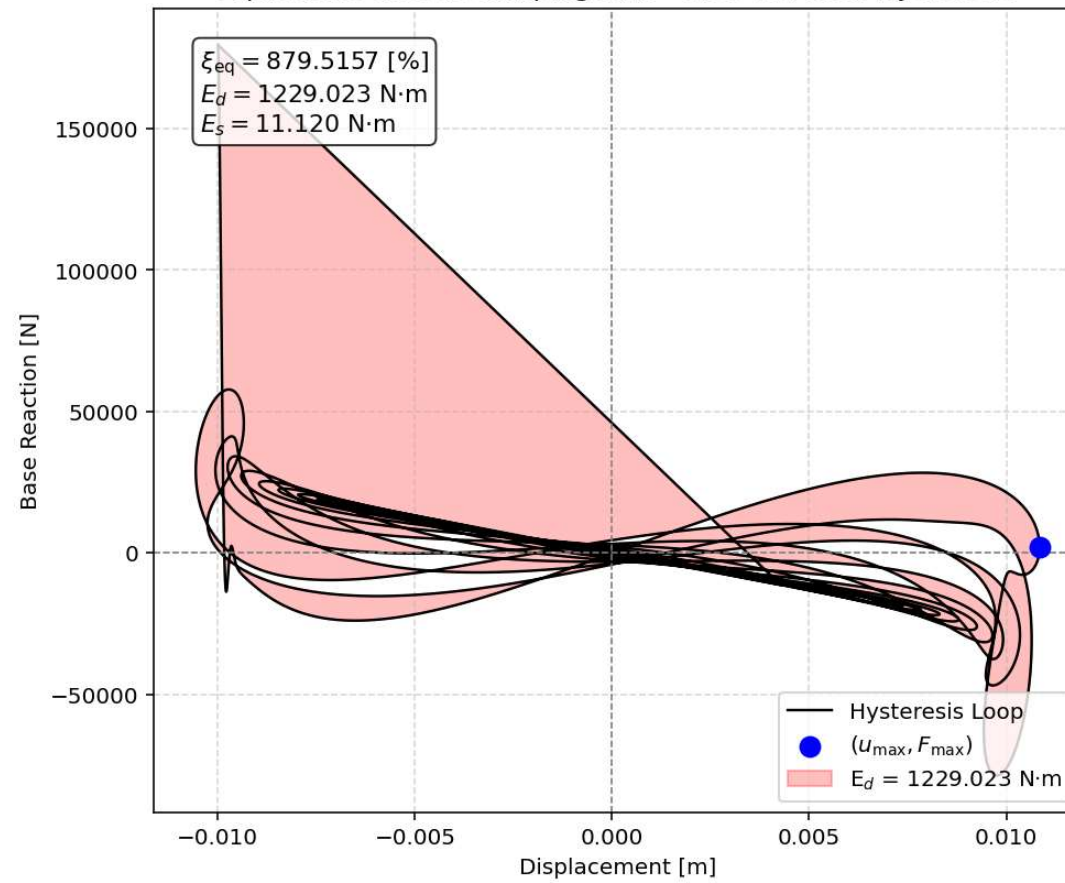
Histogram - Probability of Positive REACTION FROM FREE-VIBRATION ANALYSIS is 50.18 %

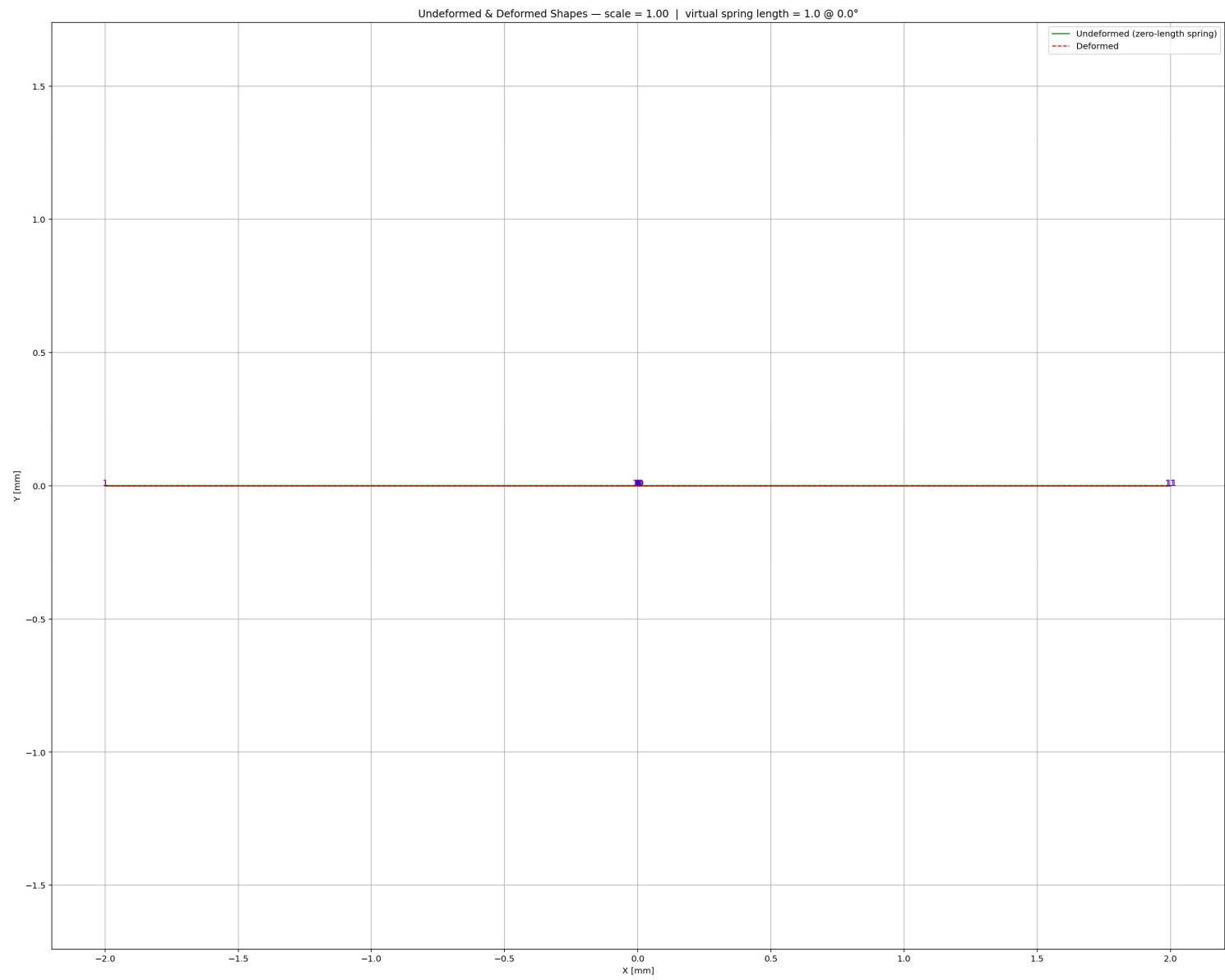


Histogram - Probability of Positive DUCTILITY DAMAGE INDEX FROM FREE-VIBRATION ANALYSIS is 0.00 %



Equivalent viscous damping ratio - Free-vibration Hysteresis





SEISMIC ANALYSIS

C:\Users\Dell\Desktop\OPENSEES_FILES\EIGHT_ANALY...IRE+\EXAMPLE_04_MDOF\P(t)_POST-FIRE_MDOF_8_ANA.py

P(t)_POST-FIRE_MDOF_8_ANA.py X COMPUTE_EFFECTIVE_..._FREE_VIBRATION.py X

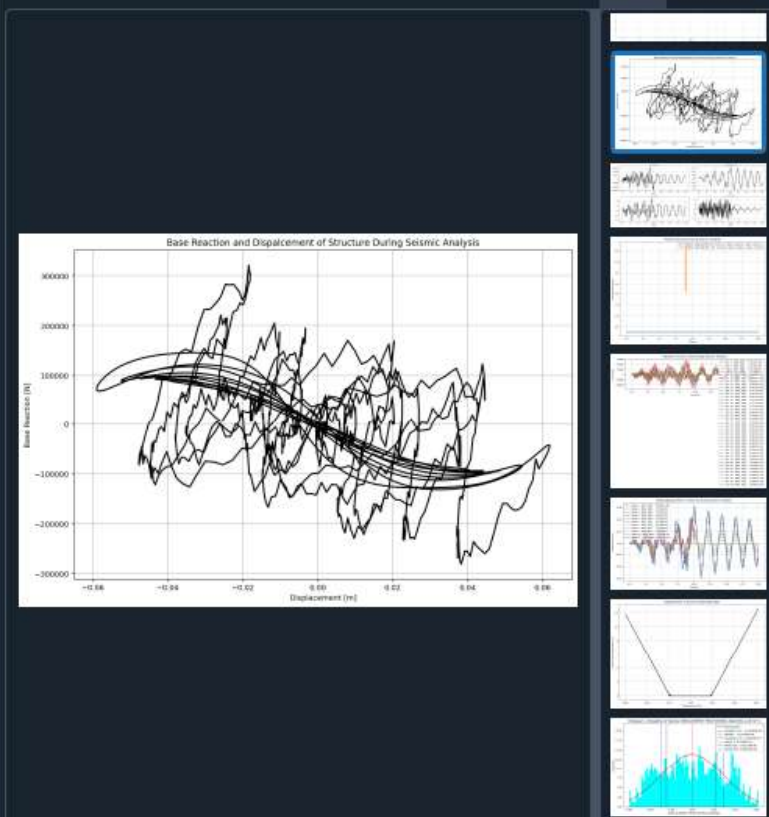
```

1446 # SEISMIC ANALYSIS (DYNAMIC TIME-HISTORY ANALYSIS)
1447 MAT_TYPE = 'INELASTIC' # 'ELASTIC' OR 'INELASTIC'
1448 ANAL_TYPE = 'SEISMIC'
1449
1450 DATA = MDOF(MAT_TYPE, TOTAL_MASS, ANAL_TYPE)
1451
1452 (time_SEI, reaction_SEI, disp_SEI, velo_SEI, acc_SEI, DI_SEI,
1453  ele_force_SEI, node_displacements_SEI,
1454  stiffness_SEI, PERIOD_SEI, damping_ratio_SEI,
1455  PERIOD_MIN_SEI, PERIOD_MAX_SEI) = DATA
1456
1457
1458 XDATA = disp_SEI
1459 YDATA = reaction_SEI
1460 XLABEL = 'Displacement [m]'
1461 YLABEL = 'Base Reaction [N]'
1462 TITLE = 'Base Reaction and Displacment of Structure During Seismic Analysis'
1463 COLOR = 'black'
1464 SEMIOLOGY = False
1465 PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMIOLOGY)
1466
1467 PLOT_TIME_HISTORY(time_SEI, reaction_SEI, disp_SEI, velo_SEI, acc_SEI)
1468
1469 # PLOT STRUCTURAL PERIOD DURING THE ANALYSIS
1470 plt.figure(0, figsize=(12, 8))
1471 plt.plot(time_SEI, PERIOD_MIN_SEI, linewidth=3)
1472 plt.plot(time_SEI, PERIOD_MAX_SEI, linewidth=3)
1473 plt.title('Period of Structure During Seismic Analysis')
1474 plt.ylabel('Structural Period [s]')
1475 plt.xlabel('Time [s]')
1476 #plt.semilogy()
1477 plt.grid()
1478 plt.legend([f'PERIOD - MIN VALUES: Min: {np.min(PERIOD_MIN_SEI):.3f} (s) - Mean: {np.mean(PERIOD_MIN_SEI):.3f} (s)',
1479            f'PERIOD - MAX VALUES: Min: {np.min(PERIOD_MAX_SEI):.3f} (s) - Mean: {np.mean(PERIOD_MAX_SEI):.3f} (s)'])

```

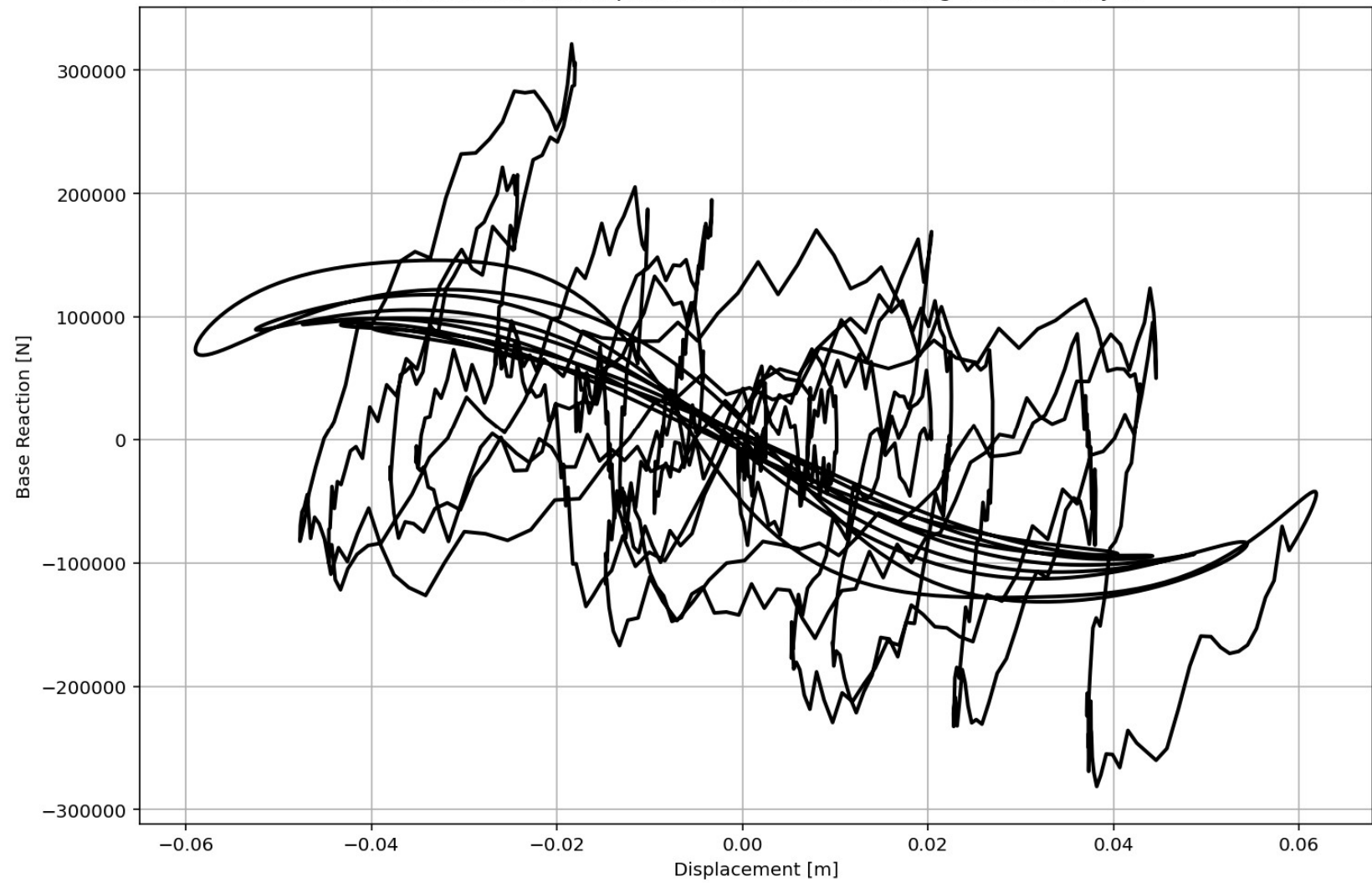
...GY DISSIPATION_CAPACITY_INDEX\57_POST_FIRE+\EXAMPLE_04_MDOF

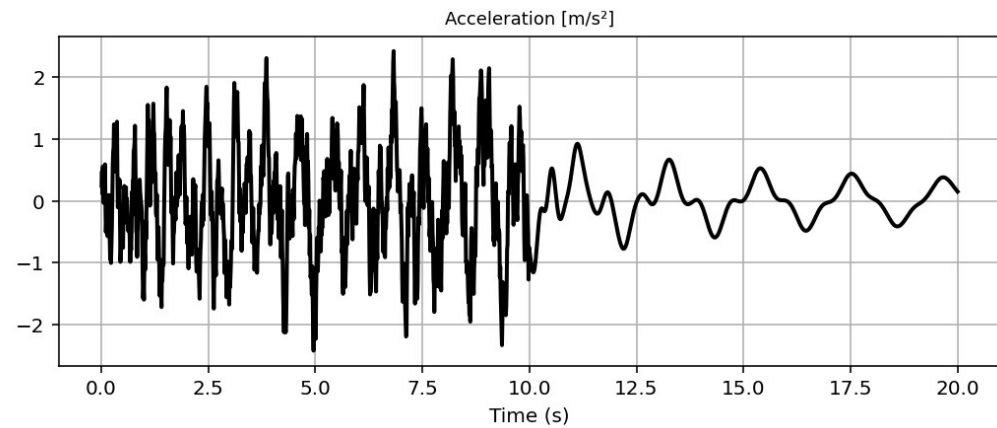
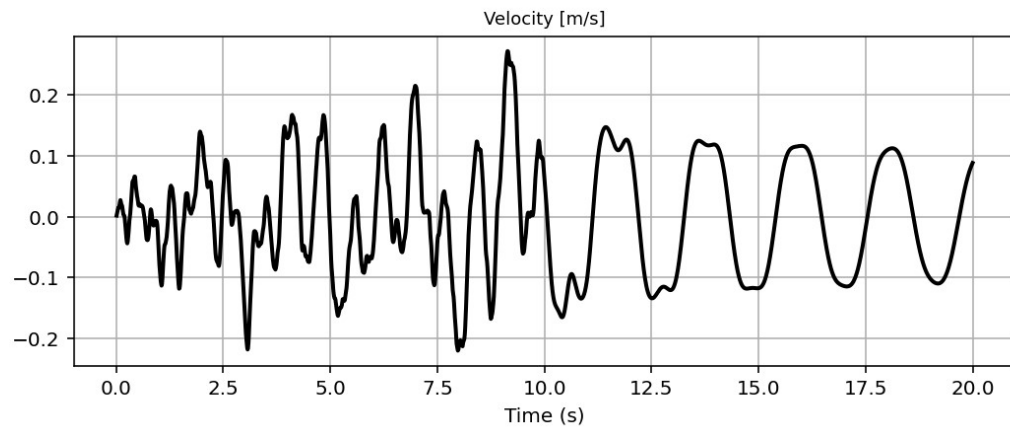
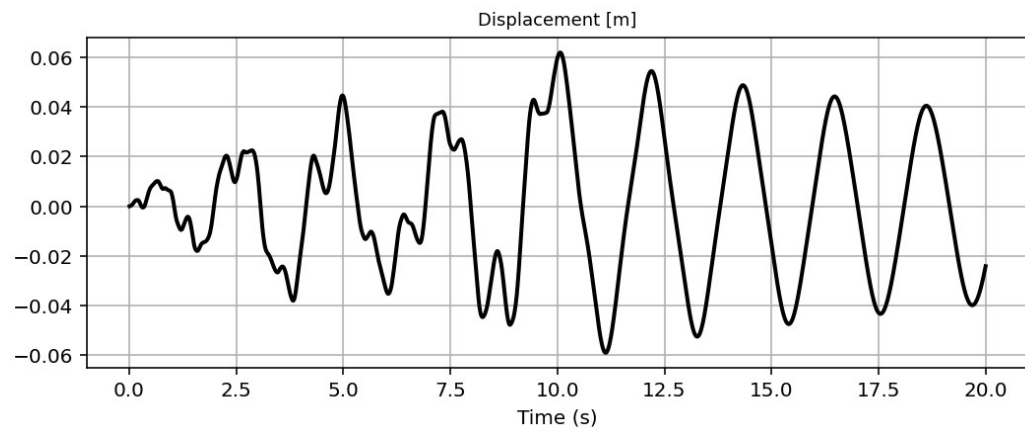
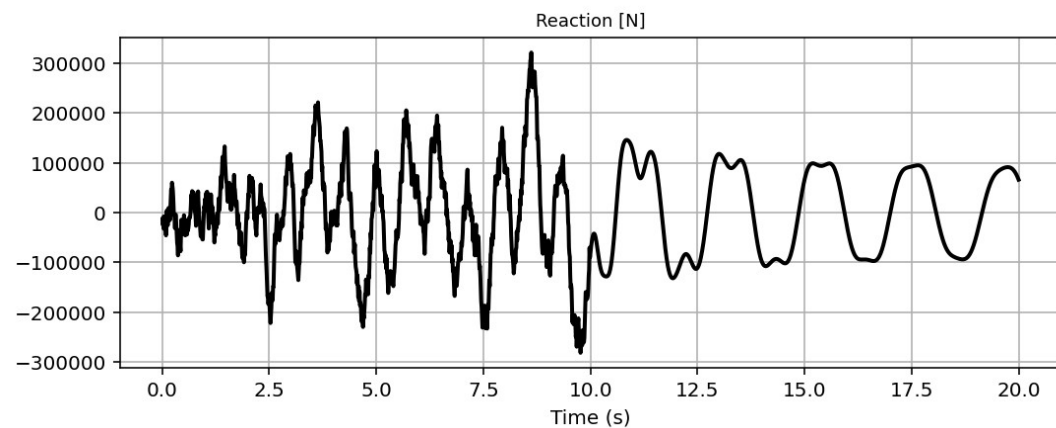
24 %



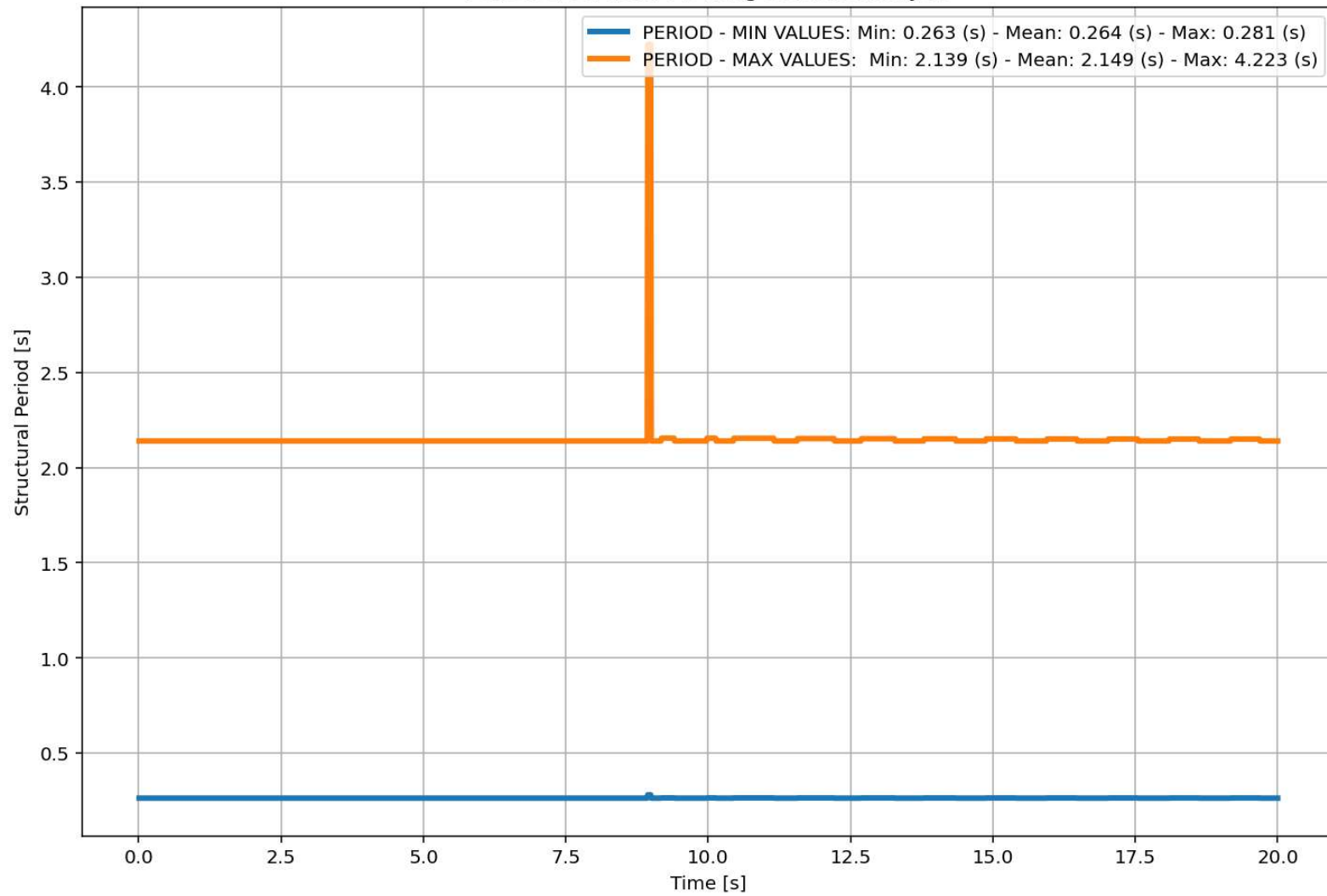
IPython Console Plots Variable Explorer Help Debugger History Files

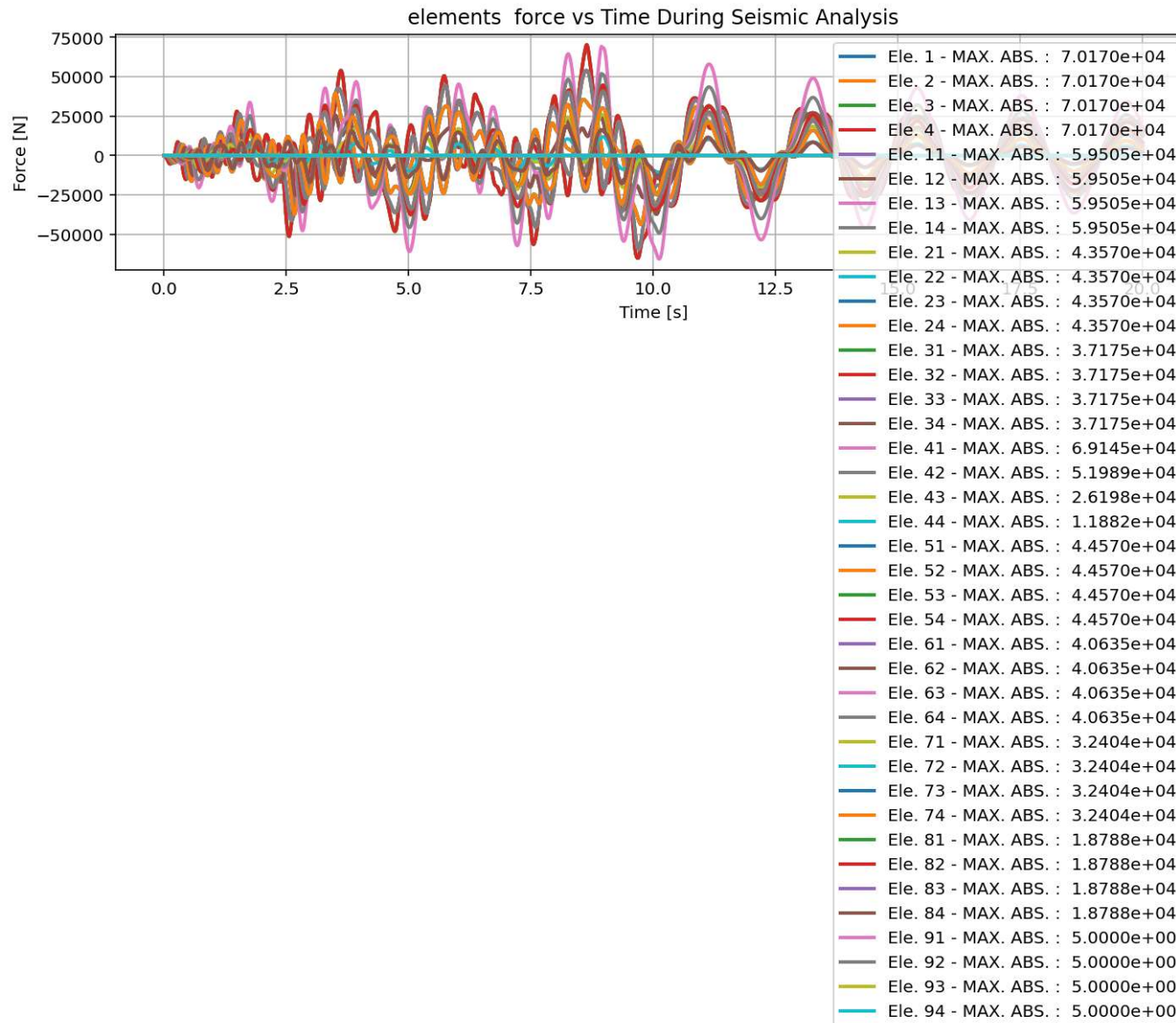
Base Reaction and Dispalcement of Structure During Seismic Analysis



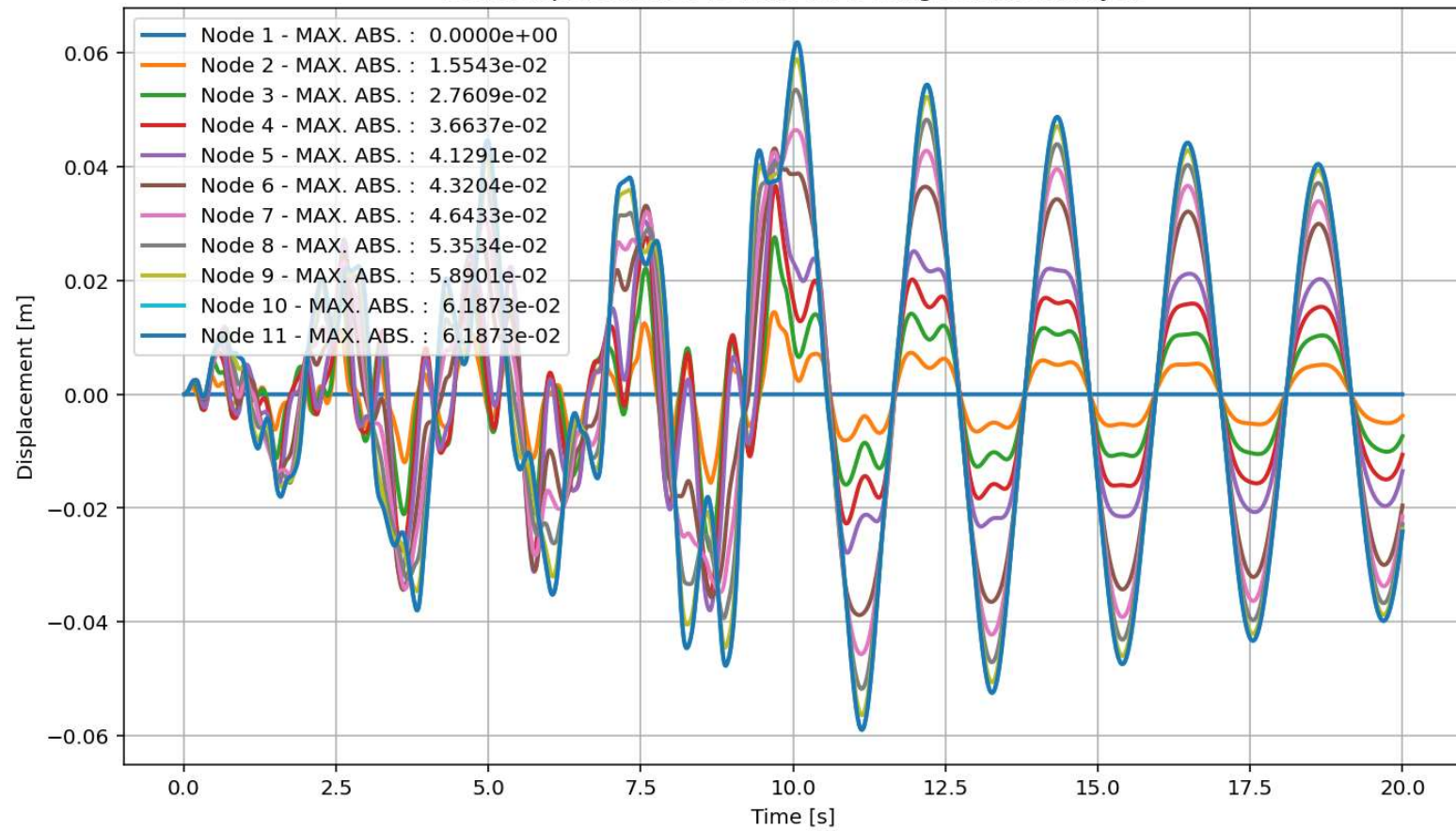


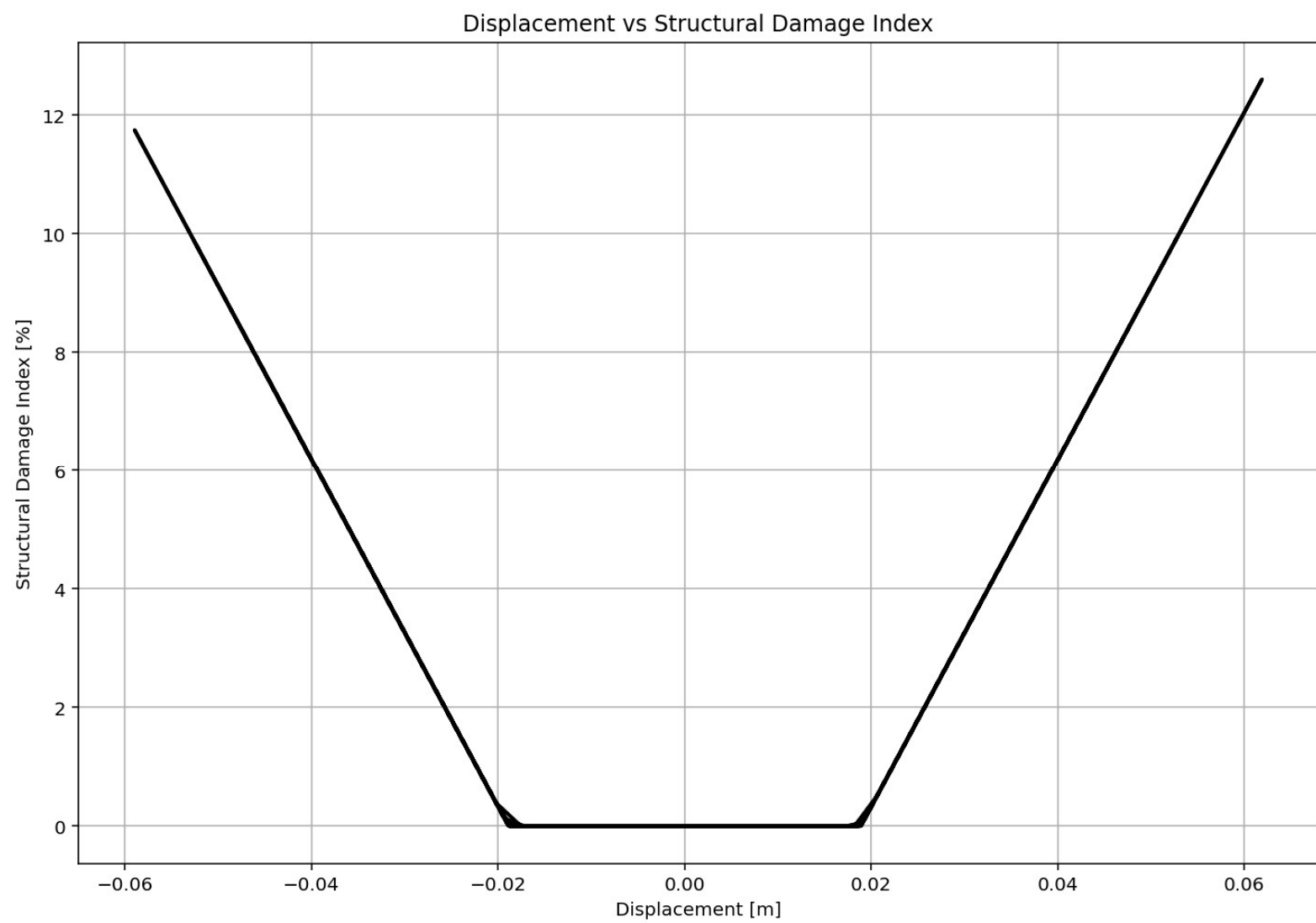
Period of Structure During Seismic Analysis



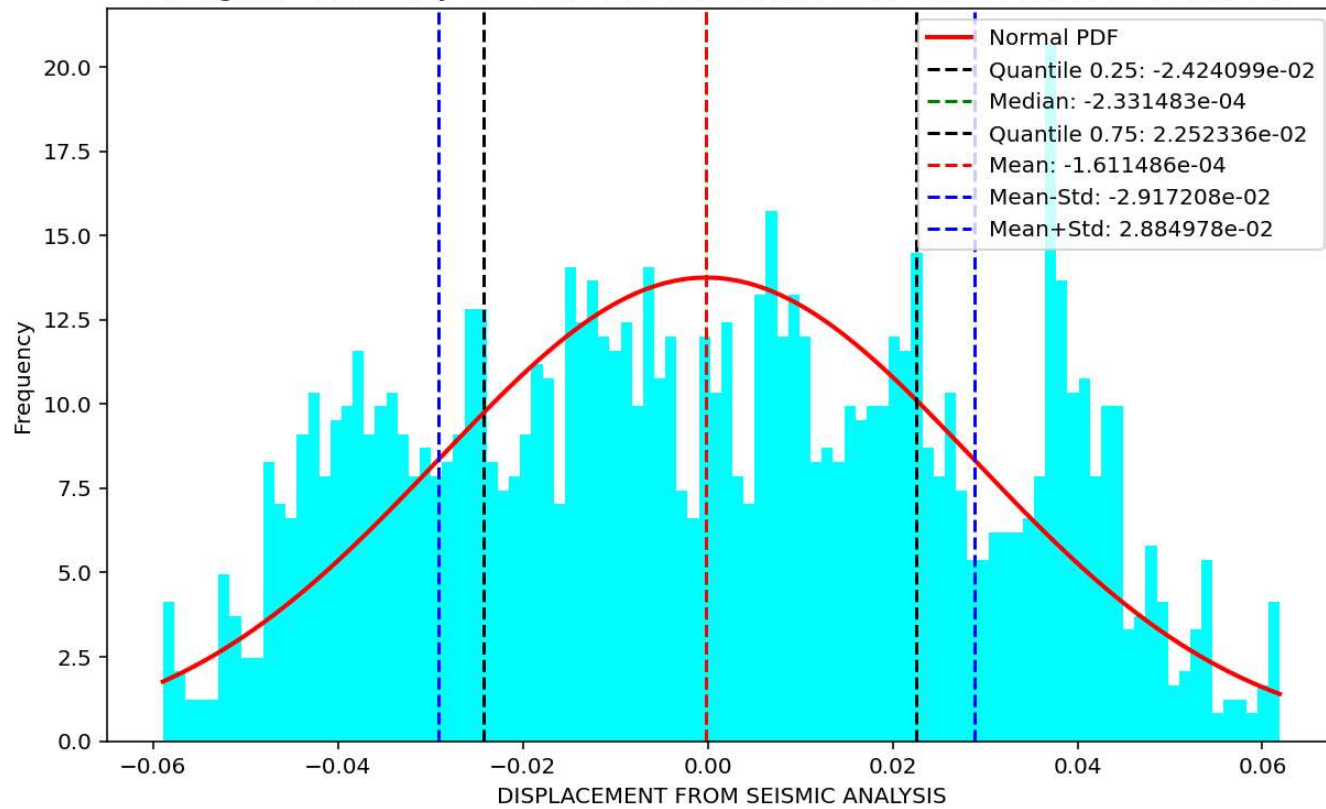


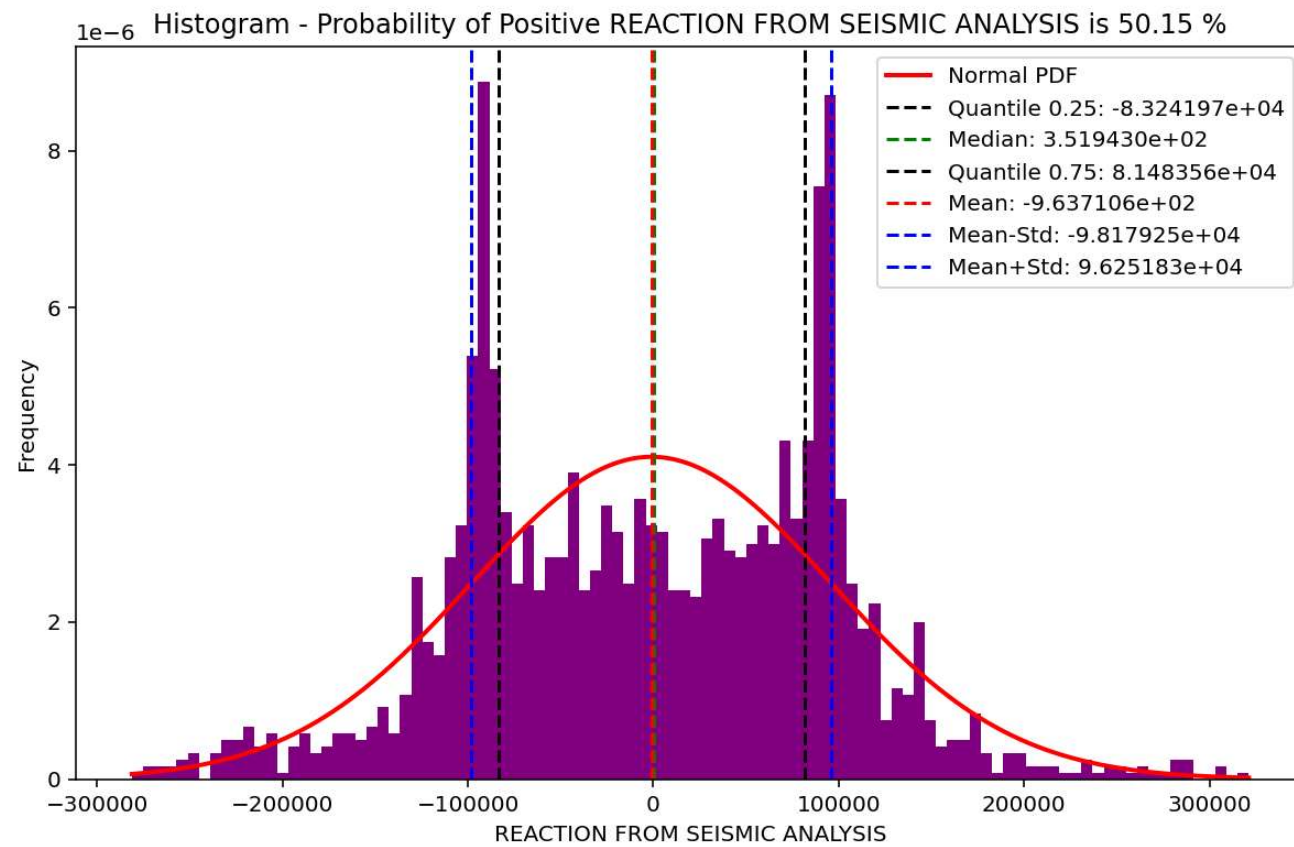
Node Displacements vs Time for During Seismic Analysis

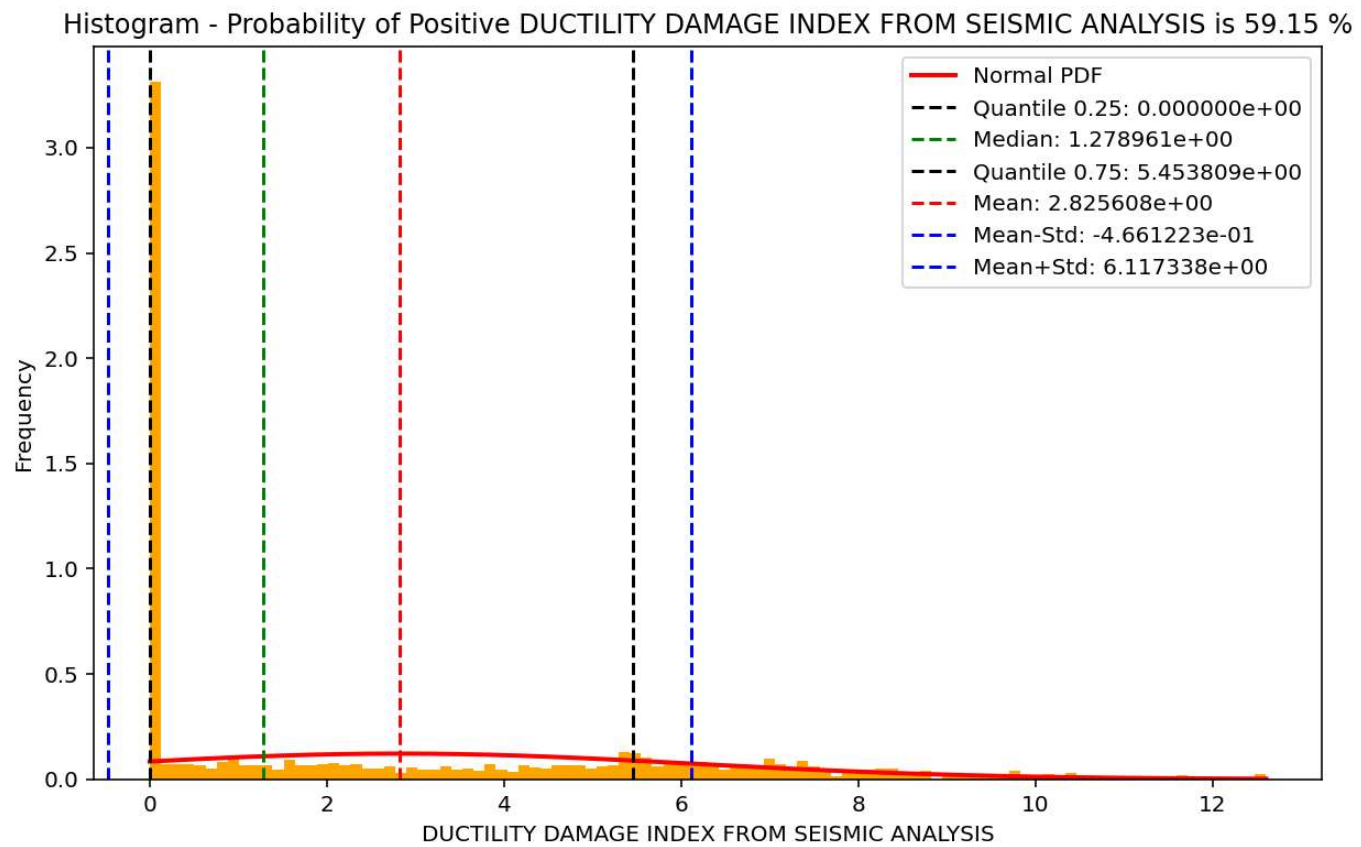




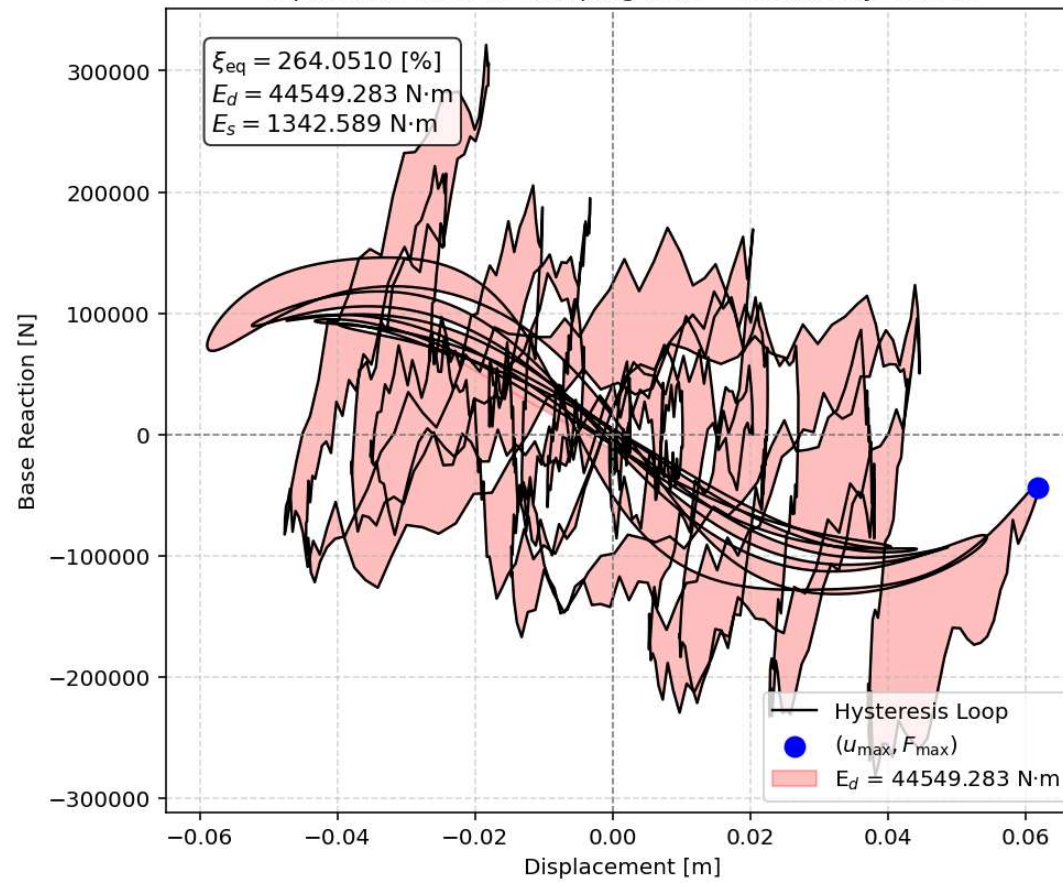
Histogram - Probability of Positive DISPLACEMENT FROM SEISMIC ANALYSIS is 49.70 %

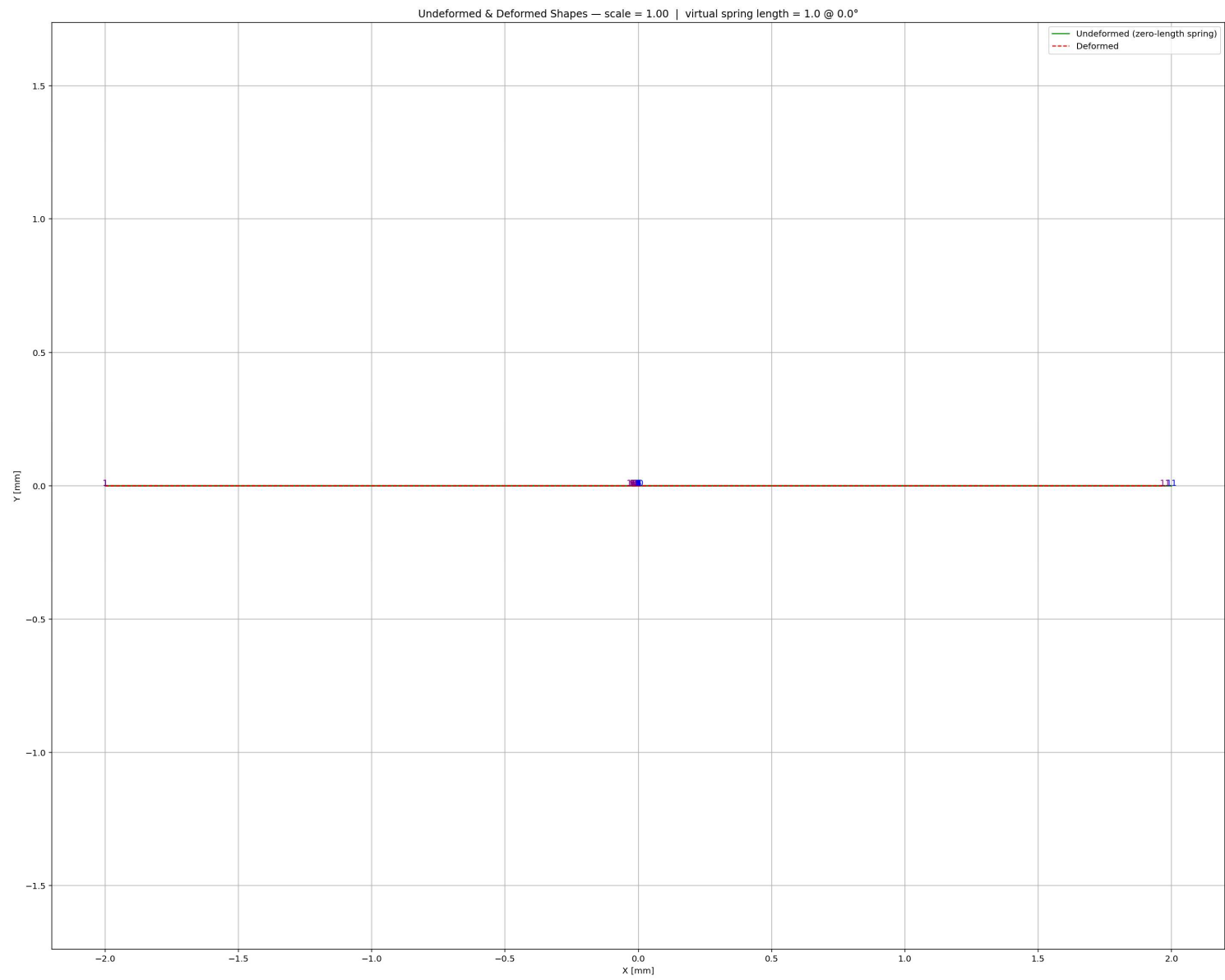


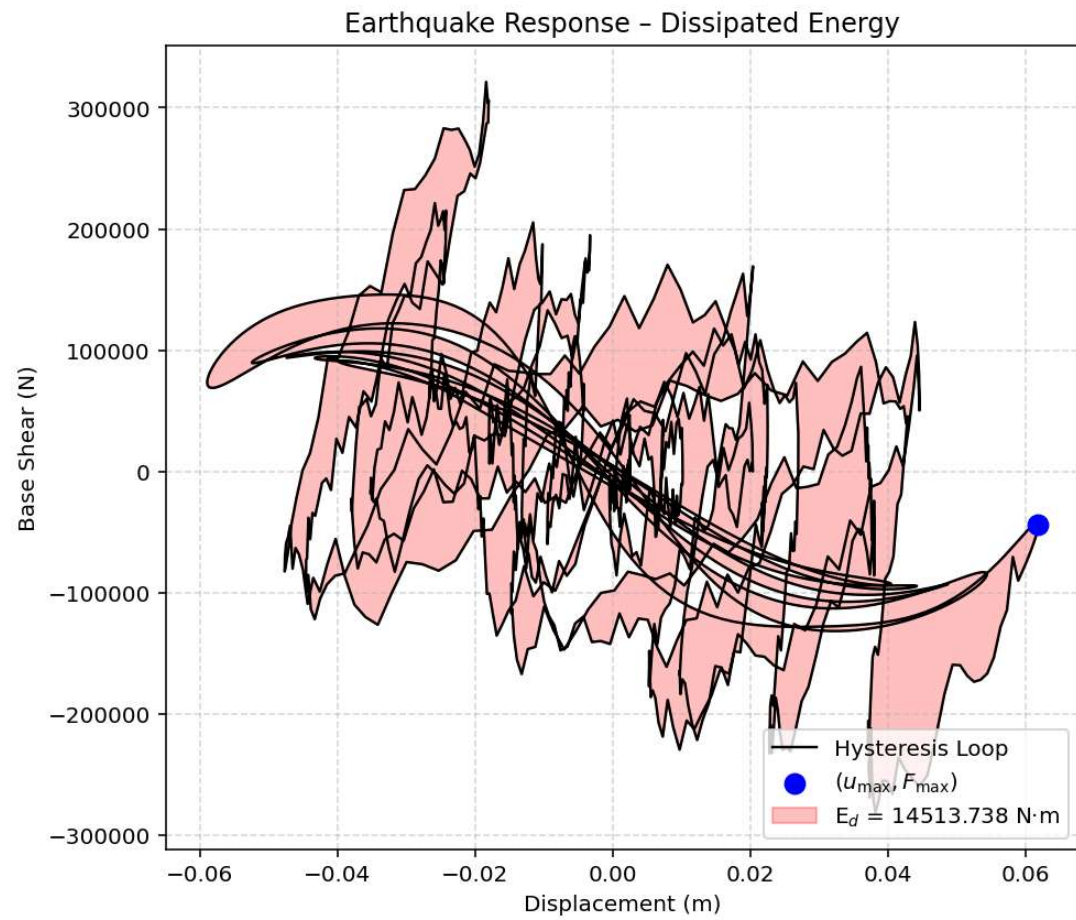


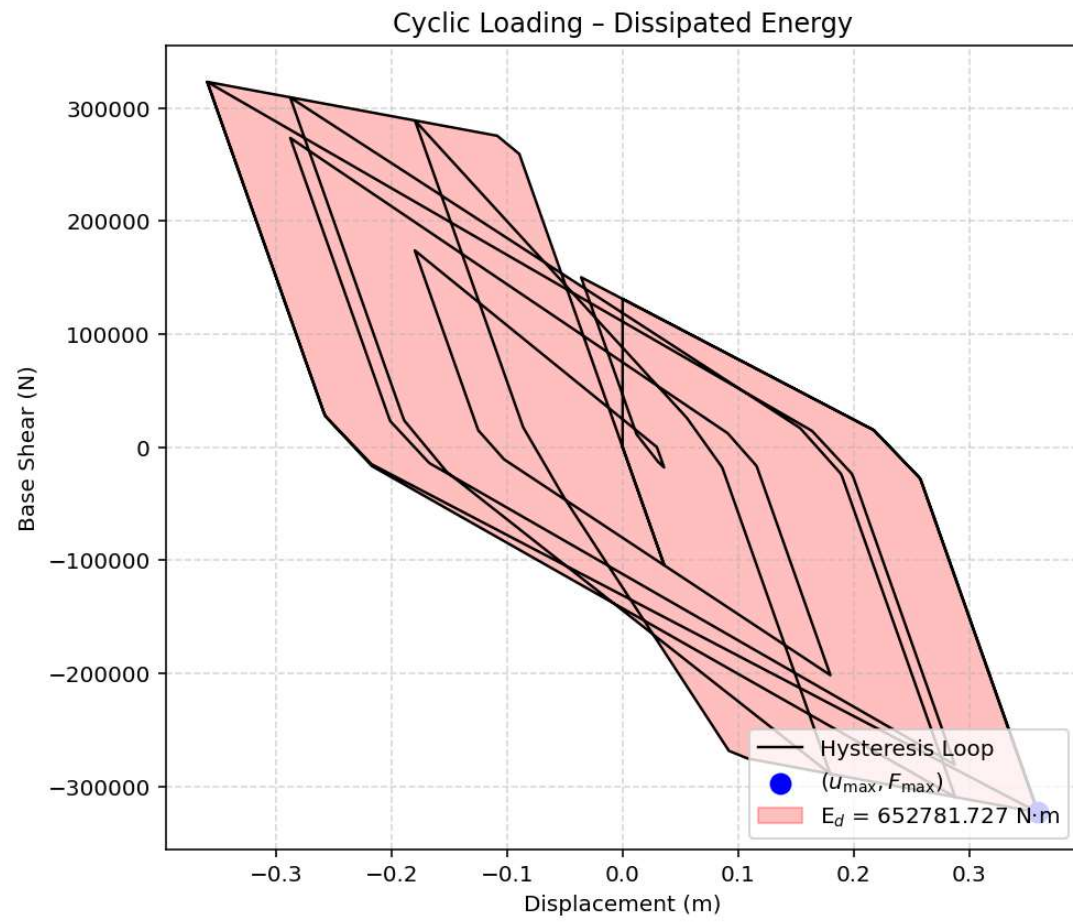


Equivalent viscous damping ratio - Seismic Hysteresis









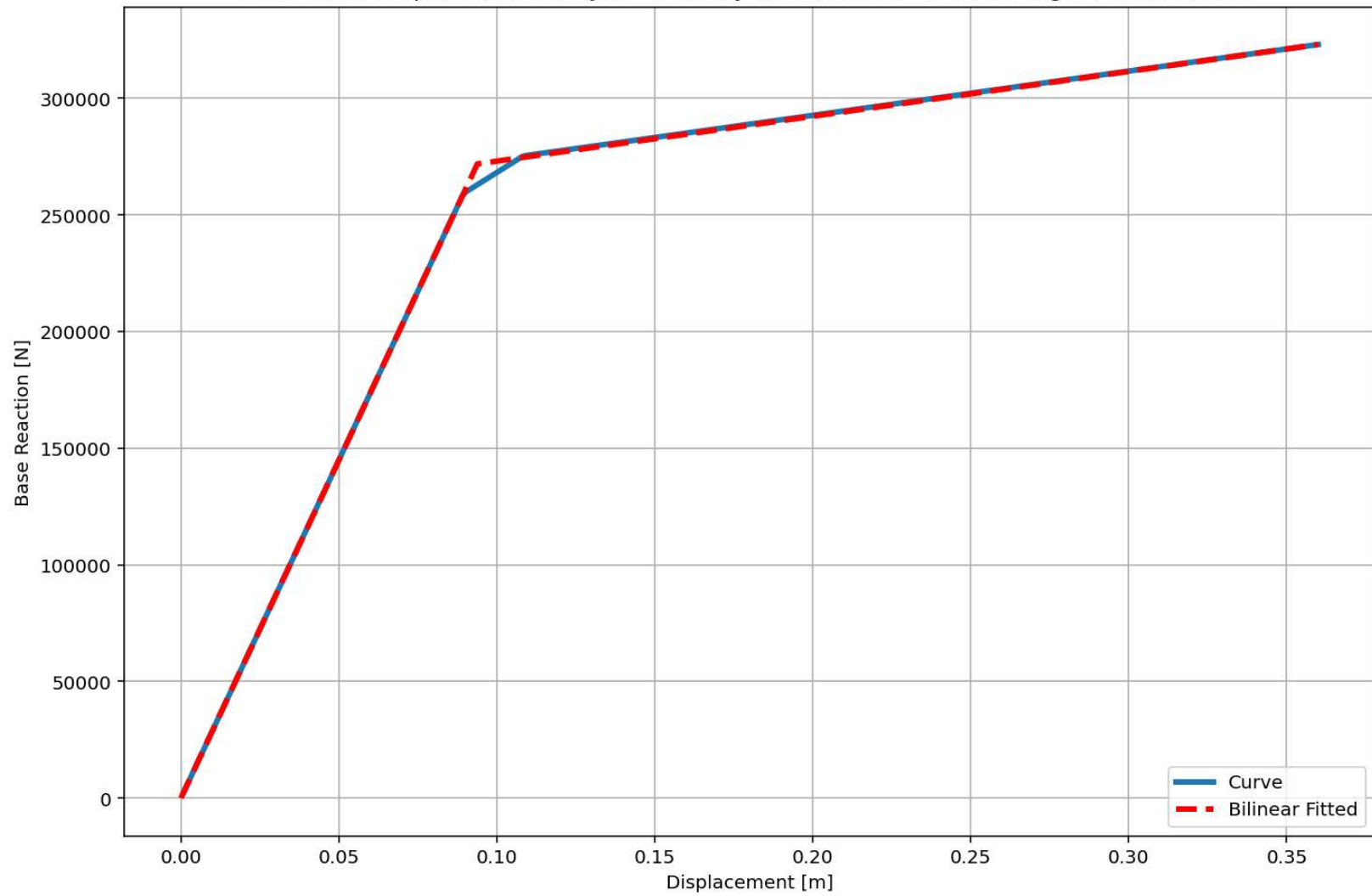
Dissipated Energy from Earthquake= 14513.74 N·m

Dissipated Energy from Cyclic Displacement= 652781.73 N·m

DISSIPATED ENERGY CAPACITY INDEX: 2.223 [%]

ZONE 1: VERY MINOR DAMAGE

Base Shear-Displacement Analysis - Ductility Ratio: 3.8370 - Over Strength Factor: 1.1883



+=====+

= Analysis curve fitted =

Disp Base Shear

[[0.00000000e+00 0.00000000e+00]

[9.38225157e-02 2.71884725e+05]

[3.60000000e-01 3.23080863e+05]]

+=====+

+-----+

Structure Elastic Stiffness : 2897862.23

Structure Plastic Stiffness : 897446.84

Structure Tangent Stiffness : 192338.35

Structure Ductility Ratio : 3.84

Structure Over Strength Factor: 1.19

+-----+

Over Strength Coefficient (Ω_0): 1.1883

Displacement Ductility Ratio (μ): 3.8370

Ductility Coefficient (R_μ): 3.8370

Structural Behavior Coefficient (R): 4.5595

Structural Ductility Damage Index in X Direction: -12.0030 (%)

Seismic Fragility Curves by Damage State from DYN. TIME-HISTORY ANA. DUCTILITY DAMAGE INDEX [%]

